Faculty of Artificial Intelligence Engineering

Department of Informatics

Software Engineering and Information Systems

الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

Language Learning Platform Using Games

Graduation Project 1 – Software Engineering and Information Systems

Prepared by

Noor Sheikha Ayat Mahmoud

Supervised by

Dr. Shadi Al-Bleidi

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إقرار المشرف

أشهد بأن إعداد هذا المشروع الموسوم...

....

و المعد من قبل الطلاب...

....

وقد تم تحت اشرافي في قسم هندسة البرمجيات ونظم المعلومات-كلية الهندسة , وهو جزء من متطلبات نيل الإجازة في هندسة البرمجيات ونظم المعلومات.

المرتبة العلمية:...

الأسم: ....

التوقيع:..

التاريخ:..

شهادة الإشراف

الملخص

يمثل هذا المشروع منصة تعليمية رقمية مبتكرة تهدف إلى إحداث تحول جذري في طريقة تعلم اللغات، من خلال اعتماد الألعاب التفاعلية كوسيلة تعليمية فعّالة تجمع بين المتعة والفائدة. فبدلاً من الأساليب التقليدية الجامدة التي قد تفتقر إلى التشويق والتحفيز، يوفر النظام تجربة تعليمية مرنة، ديناميكية، وشاملة، تتيح للمتعلمين الانخراط في أنشطة وألعاب مصممة وفق مستويات مدروسة تضمن التدرج في اكتساب المهارات اللغوية الأساسية والمتقدمة.

تتيح المنصة للمتعلمين اختيار اللغة التي يرغبون في تعلمها، ثم خوض اختبارات تحديد المستوى التي تساعد في تقييم قدراتهم الأولية وتوجيههم إلى المسار التعليمي الأنسب لأهدافهم. كما توفر آليات فورية لعرض النتائج بعد كل اختبار أو لعبة، مع إمكانية تحديث مستوى المتعلم تلقائياً بناءً على أدائه الفعلي، مما يعزز مبدأ التعلم القائم على التقدم الذاتي. إضافة إلى ذلك، يستطيع المشرفون إدارة المحتوى التعليمي، إضافة مراحل جديدة، وضبط الأنشطة بما يتوافق مع المعايير التربوية الحديثة، مما يضمن استمرارية تطوير المنصة وتحديثها.

يعتمد المشروع في تطويره على منهجية **Agile Scrum** التي تركز على العمل التعاوني بين فرق صغيرة، والتطوير المرحلي، والاستجابة السريعة لمتطلبات المستخدمين. ويسهم هذا النهج في تحسين جودة النظام وضمان توافقه مع احتياجات المتعلمين المتغيرة. كما يدعم النظام خاصية الدفع الإلكتروني لتسهيل الاشتراك في الألعاب المدفوعة، ويعرض معلومات تفصيلية حول كل لعبة مثل المدة والمحتوى والتقييمات.

ولا يقتصر دور المنصة على تقديم المحتوى التعليمي فحسب، بل تسعى أيضاً إلى بناء مجتمع تعلم تفاعلي يتيح للمتعلمين التواصل وتبادل الخبرات وطرح الأسئلة، مما يعزز روح التعاون ويثري التجربة التعليمية. وبذلك يحقق المشروع رؤيته في تقديم تجربة تعلم لغوية حديثة، محفزة، وشاملة، تراعي الفروق الفردية وتدعم الاستمرارية في التعلم.

Abstract

This project introduces an innovative digital platform designed to transform the process of language learning by integrating interactive educational games with structured pedagogical content. The system aims to shift traditional learning approaches—often characterized by rigidity, limited engagement, and passive instruction—toward a more dynamic, enjoyable, and effective experience that blends education with entertainment.

Through its comprehensive digital environment, the platform enables learners to select the language they wish to acquire and engage in a variety of interactive games and activities distributed across multiple proficiency levels. This structure ensures gradual skill development and supports learners as they progress from foundational to advanced linguistic competencies.

A key feature of the platform is its ability to assess the learner's initial level through placement tests, which evaluate core language skills and determine the most suitable learning path. Based on the results, the system guides learners toward personalized training tracks that align with their abilities, goals, and pace. The platform also provides immediate feedback after each test or game, allowing learners to view their results, track their progress, and understand areas that require improvement. Additionally, the system can update the learner's level automatically when performance indicators demonstrate readiness for advancement.

From an administrative perspective, supervisors have access to tools that allow them to add new content, manage educational stages, and adjust activities in accordance with pedagogical standards.

This ensures that the platform remains adaptable, scalable, and capable of supporting continuous content enhancement.

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- **Diagrams && images:**

List of abbreviations

Abbreviation	Meaning / Definition
UI	User Interface – واجهة المستخدم
UX	User Experience – تجربة المستخدم
KPI	Key Performance Indicator – مؤشرات قياس الأداء
ERD	Entity Relationship Diagram – مخطط العلاقات بين الكيانات
MVC	Model–View–Controller – نمط معماري لفصل مكونات النظام
UML	Unified Modeling Language – لغة النمذجة الموحدة
RTM	Requirements Traceability Matrix – مصفوفة تتبع المتطلبات
FR	Functional Requirements – المتطلبات الوظيفية
NFR	Non-Functional Requirements – المتطلبات غير الوظيفية
Gantt	Gantt Chart – مخطط جانت الزمني
HTML	HyperText Markup Language – لغة بناء صفحات الويب
CSS	Cascading Style Sheets – لغة تنسيق صفحات الويب
JS	JavaScript – لغة برمجة الواجهة الأمامية
PHP	Hypertext Preprocessor – لغة برمجة خادم الويب
API	Application Programming Interface – واجهة برمجة التطبيقات
Scrum	Scrum Agile Methodology – منهجية تطوير مرنة تعتمد على السبرنت
SRS	Software Requirements Specification – وثيقة متطلبات النظام
LLP	Language Learning Platform – منصة تعلم اللغات
GBL	Game-Based Learning – التعلم القائم على الألعاب
ERD	Entity-Relationship Diagram
TC	Test case
Req	Requirement
UC	Use case
RTM	Requirements Traceability Matrix
MVC	Model-View-Controller

Chapter (1):
Introduction

1.1 Project Overview and Background:

This project presents an advanced educational platform designed to enhance language learning through interactive digital games. The system was developed in response to the growing need for modern, engaging, and effective learning tools that move beyond traditional instructional methods. By integrating entertainment with structured educational content, the platform aims to create a dynamic learning environment that motivates learners and supports their linguistic development across different proficiency levels.

1.2 Problem Statement and Its Practical and Academic Significance:

Traditional language-learning methods often lack engagement, personalization, and interactivity, leading to reduced learner motivation and limited long-term retention. This project addresses the need for a more immersive and flexible learning experience that aligns with modern learners' expectations. Academically, the system contributes to the field of educational technology by demonstrating how game-based learning can enhance linguistic acquisition. Practically, it offers a scalable solution that supports self-paced learning and accommodates diverse learner needs.

1.3 Project Scope, Limitations, and Assumptions:

The project focuses on developing a language-learning platform that integrates educational games, interactive activities, and personalized learning paths. The scope includes user registration, content management, progress tracking, and electronic payment support. Limitations may include dependency on internet connectivity and the need for continuous content updates. The project assumes that users possess basic digital literacy and access to compatible devices.

Main Objectives:

1. Develop an interactive language-learning platform that uses educational games to create an enjoyable and effective learning experience.
2. Enhance learner motivation by integrating gamification elements such as challenges and rewards.
3. Provide personalized learning experiences based on the learner's level, progress, and individual goals.
4. Support flexible and self-paced learning, allowing users to choose their preferred time and learning path.
5. Offer diverse and continuously updated educational content, including vocabulary, grammar, conversations, and interactive exercises.

6. Build a comprehensive digital learning environment that combines education and entertainment for all proficiency levels.

Sub-Objectives:

1. Provide simple, clear, and user-friendly interfaces suitable for all age groups.
2. Achieve high performance and fast system responsiveness to ensure a smooth user experience.
3. Ensure compatibility with various devices and browsers for broader accessibility.
4. Enhance visual design quality to make the learning experience more appealing and comfortable.
5. Offer an easy and flexible registration and login process.
6. Enable users to quickly access games and learning paths that match their proficiency level.
7. Provide an accurate progress-tracking system that clearly displays achievements and levels.

1.4 Project Contribution to Knowledge and the Field:

The project contributes to the advancement of game-based learning by offering a practical model that integrates pedagogical principles with interactive digital environments. It enriches the academic discourse on adaptive learning systems and provides empirical insights into how gamification can improve learner engagement, motivation, and performance.

1.5 Research Methodology and Applied Scientific Approaches:

The project adopts a descriptive and analytical research methodology supported by modern software-engineering practices. A user-centered design approach was applied to ensure that the system meets the needs and expectations of learners, administrators, and supervisors. The development process follows the Scrum agile methodology, which enables iterative planning, continuous improvement, and incremental delivery of system components.

From a technical perspective, the system is implemented using the Laravel framework, following the Model–View–Controller (MVC) architectural pattern to ensure modularity, scalability, and maintainability. Front-end development relies on HTML, CSS, and JavaScript to deliver an interactive and responsive user experience.

Data collection and analysis involve studying user requirements, evaluating existing language-learning platforms, and conducting iterative testing cycles. The scientific approach emphasizes usability testing, performance evaluation, and refinement based on user feedback and sprint reviews.

1.6 Organizational Structure of the Document:

The document is organized into ten chapters, each addressing a major component of the project:

1. **Chapter One:** Introduces the system and outlines the project's main objectives.
2. **Chapter Two:** Presents the project management and development methodology, including the use of Scrum and the adopted technologies.
3. **Chapter Three:** Reviews related work and previous studies relevant to the system.
4. **Chapter Four:** Focuses on requirements modeling and analysis.
5. **Chapter Five:** Presents the architectural design of the system.
6. **Chapter Six:** Describes the development environment and the tools used.
7. **Chapter Seven:** Details the software implementation phase.
8. **Chapter Eight:** Covers system testing procedures and evaluation results.
9. **Chapter Nine:** Discusses the findings, results, and analytical insights.
10. **Chapter Ten:** Provides the conclusion and recommendations for future work.

1.7 Definitions and Key Concepts:

1. Game-based learning: An educational approach that integrates game mechanics to enhance engagement and motivation.
2. Interactive activities: Tasks designed to promote active learner participation.
3. Adaptive learning paths: Personalized learning sequences based on user performance and progress.
4. User progress tracking: Monitoring and analyzing learner achievements across different stages.
5. Digital educational content: Structured multimedia materials used to support language acquisition.
6. Scrum methodology: An agile framework for iterative and incremental software development.
7. MVC architecture: A software design pattern that separates the system into Model, View, and Controller components.
8. Laravel framework: A PHP-based framework used to build scalable and maintainable web applications.

Chapter (2):

Project Development and Management Methodology

2.1 Selected Development Methodology and Justifications:

The Scrum methodology was selected because it provides a flexible and iterative development framework that aligns with the dynamic nature of the project. The platform requires continuous refinement, frequent updates, and the ability to incorporate new features based on user feedback. Scrum enables the team to divide the work into manageable sprints, ensuring incremental delivery and early validation of system components. This approach reduces risks, improves communication among team members, and allows rapid adaptation to changing requirements. Additionally, Scrum supports continuous testing and evaluation, which is essential for building a stable, user-centered educational platform.

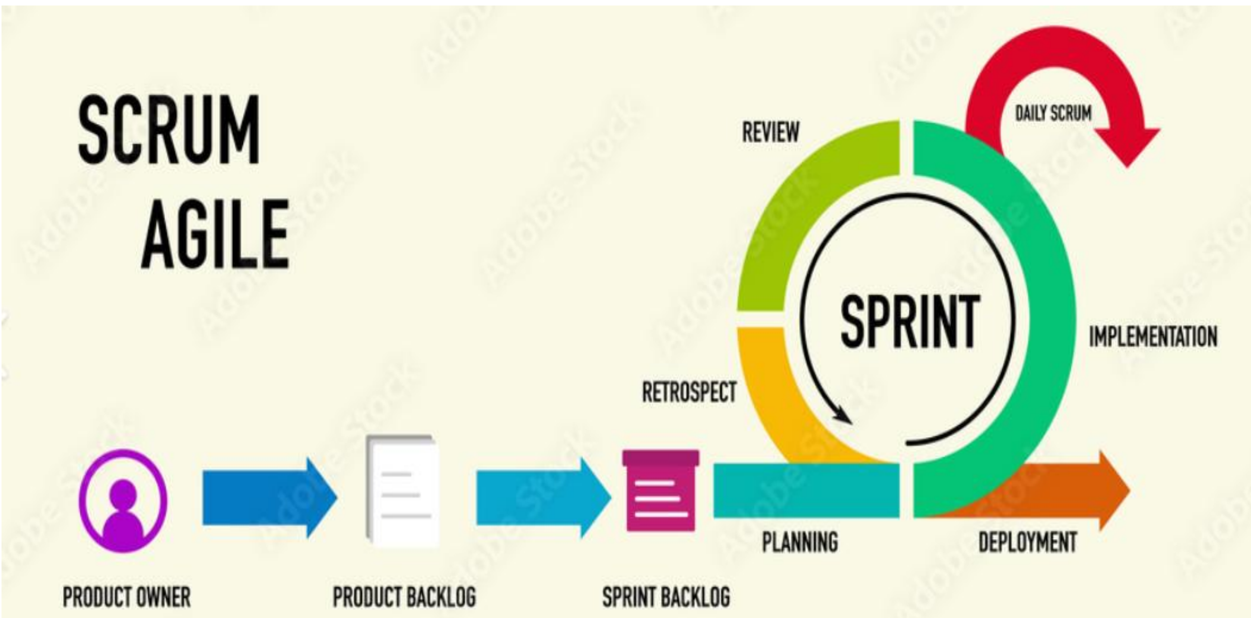


Figure 1

1. **Sprints 1–4** focused on building the foundation: planning, requirements analysis, diagrams, and implementing core access, language, and basic game features.
2. **Sprint 5** developed the core system structure and level progression to prepare the platform for expansion.
3. **Sprint 6** added user-interaction features and the rating system to enhance social engagement.
4. **Sprint 7** completed the platform by developing educational games and communication features.

2.2 Project Timeline (Gantt Chart):

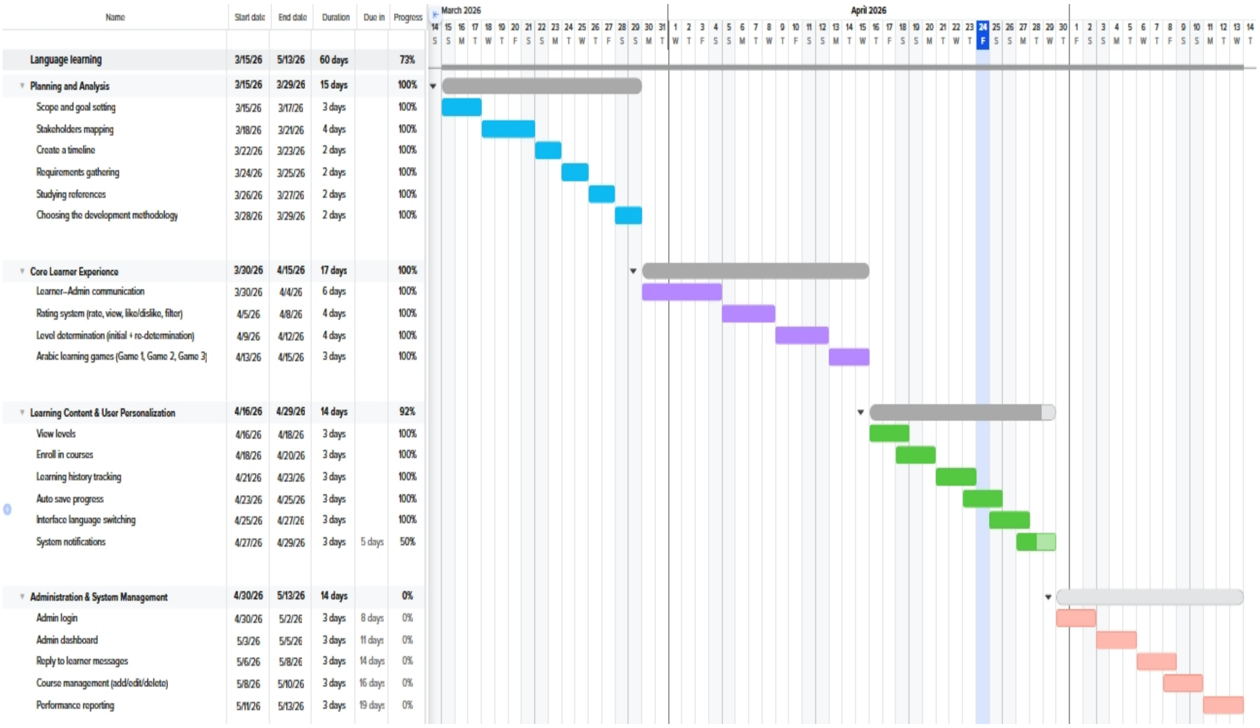


Figure 2

2.3 Risk Management:

Risk	Probability	Impact	Mitigation Strategy
Delay in development due to workload or unexpected issues	Medium	High	Allocate buffer time in each sprint, prioritize tasks, and conduct frequent progress reviews
Technical challenges in integrating system components	Medium	Medium	Perform early technical validation, use prototypes, and consult documentation and experts when needed
Changing requirements during development	High	Medium	Maintain continuous communication with stakeholders, apply change-control procedures, and adjust sprint backlog accordingly
Performance issues during testing	Medium	High	Conduct iterative testing, optimize code, and perform load and stress tests early

Insufficient user engagement or unclear feedback	Low	Medium	Increase user testing sessions, gather structured feedback, and adjust features based on user behavior
Security vulnerabilities or data protection risks	Low	High	Implement secure coding practices, conduct security testing, and apply encryption and access-control measures

Table 1

2.4 Team Structure :

Scrum Role	Responsibilities	Assigned To
Product Owner	Defining product vision, prioritizing backlog items, communicating requirements, ensuring the final product meets user needs	**Member 1**
Scrum Master	Facilitating Scrum events, removing obstacles, ensuring adherence to Scrum practices, supporting team productivity	**Member 2**
Development Team	Designing, coding, testing, documenting, and delivering increments of the product during each sprint	**Member 1 & Member 2 (shared)**

Table 2

2.5 Quality Management :

Quality management played a critical role in ensuring that the project met the required standards of performance, usability, and reliability. Throughout the development process, the team applied structured quality assurance practices to maintain consistency and detect issues early. These practices included defining acceptance criteria, conducting iterative testing at the end of each sprint, reviewing completed tasks, and applying continuous improvements based on feedback. Both team members contributed to quality activities to ensure that the final product was stable, user-friendly, and aligned with project objectives.

Key Quality Management Practices:

- ✓ Defining clear acceptance criteria for each user story
- ✓ Conducting functional and performance testing after every sprint
- ✓ Reviewing completed tasks during Sprint Review meetings
- ✓ Performing code reviews to ensure consistency and maintainability
- ✓ Applying iterative improvements based on user and team feedback
- ✓ Using both manual and automated testing techniques
- ✓ Documenting issues and resolving them promptly

2.6 Project Management Tools Used:

1. Jira – Agile Project and Sprint Management:

- ✓ Used to manage the Product Backlog and User Stories.
- ✓ Supports Sprint planning, scheduling, and task assignment.
- ✓ Provides Scrum and Kanban boards for visual task tracking.
- ✓ Enables issue reporting, bug tracking, and workflow automation.
- ✓ Offers analytics such as burndown charts and velocity reports.

2. GitHub – Version Control and Collaboration:

- ✓ Hosts the project repository and maintains source-code history.
- ✓ Supports branching and merging for parallel development.
- ✓ Enables Pull Requests for code review and quality assurance.
- ✓ Provides issue tracking for bugs and feature requests.
- ✓ Ensures secure cloud-based storage and team collaboration.

3. Google Drive – Documentation and File Management:

- ✓ Stores project documents, diagrams, and reports.
- ✓ Facilitates file sharing among team members.
- ✓ Maintains backups of important project assets.

4. WhatsApp – Communication Tools:

- ✓ Used for team communication and coordination.
- ✓ Supports quick updates, discussions, and decision-making.
- ✓ Helps schedule meetings, sprint reviews, and planning sessions.

2.7 Monitoring and Continuous Evaluation Mechanisms:

To ensure the quality, stability, and continuous improvement of the system, several monitoring and evaluation mechanisms were applied throughout the development lifecycle. These mechanisms supported early detection of issues, enhanced performance, and ensured alignment with user needs and project objectives.

1. Sprint Review and Retrospective Sessions:

- ✓ Conducted at the end of each Sprint to evaluate completed work.
- ✓ Reviewed delivered User Stories and implemented features.
- ✓ Identified challenges, workflow issues, and improvement opportunities.
- ✓ Adjusted future Sprint plans based on team performance and feedback.

2. Continuous Integration and Code Review:

- ✓ Implemented through GitHub Pull Requests to ensure code quality.
- ✓ Allowed developers to review, comment, and approve changes before merging.
- ✓ Reduced bugs and inconsistencies by enforcing coding standards.
- ✓ Ensured smooth integration of new features into the main codebase.

3. Performance Monitoring:

- ✓ Regularly measured system responsiveness and page-loading times.
- ✓ Evaluated server performance under different usage conditions.
- ✓ Identified slow queries or inefficient code segments for optimization.
- ✓ Ensured stable and fast performance across devices and browsers.

4. Error Logging and Issue Tracking:

- ✓ Used Jira and GitHub Issues to record and track bugs.
- ✓ Monitored recurring issues to identify root causes.
- ✓ Prioritized critical issues for immediate resolution.
- ✓ Ensured transparency and traceability of all reported problems.

5. Documentation Updates and Review:

- ✓ Maintained updated technical and user documentation throughout development.
- ✓ Ensured that architectural changes and new features were properly documented.
- ✓ Supported long-term maintainability and scalability of the system.

Chapter (3):

Literature Review and Related Studies

3.1 Studies on Previous Systems:

Study 1: “Adaptive Game-Based Learning System for Language Acquisition” (2021)

What it presented:

1. Developed an adaptive learning platform that adjusts difficulty based on learner performance.
2. Integrated game mechanics such as points, levels, and challenges to increase motivation.
3. Demonstrated improved vocabulary retention among learners.

What it lacked:

1. Limited support for multiple languages.
2. No personalized learning paths beyond difficulty adjustment.
3. Did not include progress analytics for teachers or administrators.

Study 2: “Intelligent Recommendation System for E-Learning Platforms Using Machine Learning” (2020)

What it presented:

1. Proposed a recommendation engine that suggests learning materials based on user behavior.
2. Used machine-learning algorithms (KNN, SVM) to classify learner preferences.
3. Improved content relevance and user engagement.

What it lacked:

1. Did not integrate gamification elements.
2. Focused only on content recommendation, not full learning experience.
3. Lacked real-time adaptation during learning sessions.

Study 3: “Gamified Mobile Application for Language Learning: A User-Centered Approach” (2022)

What it presented:

1. Designed a mobile app with interactive games for vocabulary and grammar learning.
2. Applied user-centered design principles to improve usability.

3. Showed high user satisfaction and increased learning motivation.

What it lacked:

1. No web-based version, limiting accessibility.
2. Did not include advanced tracking or analytics.
3. Content updates required manual intervention and were not automated.

Study 4: "Personalized Learning Path Generation in E-Learning Systems Using Data Analytics" (2023)

What it presented:

1. Introduced a system that generates personalized learning paths based on learner progress and goals.
2. Used data analytics to identify weaknesses and recommend targeted exercises.
3. Enhanced learning efficiency and reduced time to mastery.

What it lacked:

1. No integration with game-based learning.
2. Limited interactivity in the learning activities.
3. Required high computational resources for analytics.

Study 5: "Interactive Digital Games for Enhancing Language Skills in Online Education" (2020)

What it presented:

1. Developed a set of digital games targeting listening, reading, and vocabulary skills.
2. Demonstrated significant improvement in learner engagement and participation.
3. Provided a flexible environment suitable for self-paced learning.

What it lacked:

1. Did not include a unified platform or dashboard.
2. Lacked adaptive difficulty or personalized recommendations.
3. No support for payment systems or content management.

3.2 Modern Technologies and Frameworks in the Field:

1. Artificial Intelligence and Machine Learning:

- ✓ Core technologies used to build recommendation engines through collaborative filtering, content-based filtering, and deep learning models.
- ✓ Enable adaptive learning, personalized content delivery, and prediction of user behavior.

2. Modern Web and Front-End Technologies:

- ✓ HTML5, CSS3, and JavaScript frameworks such as React.js and Vue.js support interactive, responsive, and user-friendly interfaces.
- ✓ Essential for building dynamic educational activities, dashboards, and gamified components.

3. Backend Frameworks and Databases:

- ✓ Frameworks like Laravel, Django, and Node.js provide secure, scalable architectures for managing user data, authentication, and APIs.
- ✓ Databases such as MySQL, PostgreSQL, MongoDB, and SQLite store user progress, learning materials, and recommendation data

4. Gamification and Interactive Engines:

- ✓ Technologies like Phaser.js, Unity, and gamification APIs enable the integration of points, badges, levels, and interactive game-based learning.
- ✓ Enhance motivation, engagement, and overall learning effectiveness.

3.3 Comparative Analysis of Existing Systems and Products:

Platform	Strengths	Weaknesses
Duolingo	Gamified experience – AI personalization – Free content – Motivating rewards – Multi-platform	Shallow content – Inaccurate translations – Limited voice interaction – Weak grammar depth
Memrise	Natural pronunciation videos – Smart repetition – Daily challenges – Free content	Limited grammar – Minimal voice interaction – Weak

		advanced content – Literal translations
freeCodeCamp	Fully free – Real projects – Supportive community – Structured paths	Needs external resources – No instructor support – Simple UI – No instant feedback
edX	High-quality academic content – Accredited certificates – Diverse subjects – Multilingual support	High fees – Limited interaction – Not personalized – Rigid schedules
Busuu	Native-speaker feedback – Personalized plans – Strong grammar coverage	Many paid features – Limited free content – Interaction depends on user availability

Table 3

3.4 Research Gap and Proposed Innovation:

Despite the significant progress achieved by existing learning platforms such as Duolingo, Memrise, freeCodeCamp, edX, and Busuu, several gaps remain unaddressed in the current landscape. Most platforms offer strong features—such as gamification, AI-driven personalization, native-speaker feedback, or high-quality academic content—yet they still suffer from limitations that reduce their effectiveness and inclusiveness.

A major gap lies in the lack of a unified system that combines deep grammar coverage, accurate content, real-time interaction, and adaptive learning within a single platform. For example, Duolingo and Memrise provide engaging gamified experiences but fall short in grammar depth, voice interaction, and translation accuracy. Academic platforms like edX offer high-quality structured content but lack personalization, flexibility, and interactive learning. Meanwhile, Busuu provides personalized plans and native-speaker feedback, yet much of its content is locked behind paid features, limiting accessibility.

Another gap is the absence of instant, intelligent feedback that supports learners during practice. Platforms such as freeCodeCamp demonstrate the value of real projects and community support but still lack real-time guidance and rely heavily on external resources.

How We Used the Reference Study in Building Our Platform:

1. We adopted gamified interactive learning inspired by Duolingo and Memrise, where engagement proved highest.
2. We implemented intelligent content personalization based on the adaptive models used in Duolingo and Busuu.

3. We strengthened advanced and grammar-focused content to overcome the weaknesses seen in Duolingo and Memrise.
4. We integrated a social learning community inspired by the strong peer-interaction benefits seen in freeCodeCamp and Busuu.

Based on these gaps, the proposed innovation aims to develop a hybrid intelligent learning system that integrates:

1. Gamification for motivation
2. AI-based personalization for adaptive learning
3. Accurate, structured grammar content
4. Real-time feedback and interactive exercises
5. Accessible features without heavy paywalls

This integrated approach addresses the weaknesses of existing platforms and provides a more balanced, comprehensive, and engaging learning experience.

3.5 Theoretical Framework:

The proposed system is grounded in several computational and mathematical theories that form the foundation of modern recommendation engines, adaptive learning systems, and gamified educational platforms. These theories guide how user behavior is analyzed, how content is recommended, and how learning experiences are personalized.

1. Content-Based Filtering Theory:

This theory recommends items based on the characteristics of the content itself and the user's past interactions. It uses:

- ✓ Feature extraction
- ✓ Vector representations
- ✓ Similarity computations

This allows the system to suggest learning materials that match the learner's skill level, topic preferences, or performance history.

2. Reinforcement Learning Theory:

Reinforcement Learning (RL) provides a mathematical framework for adaptive learning. The system treats the learner as an "agent" and the learning environment as a "state," where:

- ✓ Correct answers = positive reward
- ✓ Mistakes = negative reward
- ✓ Progress = optimized policy

This theory supports dynamic difficulty adjustment and personalized learning paths.

3. Gamification Theory (Behavioral Motivation Models):

Gamification relies on psychological and mathematical models such as:

- ✓ The Reward System Model
- ✓ Points–Badges–Leaderboards (PBL) Framework
- ✓ Flow Theory

These models explain how rewards, challenges, and progression systems increase motivation and engagement.

3.6 Future Directions for the Proposed System:

1. **Advanced AI-Based Personalization** :Expanding the system to include deeper machine-learning models that adapt content, difficulty, and recommendations based on long-term user behavior.
2. **Real-Time Intelligent Feedback:** Integrating automated feedback for grammar, pronunciation, and performance to support learners instantly during activities.
3. **Gamification Expansion and Interactive Challenges:** Adding more game-based elements such as levels, badges, leaderboards, and competitive or collaborative challenges.
4. **Mobile Application and Offline Learning Support** :Developing a mobile version of the platform with offline access to lessons, quizzes, and progress tracking.

Chapter (4):

Requirements Modeling and Analysis

4.1 Feasibility Study:

The project aims to create a significant shift in language learning by integrating educational games and interactive activities, making the learning process enjoyable, motivating, and effective—moving away from rigid traditional methods.

Financial Aspects:

- ✓ Platform development (UI/UX design, building the interactive gaming system, content management).
- ✓ Technical infrastructure (servers, databases, user management systems).
- ✓ Digital marketing costs and user acquisition.
- ✓ Ongoing operation and maintenance, including technical support and content updates.
- ✓ Revenue streams: paid subscriptions, educational partnerships, premium content and learning tools, educational advertisements.

Technical Aspects:

- ✓ Technologies: databases for managing content and learning stages, a user management system with support for online payments.
- ✓ Human resources: a team of software developers, language and educational experts for content creation, UX/UI specialists, and a technical support and digital marketing team.

Market Aspects:

- ✓ The project targets school and university students, professionals seeking to develop new language skills, beginners, and educational institutions looking for innovative solutions.
- ✓ With the growing interest in educational games and the rising demand for flexible self-learning platforms, the project has strong potential for success and wide adoption.

4.2 Stakeholder:

Stakeholder	Main Needs	Expectations
Learner	- Create/manage account - Access languages, levels, games, and courses - Take placement tests and update level - Track progress and view results - Interact with ratings and send messages - Receive notifications and auto-save progress	- Smooth and enjoyable learning experience - Accurate level assessment - Fast access to content - Clear progress tracking - Secure and multilingual interface - Responsive support

	- Change interface language	
Admin	- Secure login and dashboard access - Manage messages and course content - View system statistics and performance reports	- Clear admin tools - Accurate - Easy content management - Efficient communication with learners
Developers	- Clear functional requirements - Stable modules for users, games, ratings, and admin tools	- Maintainable architecture - Minimal bugs - Scalable backend and clean APIs

Table 4

4.3 Functional Requirements:

RQ_ID	Requirement	actor	Description	Scenario	MoSCoW
RQ_1	Sign up	Learner	The learner creates a new account by entering their personal information such as email and username	The system should allow the learner to sign up using their details	Must
RQ_2	Log in	Learner	The learner enters their credentials (email/username and password) to access the system	The system should allow the learner to log in	Must
RQ_3	Managing Accounts	Learner	Learners can update personal details, change passwords, and adjust preferences to maintain secure and flexible accounts.	The system should allow learners to manage their accounts, including updating personal information, changing passwords, and adjusting preferences for flexibility and security.	Should
RQ_4	Logout	Learner	The learner can securely exit their account	The system should allow the learner to log out	Must
RQ_5	Show available languages	Learner	The learner can view a list of supported languages	The system should show available languages	Should

RQ_6	Show available games	Learner	The learner can browse games categorized by language (English, Arabic, Turkish)	The system should show available games by language	Should
RQ_7	Determine level	Learner	The learner completes a placement test to identify their level	The system should determine the learner’s level in the language	Must
RQ_8	Game 1 English	Learner	The learner can access and play the first English game	The system should allow the learner to play English Game 1	Must
RQ_9	Game 2 English	Learner	The learner can access and play the second English game	The system should allow the learner to play English Game 2	Should
RQ_10	Game 1 Arabic	Learner	The learner can access and play the first Arabic game	The system should allow the learner to play Arabic Game 1	Must
RQ_11	Game 2 Arabic	Learner	The learner can access and play the second Arabic game	The system should allow the learner to play Arabic Game 2	Should
RQ_12	Game 1 Turkish	Learner	The learner can access and play the first Turkish game	The system should allow the learner to play Turkish Game 1	Must
RQ_13	Game 2 Turkish	Learner	The learner can access and play the second Turkish game	The system should allow the learner to play Turkish Game 2	Should
RQ_14	Show current results	Learner	The learner can view their performance and scores across all played games	The system should show current results for all games	Must
RQ_15	Update level	Learner	The learner can retake the placement test to update their level	The system should allow the learner to update their level	Should

RQ_16	Contact admin	Learner	The learner can send a message to the admin	The system should allow the learner to contact admin	Could
RQ_17	Rate	Learner	The learner can submit a rating	The system should allow the learner to rate	Could
RQ_18	View ratings	Learner	The learner can browse ratings from other learners	The system should allow the learner to view other ratings	Could
RQ_19	Like/Dislike rating	Learner	The learner can react to ratings	The system should allow the learner to like/dislike a rating	Could
RQ_20	Filter ratings	Learner	The learner can filter ratings by likes or date	The system should allow filtering ratings	Could
RQ_21	Redetermine level	Learner	The learner can retake a new level test	The system should allow the learner to re-determine level	Should
RQ_22	Determine Arabic level	Learner	The learner can take an Arabic placement test	The system should determine Arabic level	Must
RQ_23	Arabic Game 1	Learner	The learner can play Arabic game 1	The system should allow the learner to play Arabic Game 1	Must
RQ_24	Arabic Game 2	Learner	The learner can play Arabic game 2	The system should allow the learner to play Arabic Game 2	Should
RQ_25	Arabic Game 3	Learner	The learner can play Arabic game 3	The system should allow the learner to play Arabic Game 3	Could
RQ_26	Show levels	Learner	The learner can view all available levels	The system should show levels	Should
RQ_27	Take courses	Learner	The learner can enroll in courses	The system should allow the learner to take courses	Must
RQ_28	Admin login	Admin	Admin can access the system	The system should allow the admin to log in	Must
RQ_29	Show dashboard	Admin	Admin can view system statistics and controls	The system should show admin dashboard	**Must**

RQ_30	Reply to message	Admin	Admin can respond to learner messages	The system should allow admin to reply to messages	**Should**
RQ_31	Learning history	Learner	The learner can view a detailed list of completed games, tests, and courses	The system should allow the learner to view their learning history	**Should**
RQ_32	Auto-save progress	Learner	The system saves progress automatically at checkpoints or upon exit	The system should automatically save the learner's progress in games and courses	**Must**
RQ_33	Change interface language	Learner	The learner can switch the system interface to another supported language	The system should allow the learner to change the interface language	**Could**
RQ_34	Notifications	Learner	The learner receives system-generated notifications about updates and activities	The system should provide notifications for new courses, updates, and test results	**Should**
RQ_35	Manage courses	Admin	The admin can modify course materials and update the course list	The system should allow the admin to manage course content (add, edit, delete)	**Must**
RQ_36	Performance reports	Admin	The system provides detailed reports about learner progress and activity	The system should generate performance reports for the admin	**Should**

Table 5

4.4 Non-Functional Requirements:

Category	Non-Functional Requirement	Description
Performance	Response Time < 3 seconds	The system must respond to user actions within less than 3 seconds to ensure smooth interaction.
Security	Data Validation	All user inputs must be validated to prevent incorrect or malicious data entry.

Security	Data Protection (Hashing & Encryption)	Sensitive data such as passwords must be hashed, and communication must be encrypted to ensure confidentiality.
Security	Authentication & Access Control	The system must enforce secure login and role-based access to protect user accounts and admin functions.
Usability	Easy Interface (3-Step Navigation)	The user interface must be simple, requiring no more than three steps to reach any main function.
Usability	User-Friendly Design	The system should provide clear icons, readable text, and intuitive navigation for all learners.

Table 6

4.5 USECASS:

Use case(High level):

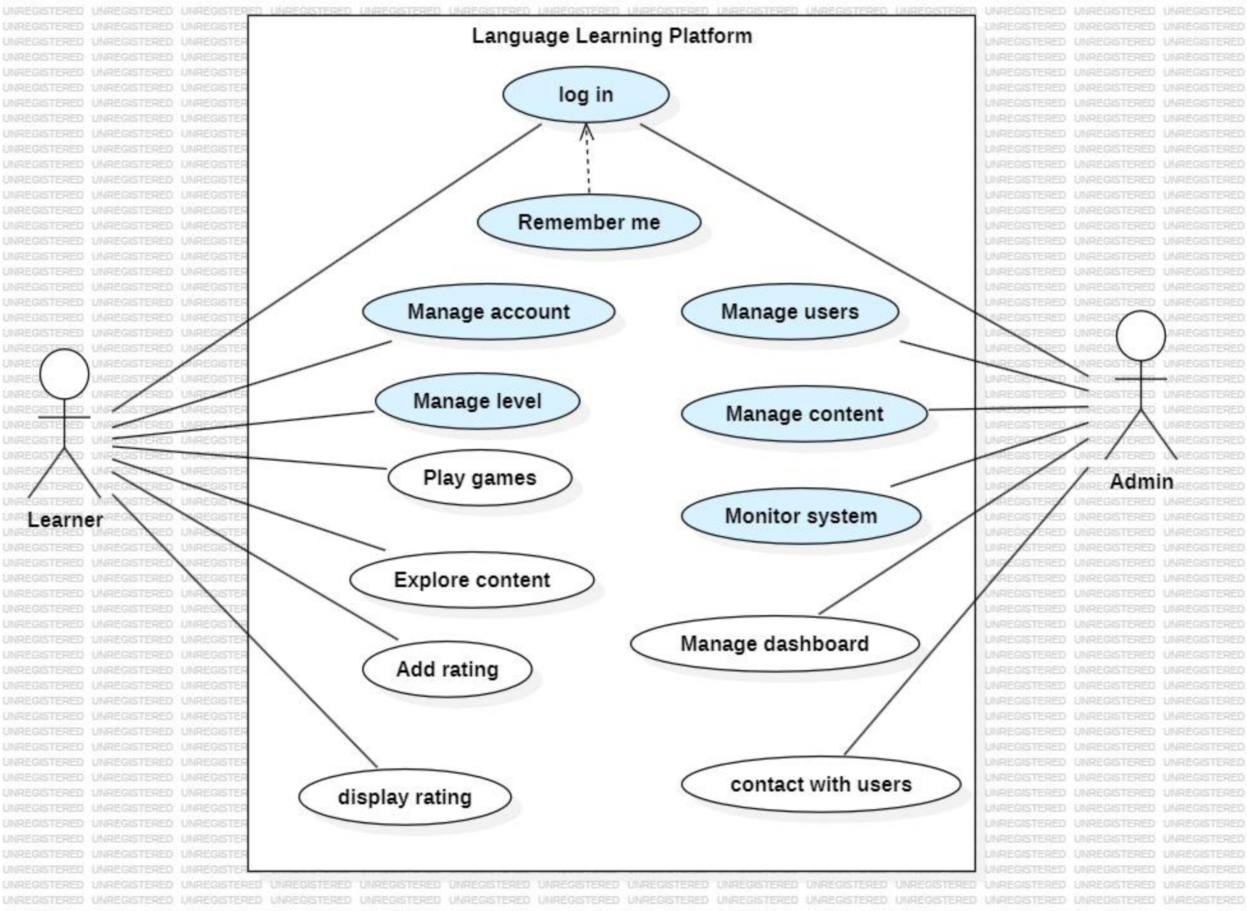


Figure 3

Use case(Low level):

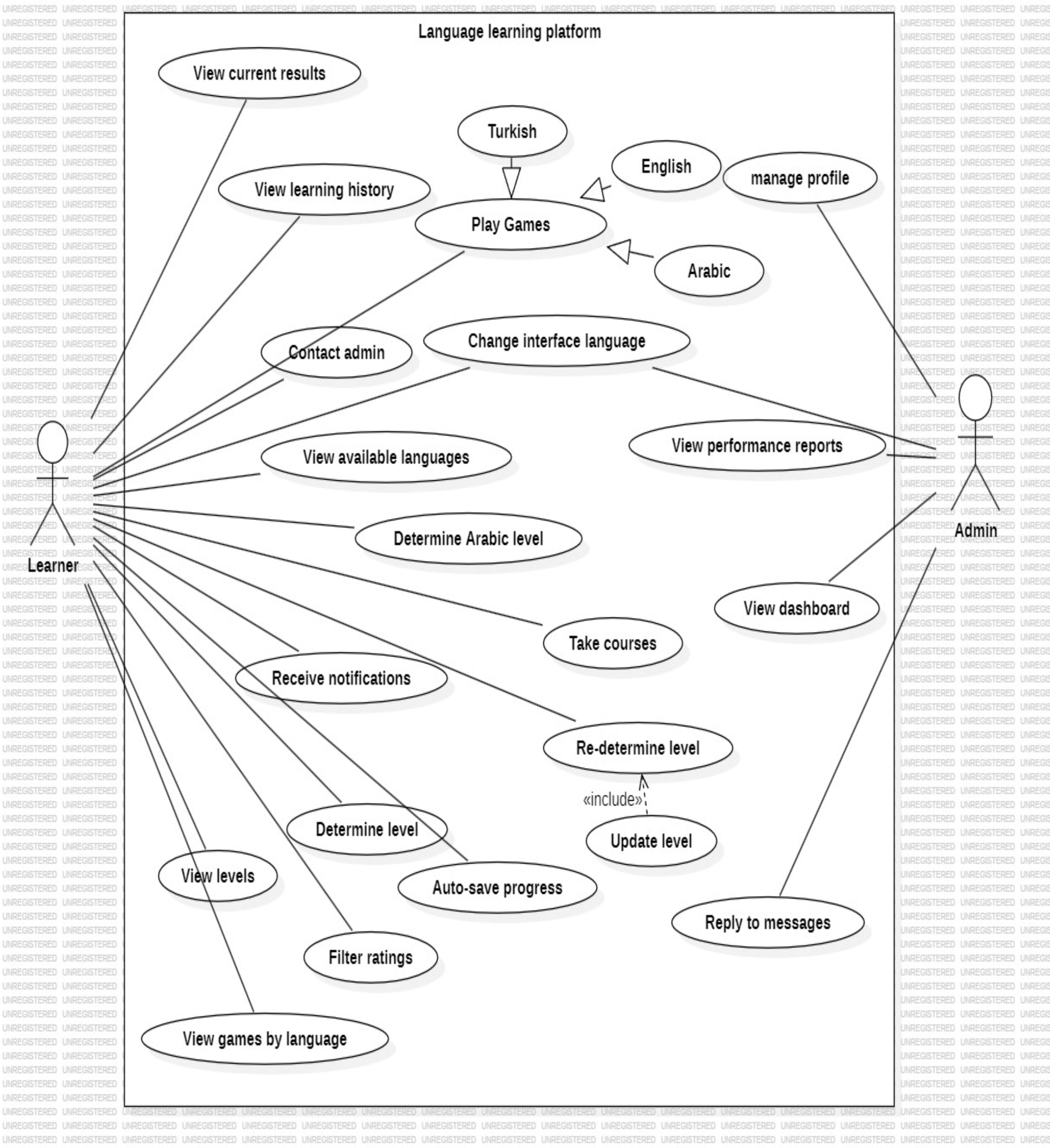


Figure 4

4.6 Requirements Specification Tables:

Use Case Name	Sign Up
Participating Actors	Learner
Entry Condition	The learner does not have an existing account and opens the sign-up page.
The Flow of Events	1. Learner enters personal details (email, username, password). 2. System validates the input. 3. System creates a new account. 4. Confirmation message is displayed.
Alternative Flows	- Email already exists → System shows error and asks for another email. - Invalid or incomplete input → System prompts learner to correct the data. - System error (e.g., database connection) → Registration fails with error message.
Exit Condition	A new learner account is successfully created and ready for login.

Table 7

Use Case Name	Log In
Participating Actors	Learner
Entry Condition	The learner has an existing account and opens the login page.
The Flow of Events	1. Learner enters credentials (email/username and password). 2. System validates the credentials. 3. If valid, the learner is granted access to the system.
Alternative Flows	- Invalid credentials → System shows error and prompts learner to retry. - Account locked or inactive → System displays appropriate message. - System error (database connection issue) → Login fails with error message.
Exit Condition	Learner successfully logs in and gains access to the system’s features.

Table 8

Use Case Name:	Managing Accounts
Participating Actors	User
Entry Condition:	The learner has an existing account and opens the dashboard page.
The Flow of Events:	1. The system displays the account management interface. 2. The manager views, updates, and manage accounts. 3. The manager clicks the "Save" button. 4. The system verifies and saves the updated account details.
Alternative Flows:	1. Missing required data: The system asks the manager to re-enter all required information.
Exit Condition:	Successful management of user accounts.

Table 9

Use Case Name	Logout
Participating Actors	Learner
Entry Condition	Learner is logged into the system.
The Flow of Events	1. Learner selects “Logout” option. 2. System terminates the active session. 3. System redirects learner to the login page.
Alternative Flows	- System error during logout → Session may persist until timeout.
Exit Condition	Learner is securely logged out and cannot access system features without logging in again.

Table 10

Use Case Name	Show Available Languages
Participating Actors	Learner
Entry Condition	Learner is logged into the system.
The Flow of Events	1. Learner navigates to language settings or preferences. 2. System displays a list of supported languages. 3. Learner can view and select a preferred language.
Alternative Flows	- No languages available → System defaults to the primary language. - Learner cancels selection → System keeps current language.
Exit Condition	Learner sees the list of supported languages and optionally selects one.

Table 11

Use Case Name	Show Available Games
Participating Actors	Learner
Entry Condition	Learner is logged into the system and navigates to the games section.
The Flow of Events	1. Learner opens the games menu. 2. System displays games categorized by language (English, Arabic, Turkish). 3. Learner browses the available options.
Alternative Flows	- No games available for a selected language → System shows a message.
Exit Condition	Learner views the list of available games by language.

Table 12

Use Case Name	Determine Level
Participating Actors	Learner
Entry Condition	Learner is logged into the system and chooses to take a placement test.
The Flow of Events	1. Learner starts the placement test. 2. System presents test questions. 3. Learner completes the test. 4. System evaluates answers and determines the learner's level.
Alternative Flows	- Learner exits test before completion → No level is determined. - System error → Test results not saved.
Exit Condition	Learner's language level is identified and stored in the system.

Table 13

Use Case Name	Game 1 English
Participating Actors	Learner
Entry Condition	Learner is logged in and selects English Game 1.
The Flow of Events	1. Learner chooses English Game 1. 2. System loads the game. 3. Learner plays the game.
Alternative Flows	- Game fails to load → System shows error message.
Exit Condition	Learner completes or exits English Game 1.

Table 14

Use Case Name	Game 2 English
Participating Actors	Learner
Entry Condition	Learner is logged in and selects English Game 2.
The Flow of Events	1. Learner chooses English Game 2. 2. System loads the game. 3. Learner plays the game.
Alternative Flows	- Game fails to load → System shows error message.
Exit Condition	Learner completes or exits English Game 2.

Table 15

Use Case Name	Game 1 Arabic
Participating Actors	Learner
Entry Condition	Learner is logged in and selects Arabic Game 1.
The Flow of Events	1. Learner chooses Arabic Game 1. 2. System loads the game. 3. Learner plays the game.
Alternative Flows	- Game fails to load → System shows error message.
Exit Condition	Learner completes or exits Arabic Game 1.

Table 16

Use Case Name	Game 2 Arabic
Participating Actors	Learner
Entry Condition	Learner is logged in and selects Arabic Game 2.
The Flow of Events	1. Learner chooses Arabic Game 2. 2. System loads the game. 3. Learner plays the game.
Alternative Flows	- Game fails to load → System shows error message.
Exit Condition	Learner completes or exits Arabic Game 2.

Table 17

Use Case Name	Game 1 Turkish
Participating Actors	Learner
Entry Condition	Learner is logged in and selects Turkish Game 1.
The Flow of Events	1. Learner chooses Turkish Game 1. 2. System loads the game. 3. Learner plays the game.
Alternative Flows	- Game fails to load → System shows error message.
Exit Condition	Learner completes or exits Turkish Game 1.

Table 18

Use Case Name	Game 2 Turkish
Participating Actors	Learner
Entry Condition	Learner is logged in and selects Turkish Game 2.
The Flow of Events	1. Learner chooses Turkish Game 2. 2. System loads the game. 3. Learner plays the game.
Alternative Flows	- Game fails to load → System shows error message.
Exit Condition	Learner completes or exits Turkish Game 2.

Table 19

Use Case Name	Show Current Results
Participating Actors	Learner
Entry Condition	Learner is logged in and has completed at least one test/game.
Flow of Events	1. Learner completes the test. 2. System automatically calculates the number of correct and incorrect answers. 3. System generates performance metrics. 4. System immediately displays the results screen showing: • Number of correct answers • Number of incorrect answers
Alternative Flows	- No Questions Attempted**:** System displays message “No results available.” - System Error**:** System displays message “Results cannot be displayed.”
Exit Condition	Learner views their performance results (correct/incorrect answers) directly after completing the test.

Table 20

Use Case Name	Update Level
Participating Actor(s)	Learner
Entry Condition	Learner is logged in, has completed a level, and chooses to take the level test.
Flow of Events	1. Learner completes the level. 2. System presents the level test. 3. Learner answers the test questions. 4. System evaluates the answers and determines performance. 5. If performance meets threshold → System updates learner's level.
Alternative Flows	- Learner exits test before completion → Level remains unchanged. - System error → Test results not saved.
Exit Condition	Learner's level is updated in the system immediately after successful test completion.

Table 21

Use Case Name	Contact Admin
Participating Actors	Learner
Entry Condition	Learner is logged in and navigates to the "Contact Admin" page
Flow of Events	1. Learner opens the Contact Admin page. 2. Learner writes a message in the message box. 3. Learner clicks "Send". 4. System validates the message. 5. System sends the message to the admin inbox. 6. System displays confirmation: "Your message has been sent."
Alternative Flows	• Empty Message**:** System displays "Message cannot be empty." • System Error**:** System displays "Message could not be sent. Please try again later."

Exit Condition	The learner successfully sends a message to the admin and receives

Table 22

Use Case Name	Rate
Participating Actors	Learner
Entry Condition	Learner is logged in and has accessed a game, course, or feature that supports rating
Flow of Events	1. Learner navigates to the rating section. 2. Learner selects a rating value (e.g., stars or score). 3. Learner optionally writes a comment. 4. Learner submits the rating. 5. System validates the input. 6. System saves the rating and displays a confirmation message.
Alternative Flows	• Empty Rating**:** System displays "Please select a rating." • System Error**:** System displays "Rating could not be submitted. Please try again later."
Exit Condition	The learner successfully submits a rating and receives confirmation

Table 23

Use Case Name	Filter Ratings
Participating Actors	Learner
Entry Condition	Learner is logged in and has accessed the ratings page
Flow of Events	1. Learner opens the ratings section. 2. Learner selects a filter option (e.g., by likes or by date). 3. System retrieves ratings based on the selected filter. 4. System displays the filtered list of ratings.

Alternative Flows	• No Ratings Available**:** System displays “No ratings found.” • System Error**:** System displays “Unable to filter ratings at the moment.”
Exit Condition	The learner views a filtered list of ratings according to the selected criteria

Table 24

Use Case Name	Redetermine Level
Participating Actors	Learner
Entry Condition	Learner is logged in and chooses to retake the level assessment
Flow of Events	1. Learner navigates to the “Level Test” section. 2. Learner selects “Retake Level Test”. 3. System loads a new or updated level test. 4. Learner completes the test. 5. System evaluates the answers. 6. System updates the learner’s level based on the new score. 7. System displays the updated level to the learner.
Alternative Flows	• Test Not Completed**:** System displays “Please complete the test to update your level.” • System Error**:** System displays “Unable to update level at the moment.”
Exit Condition	The learner receives an updated level based on the new test results

Table 25

Use Case Name	Play Arabic Game 1
Participating Actors	Learner
Entry Condition	Learner is logged in and has selected Arabic as the target language

Flow of Events	1. Learner navigates to the Arabic games section. 2. Learner selects “Arabic Game 1”. 3. System loads the game interface. 4. Learner interacts with the game (questions, tasks, or activities). 5. System records the learner’s answers and progress. 6. System displays the final score or feedback at the end of the game.
Alternative Flows	• Game Not Loaded**:** System displays “Unable to load the game.” • No Internet Connection**:** System displays “Connection required to start the game.”
Exit Condition	Learner completes Arabic Game 1 and receives results or feedback

Table 26

Use Case Name	Take Courses
Participating Actors	Learner
Entry Condition	Learner is logged in and has navigated to the courses section
Flow of Events	1. Learner opens the “Courses” section. 2. System displays a list of available courses. 3. Learner selects a course. 4. System displays course details (description, lessons, duration). 5. Learner clicks “Enroll”. 6. System enrolls the learner in the course. 7. Learner starts accessing lessons and course materials.
Alternative Flows	• Course Full / Not Available**:** System displays “Course is not available at the moment.” • Enrollment Error**:** System displays “Unable to enroll. Please try again later.”
Exit Condition	Learner is successfully enrolled in the selected course and can begin learning

Table 27

Use Case Name	View Learning History
Participating Actors	Learner
Entry Condition	Learner is logged in and has completed at least one activity (game, test, or course)
Flow of Events	1. Learner navigates to the "Profile" or "History" section. 2. System retrieves the learner's completed activities. 3. System displays a list of completed games, tests, and courses with dates and results. 4. Learner reviews the displayed history.
Alternative Flows	• No History Available**:** System displays "No learning history found." • System Error**:** System displays "Unable to load learning history."
Exit Condition	The learner successfully views their learning history

Table 28

Use Case Name	Generate Performance Reports
Participating Actors	Admin
Entry Condition	Admin is logged in and has accessed the dashboard or reporting section
Flow of Events	1. Admin navigates to the "Reports" or "Performance Reports" section. 2. System retrieves learner activity data (games, tests, courses). 3. System processes and analyzes performance metrics. 4. System generates a detailed performance report. 5. System displays the report to the admin with options to view, filter, or download.
Alternative Flows	• No Data Available**:** System displays "No performance data available." • System Error**:** System displays "Unable to generate report at the moment."
Exit Condition	Admin successfully views the generated performance report

Table 29

4.7 Activity Diagram:

- *UC_ Sign up:*

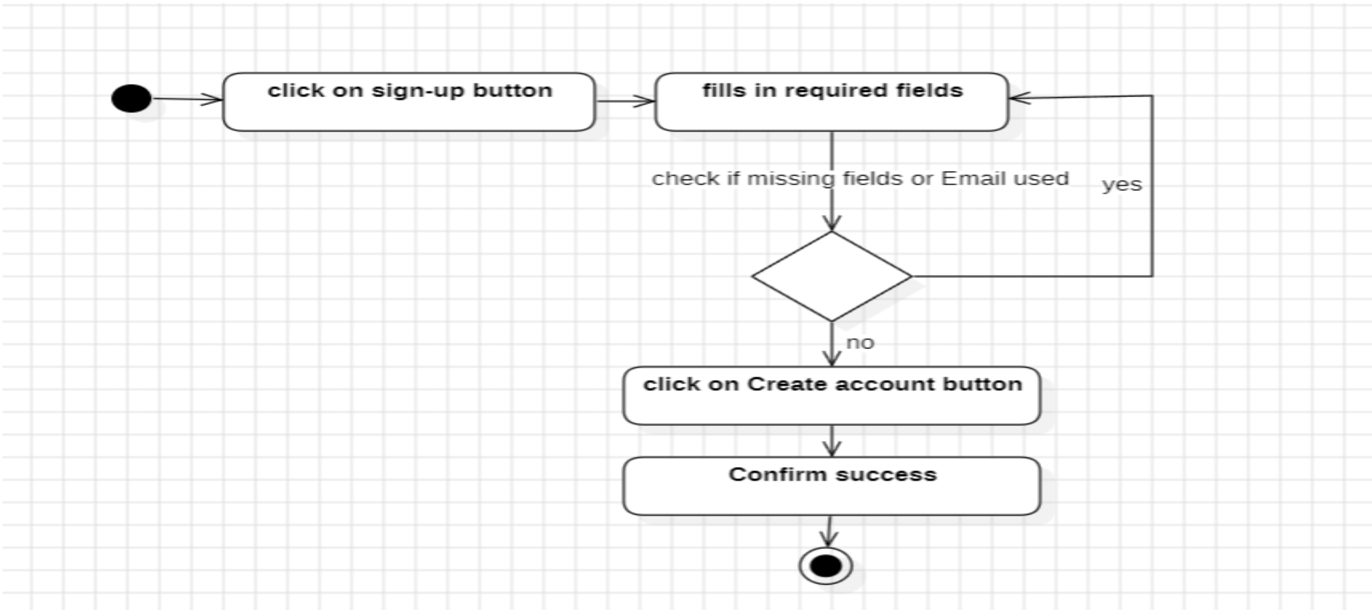


Figure 5

- *UC_ Log in:*

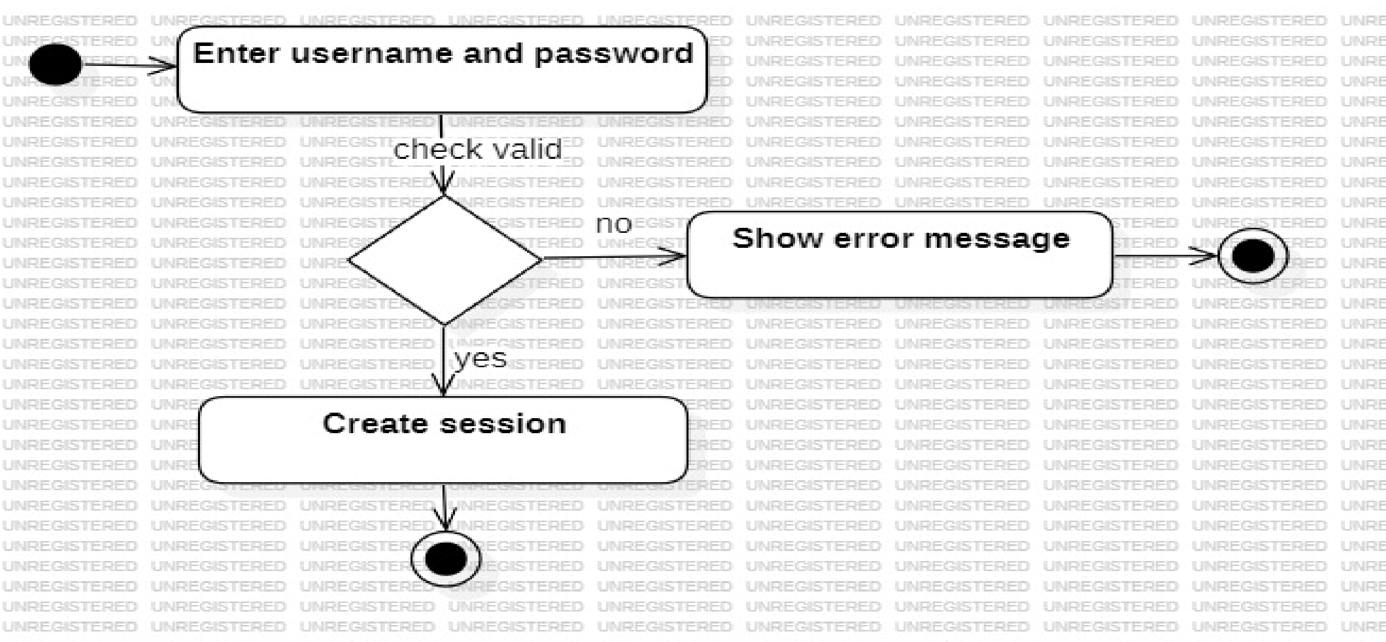


Figure 6

➢ *UC_ Managing Accounts:*

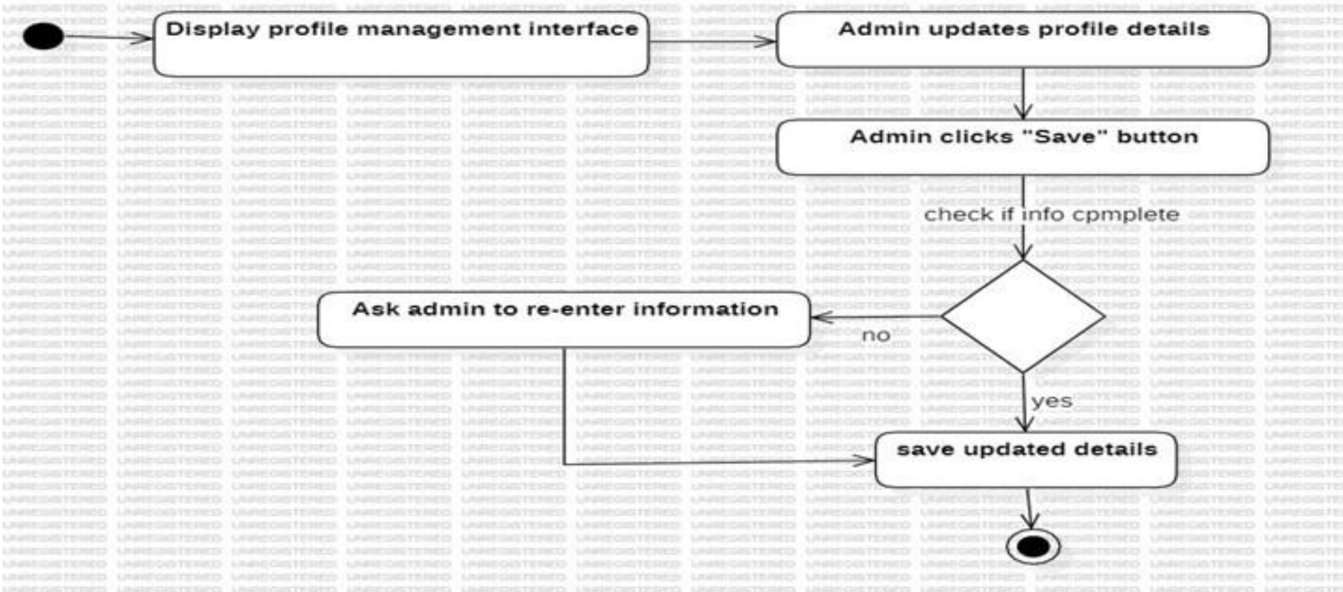


Figure 7

➢ *UC_ Log out:*

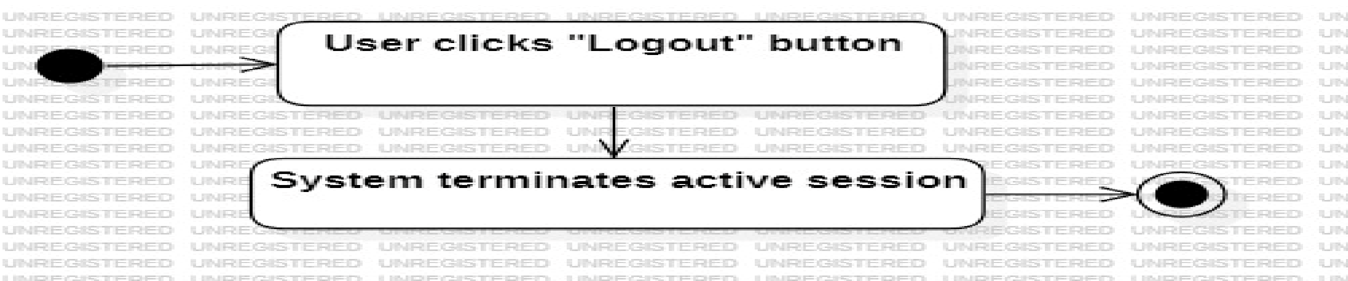


Figure 8

➢ *Show available languages*

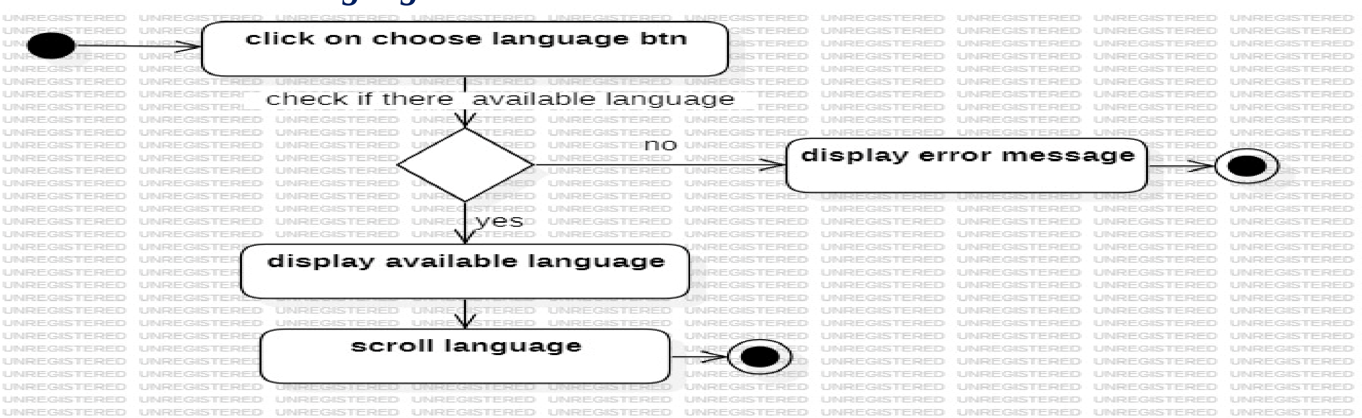


Figure 9

➢ *Show available games*

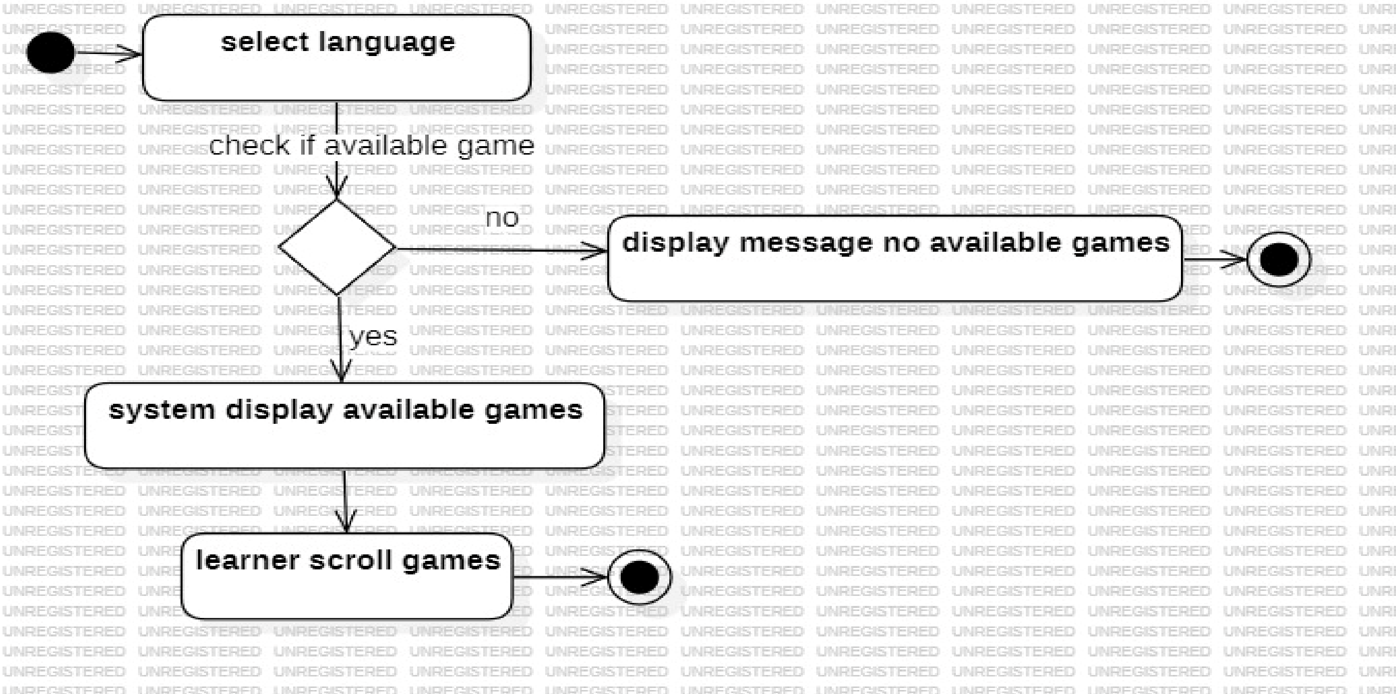


Figure 10

➢ *Determine level*

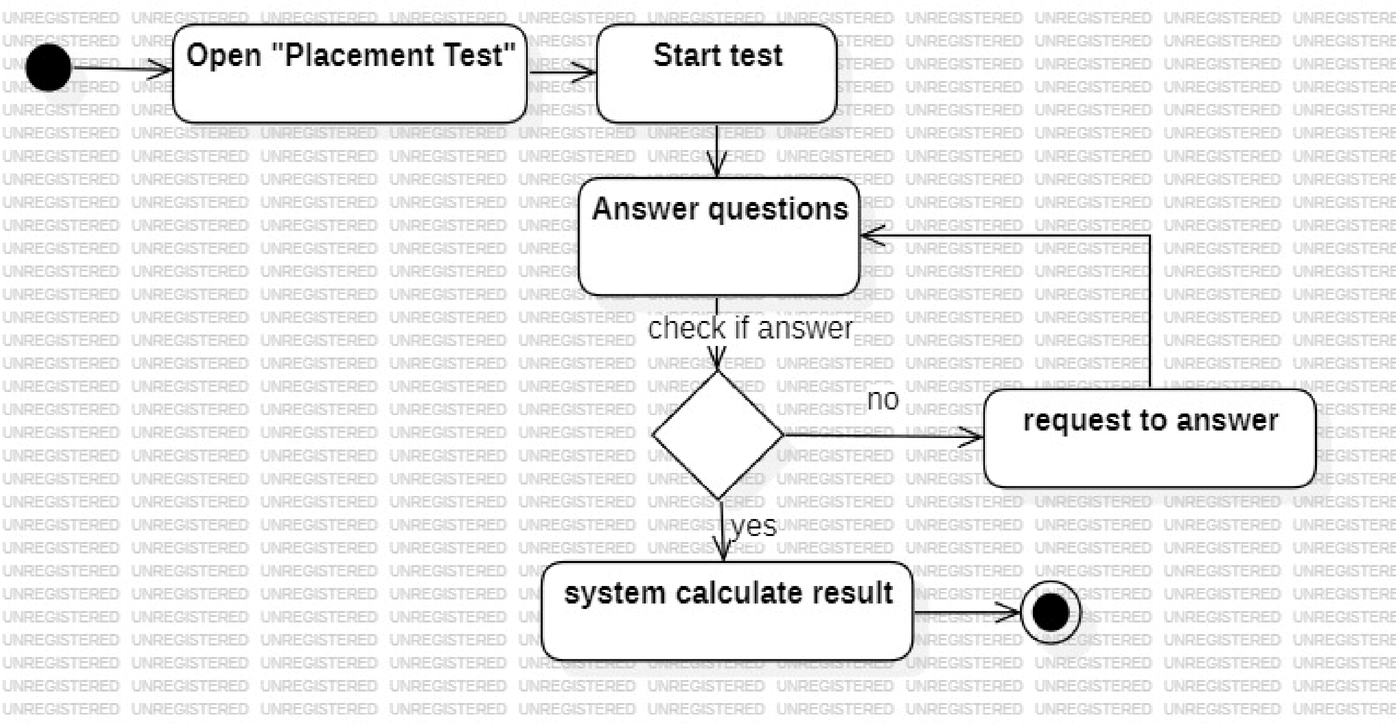


Figure 11

➢ *Game 1 English*

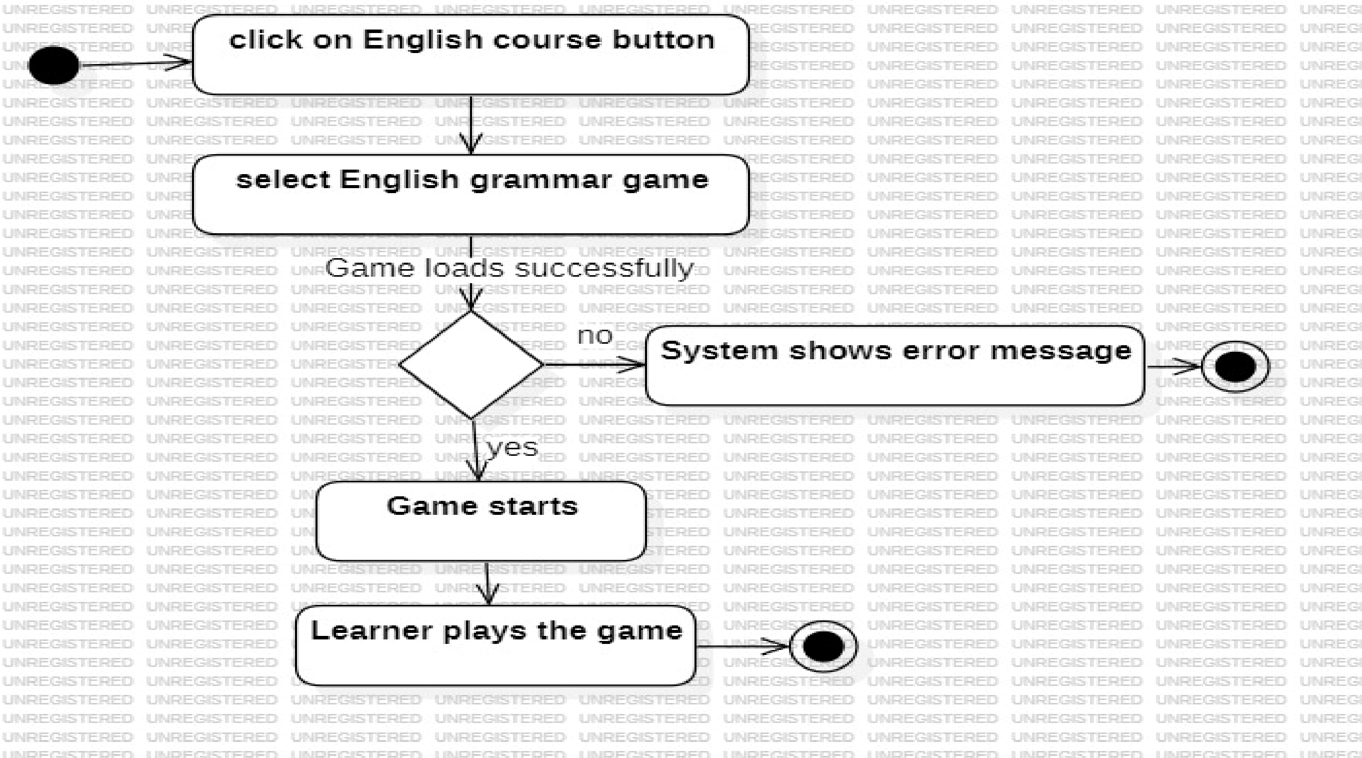


Figure 12

➢ *Game 2 English*

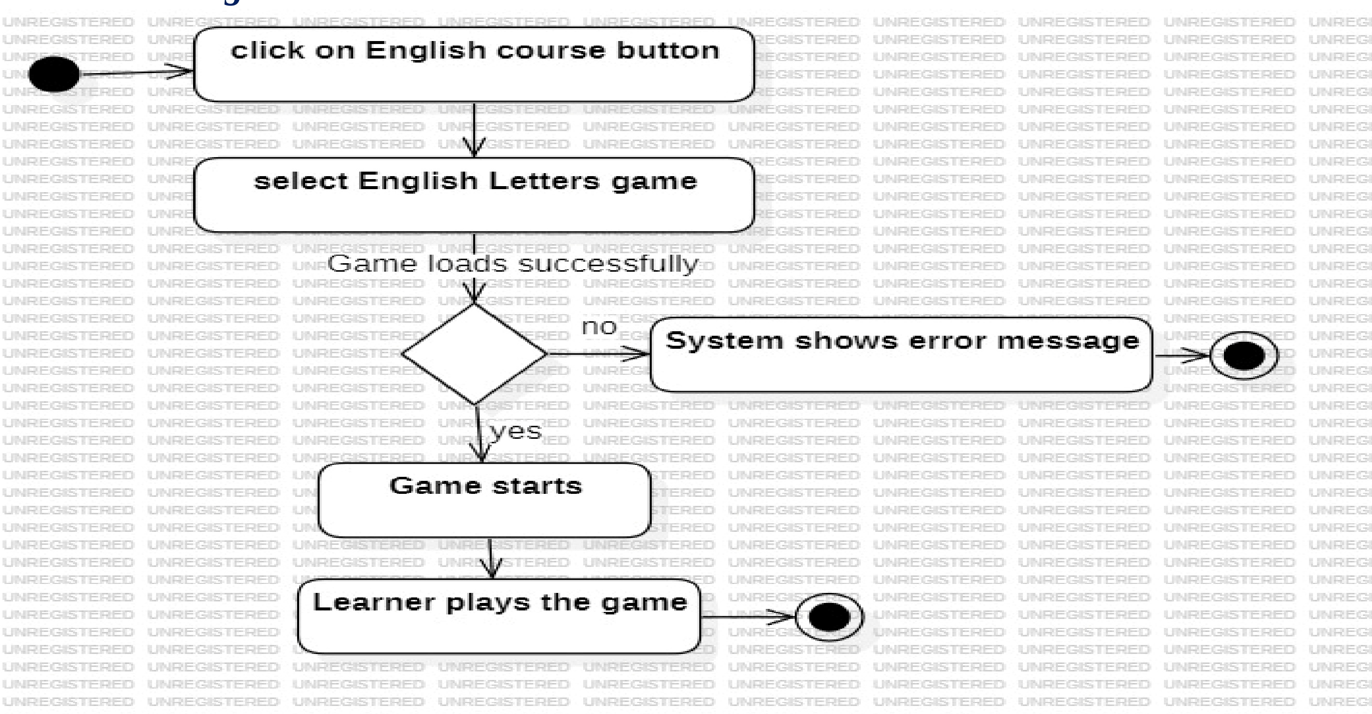


Figure 13

➢ *Game 1 Arabic*

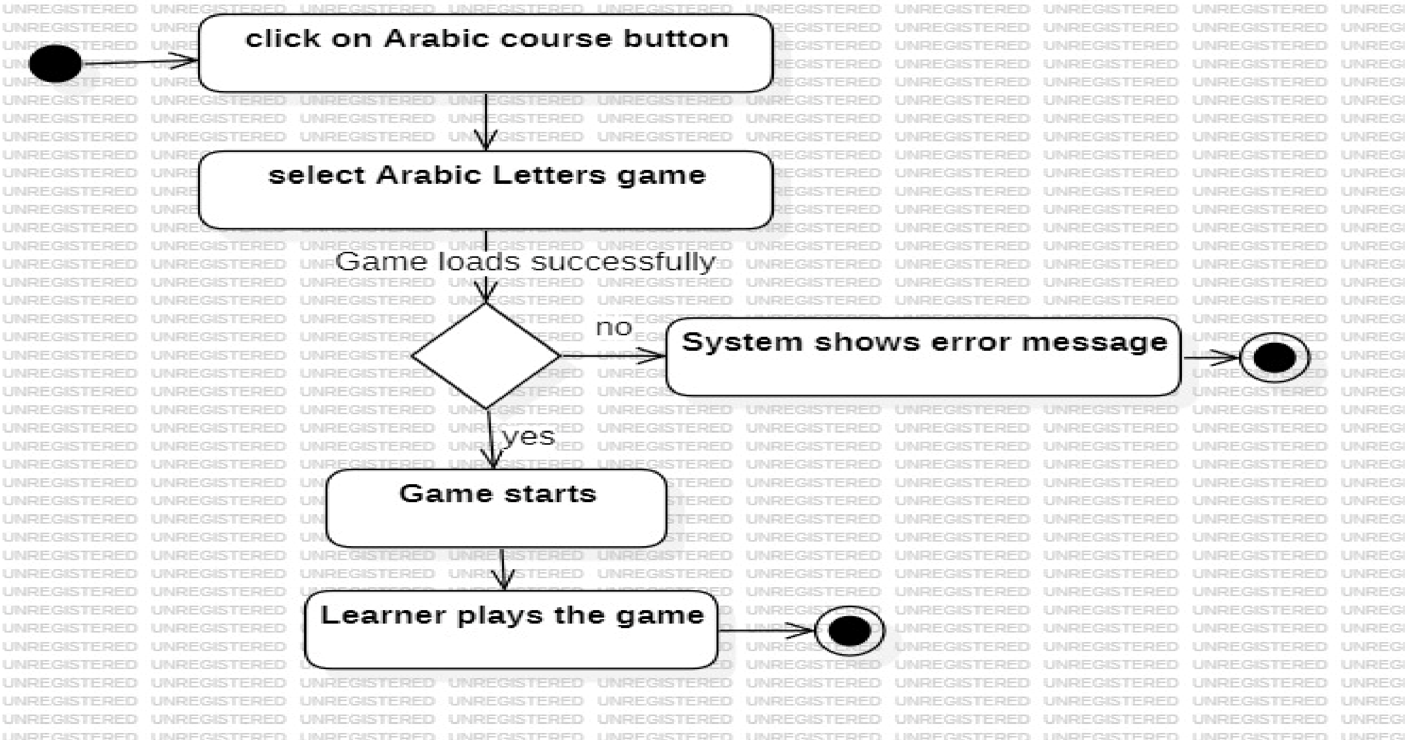


Figure 14

➢ *Game 2 Arabic*

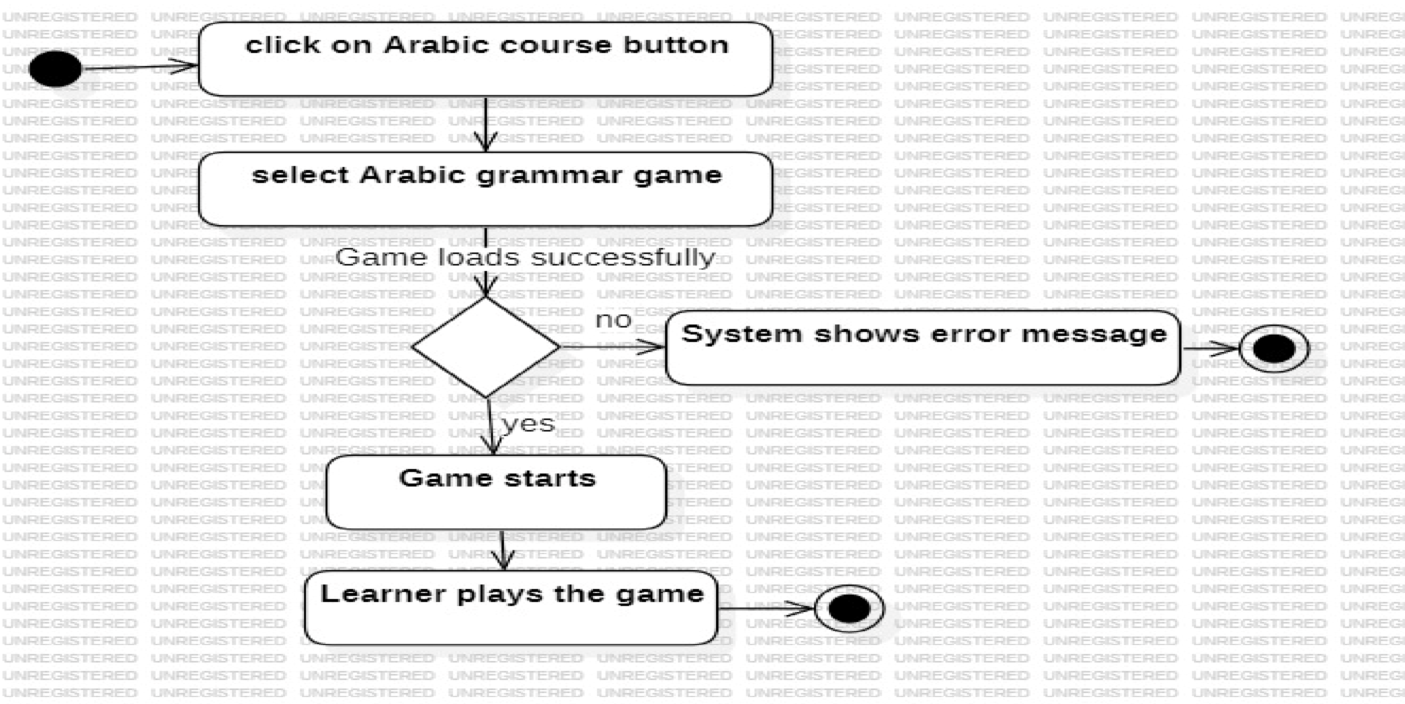


Figure 15

➢ *Game 1 Turkish*

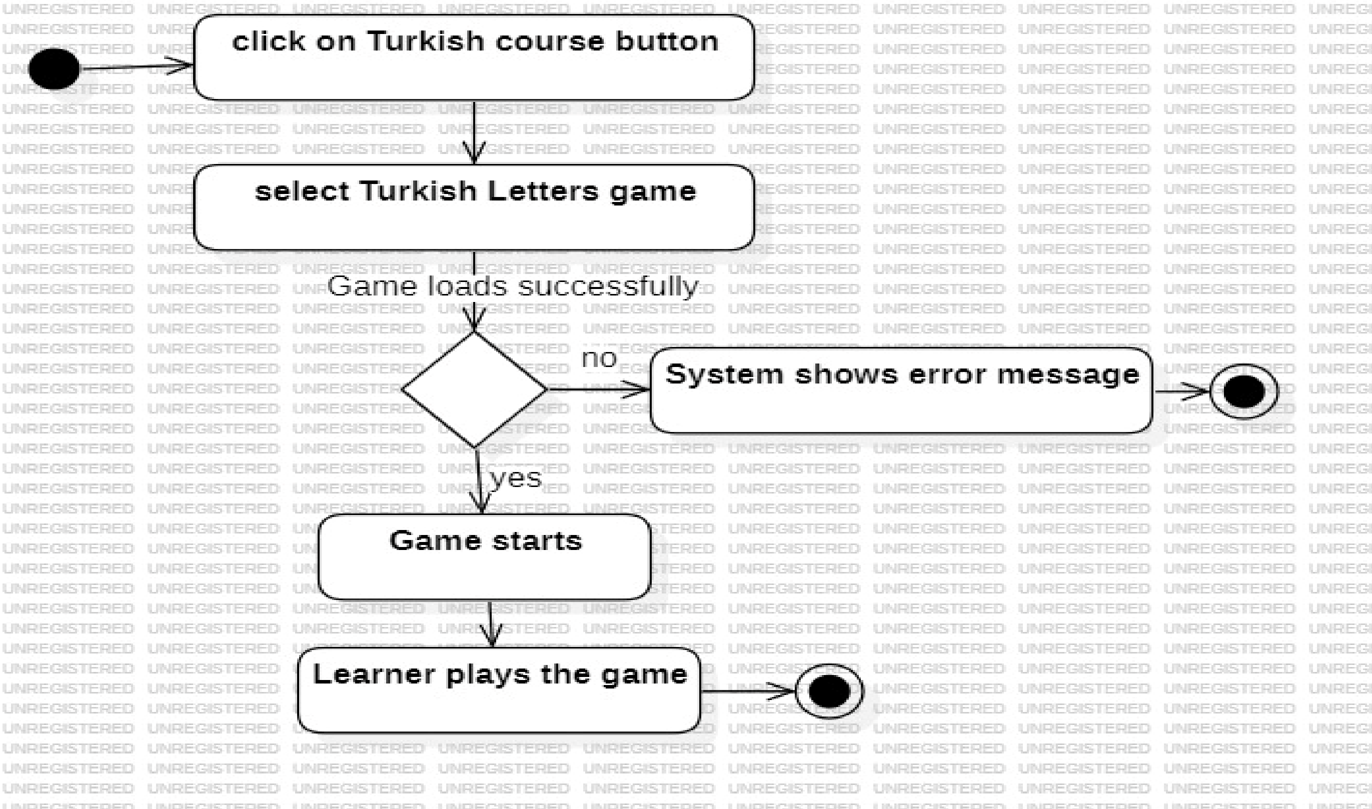


Figure 16

➢ *Game 2 Turkish*

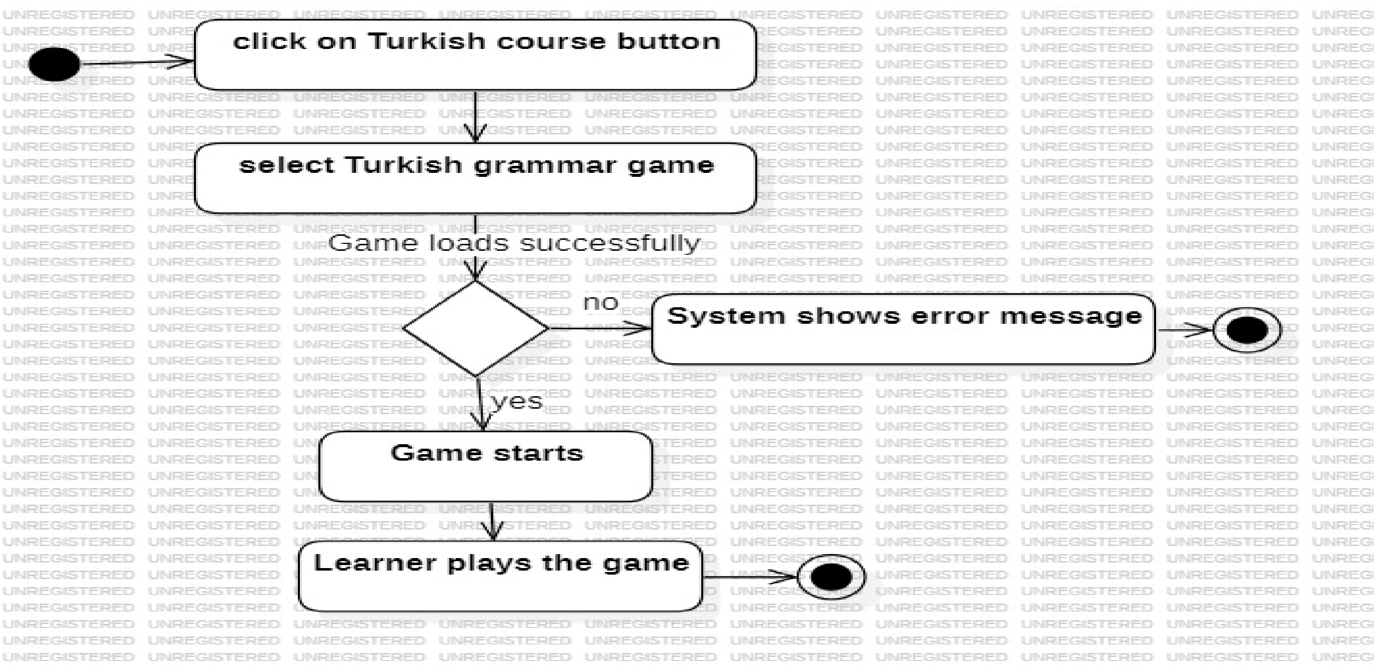


Figure 17

➢ *Show current results*

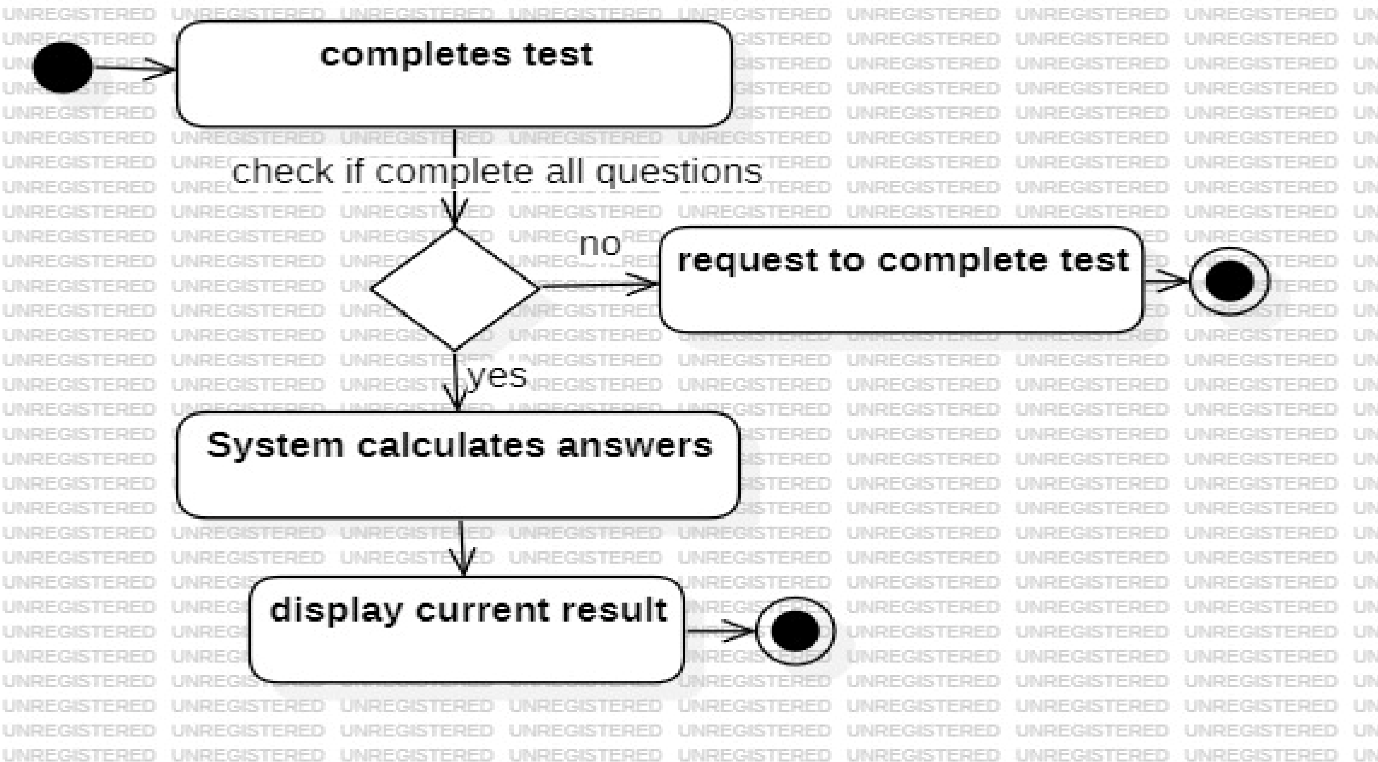


Figure 18

➢ *Update level*

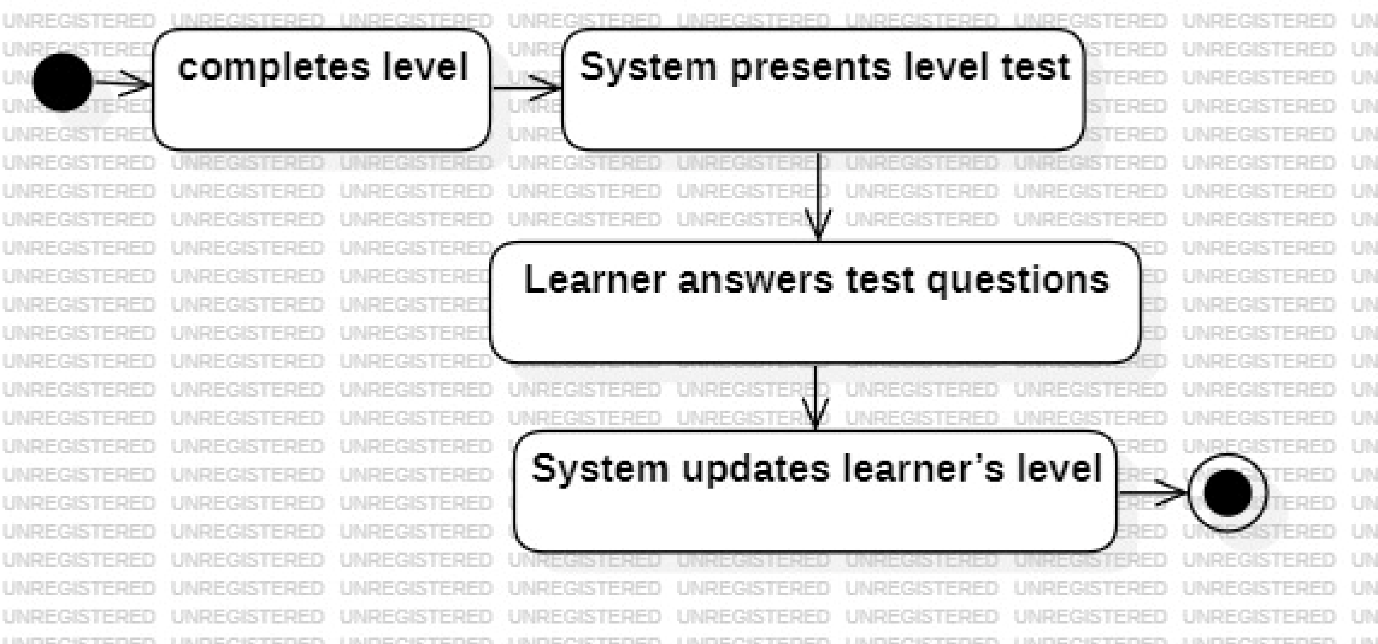


Figure 19

➢ *Contact Admin*

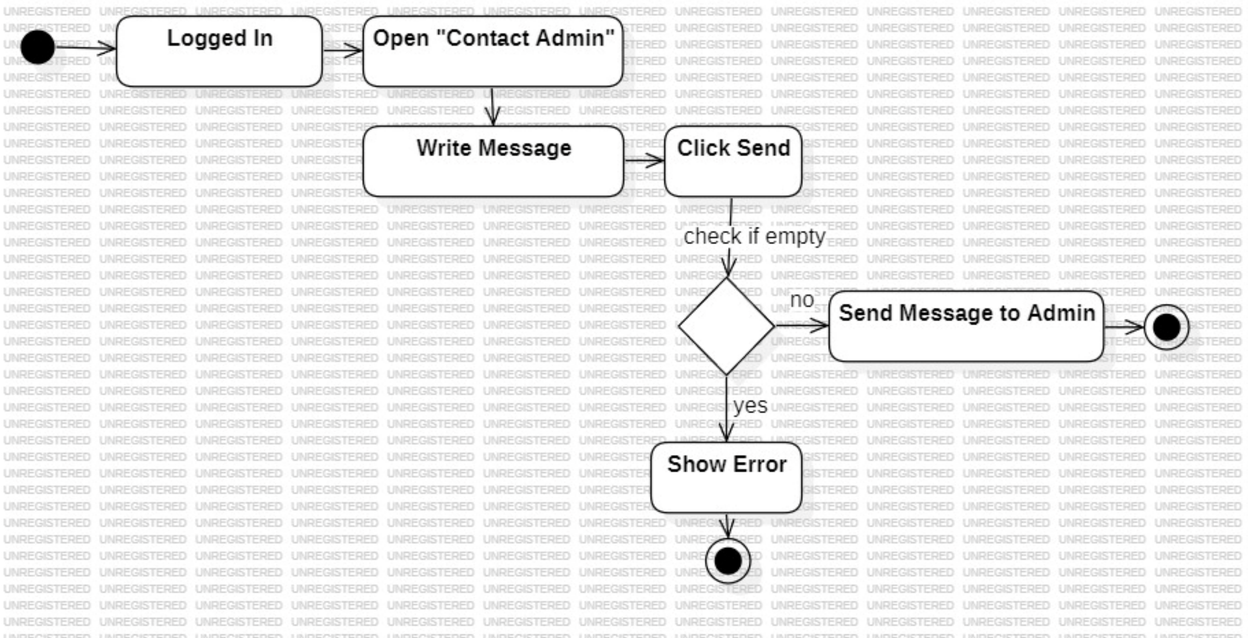


Figure 20

➢ *Rate*

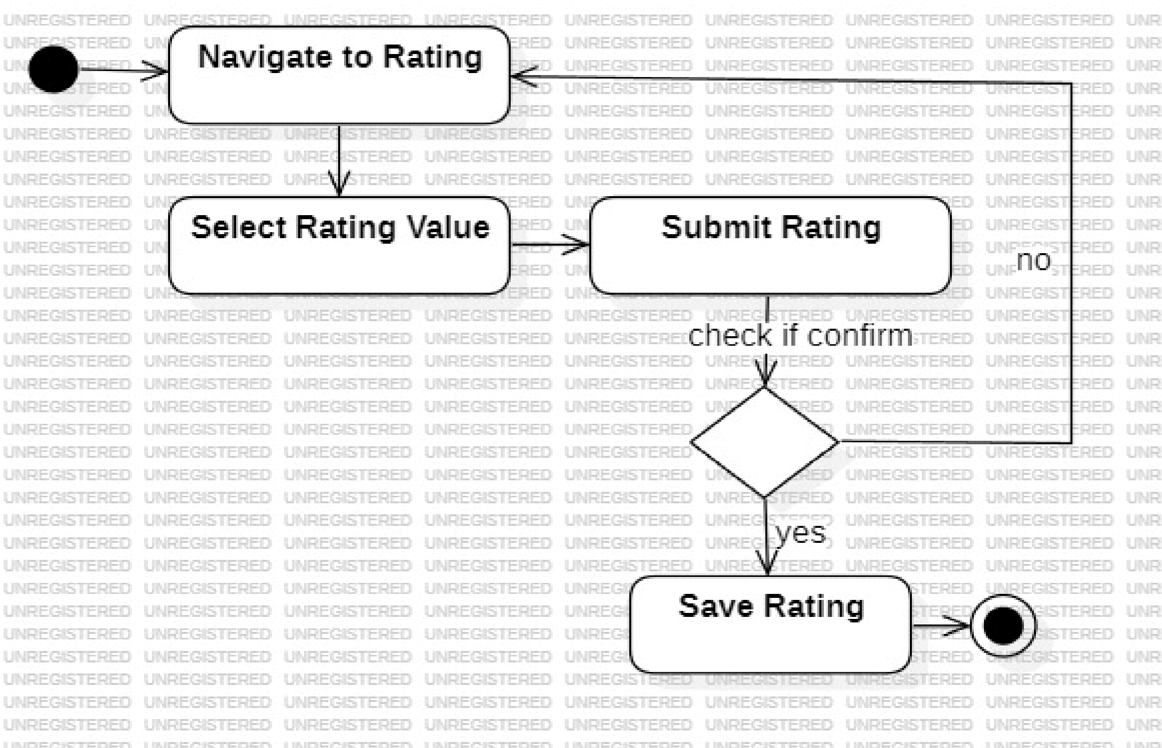


Figure 21

➢ *Filter Ratings*

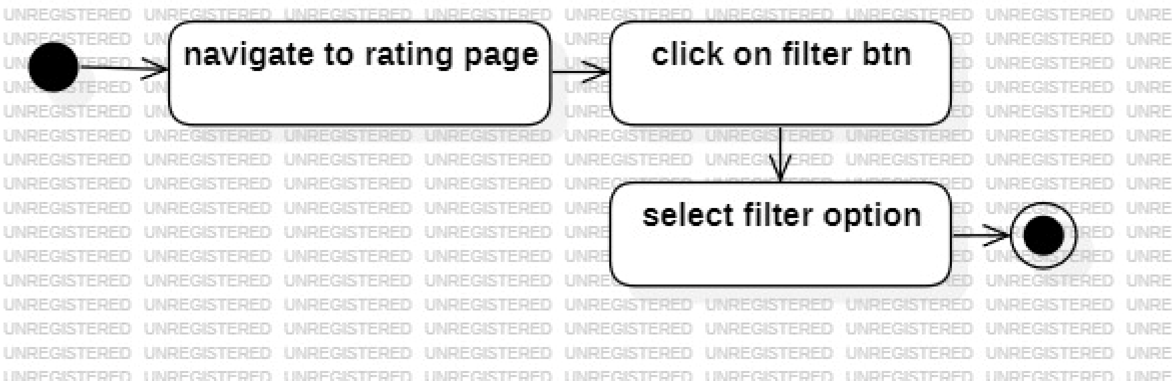


Figure 22

➢ *Redetermine Level*

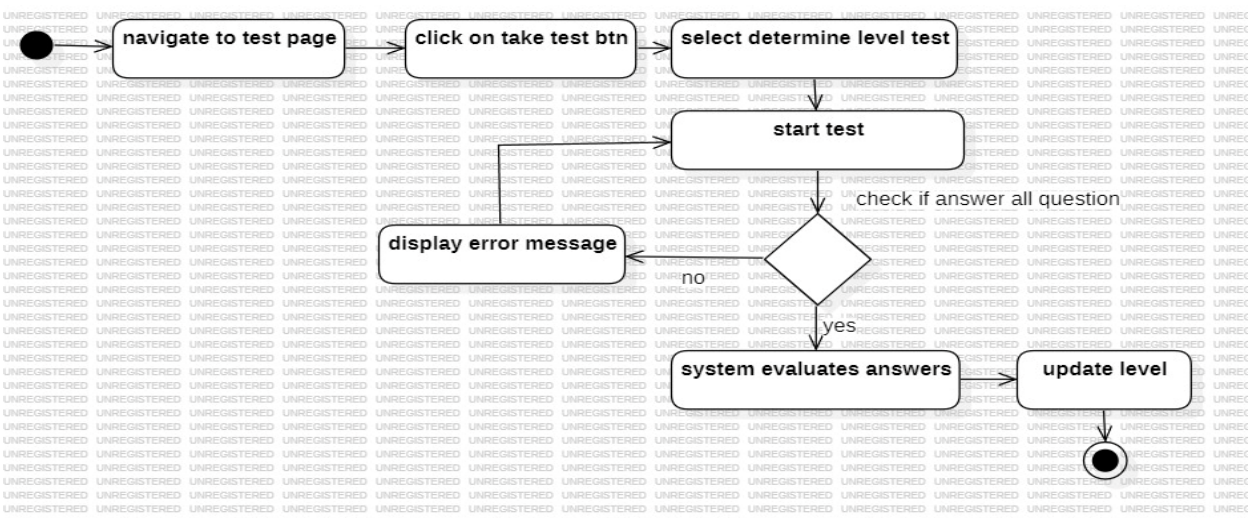


Figure 23

➢ *Play Arabic Game 1*

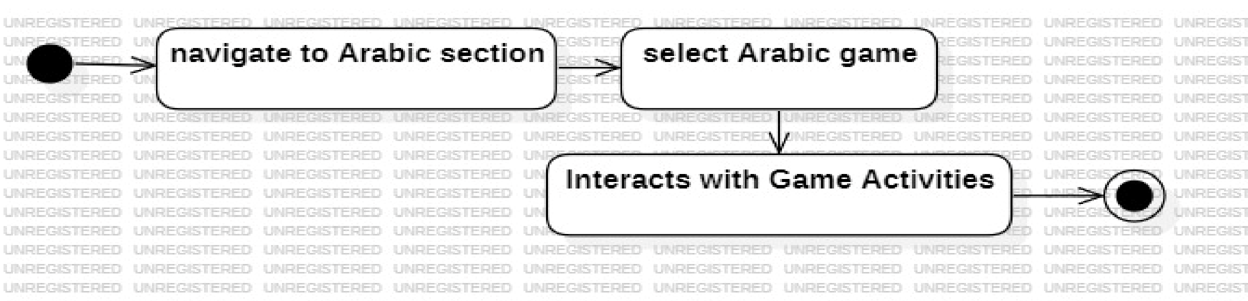


Figure 24

➢ *Take Courses*

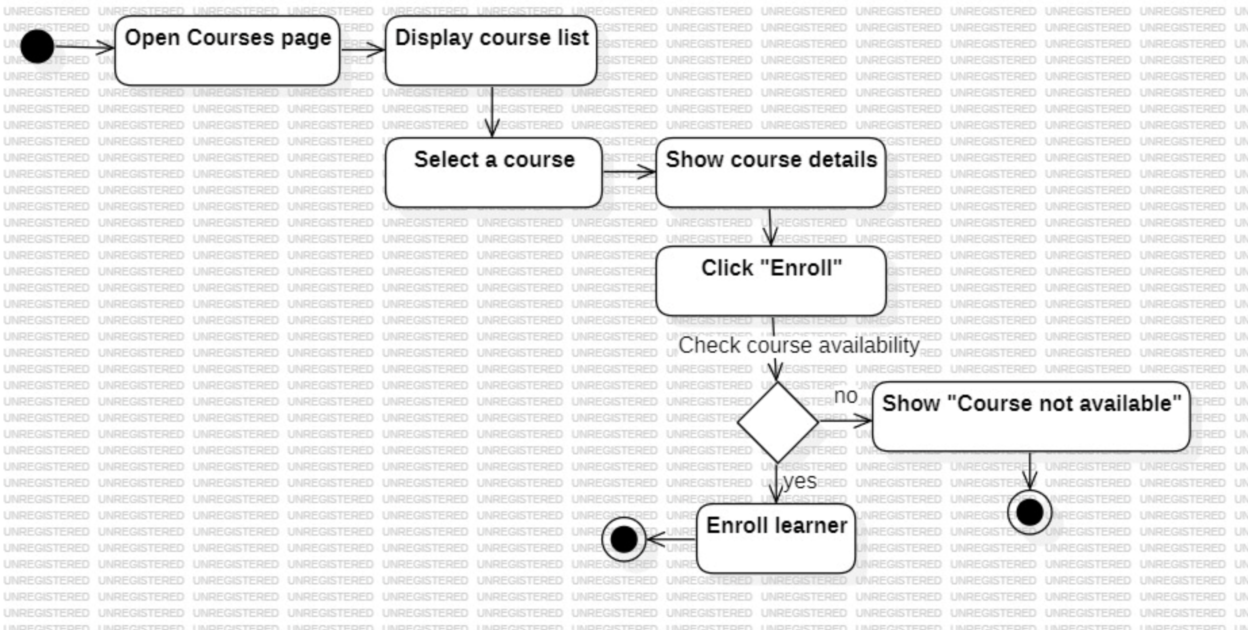


Figure 25

➢ *View Learning History*

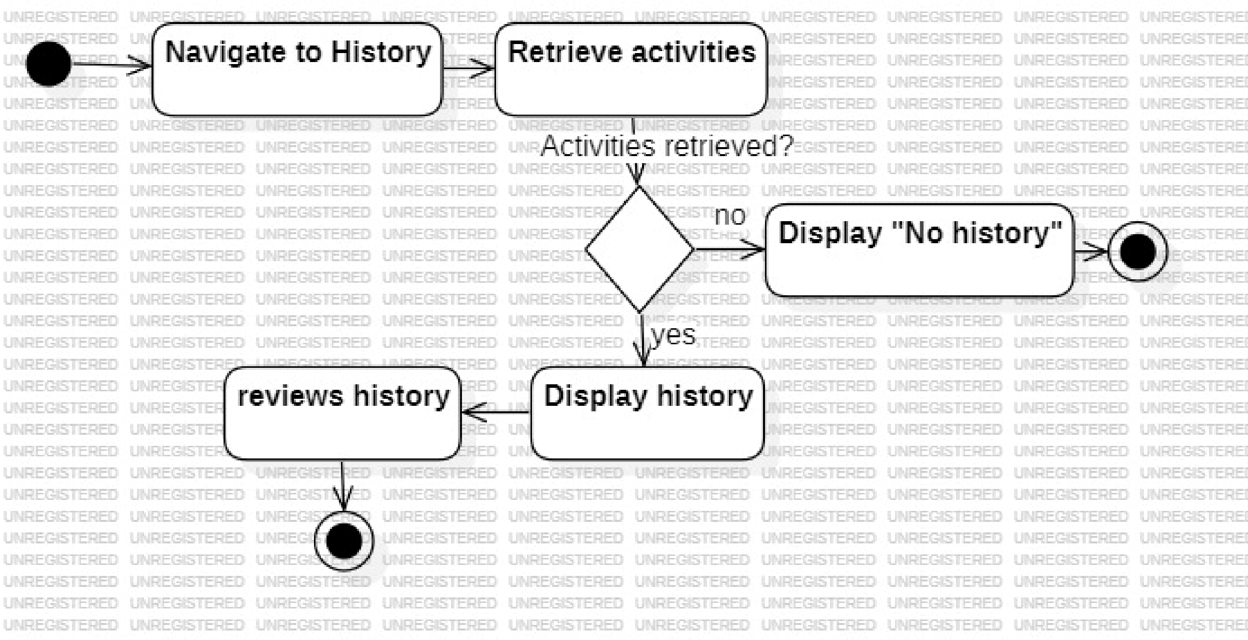


Figure 26

➢ *Generate Performance Reports*

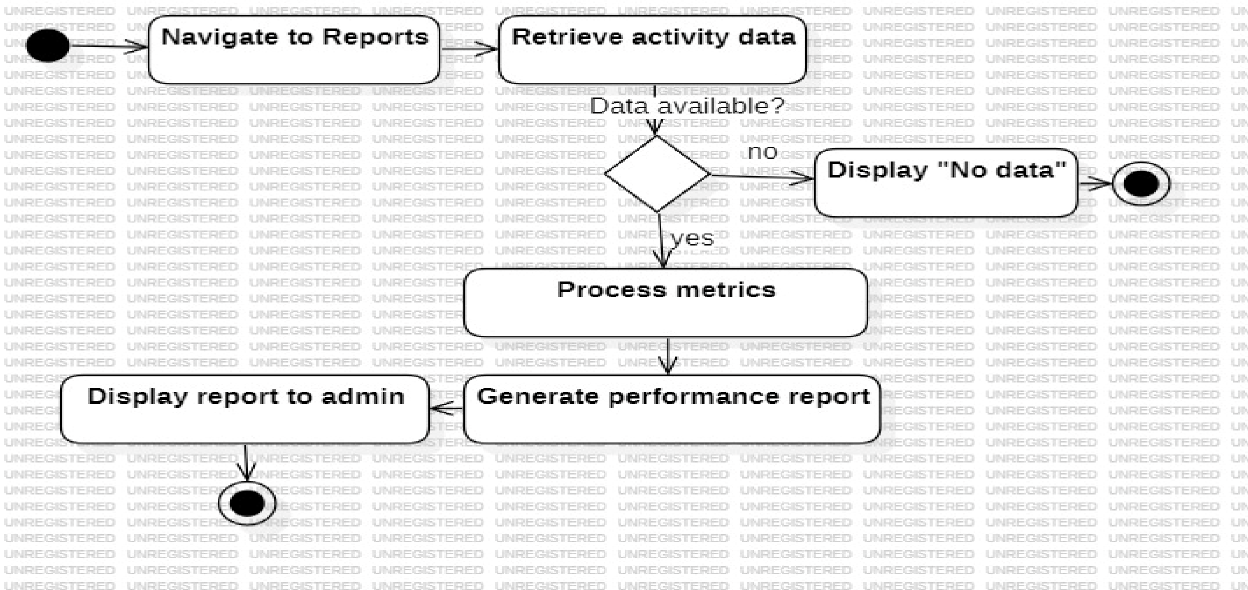


Figure 27

4.8 Sequence Diagram:

➢ *UC_ Sign up:*

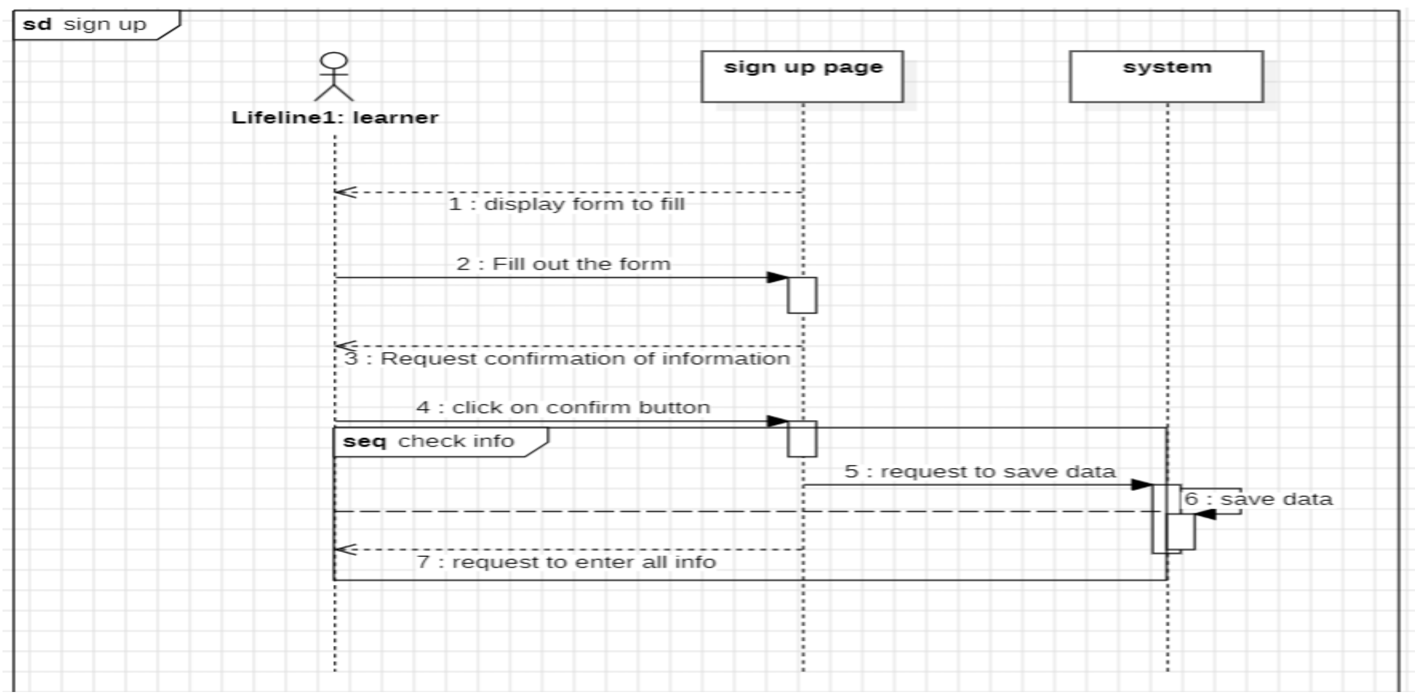


Figure 28

➢ *UC_ Managing Accounts:*

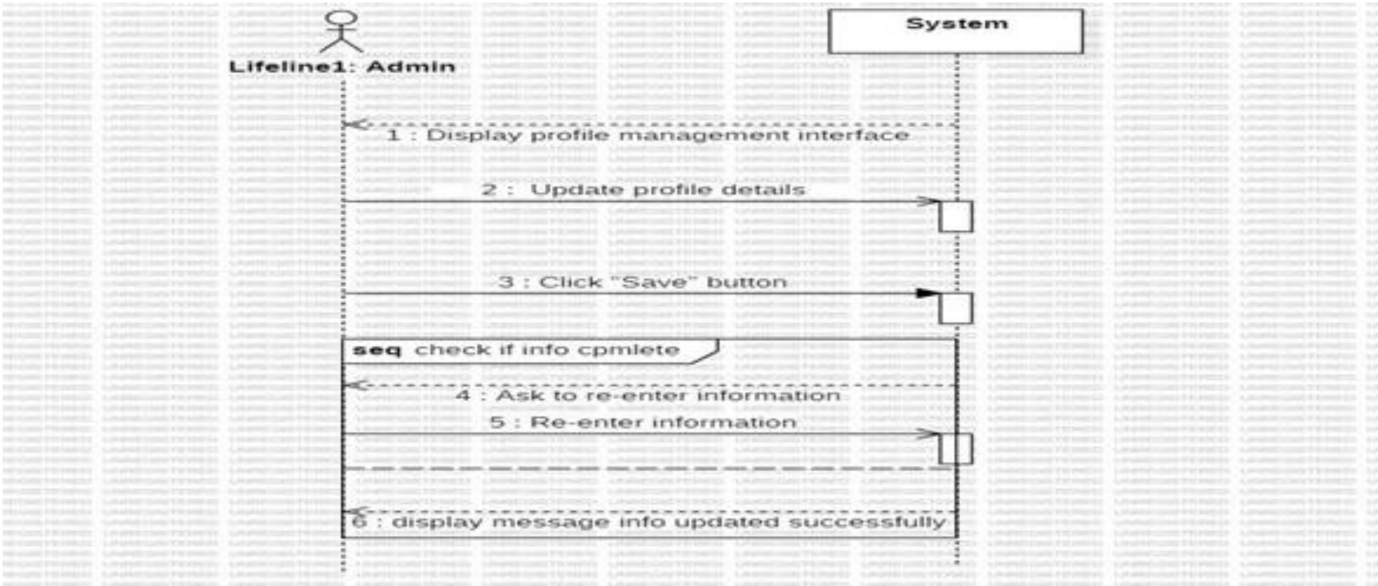


Figure 29

➢ *UC_ Log out:*

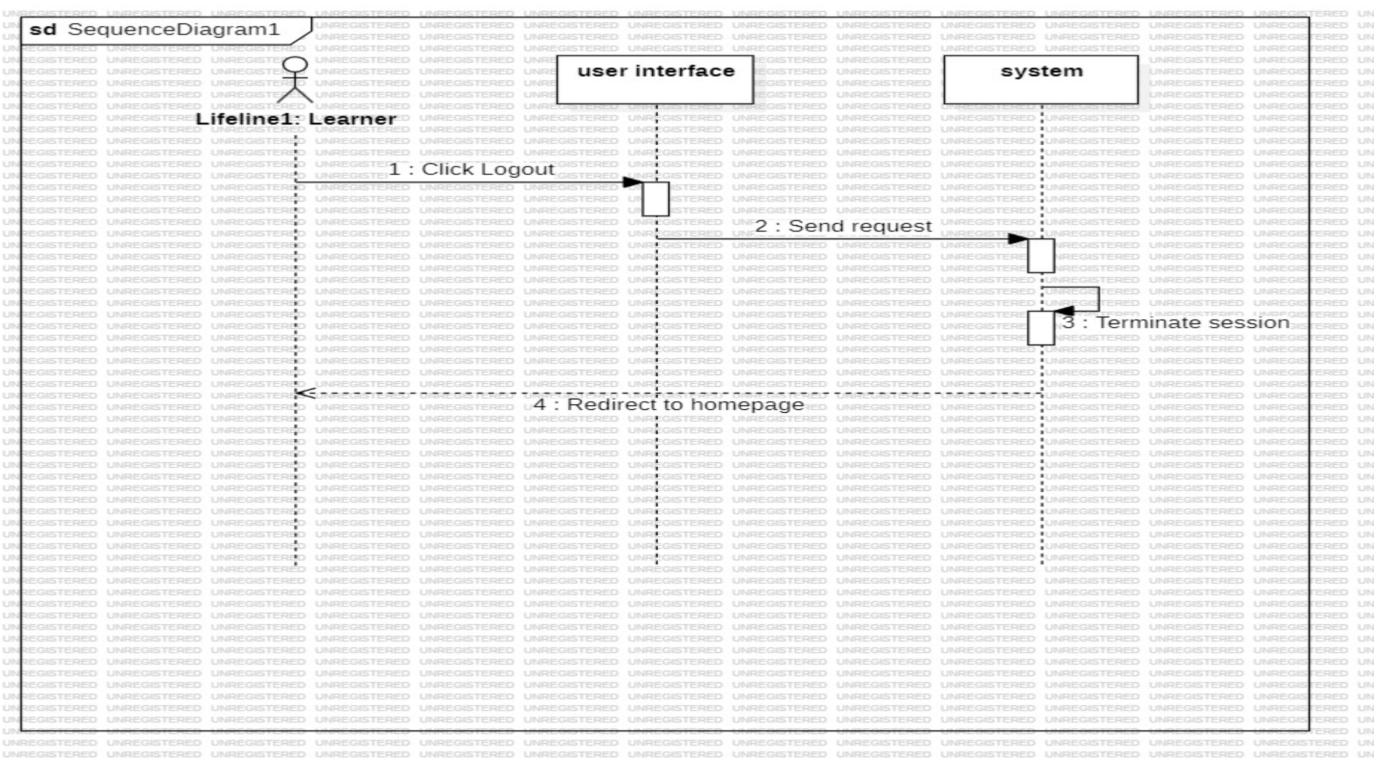


Figure 30

➢ *UC_ Log in:*

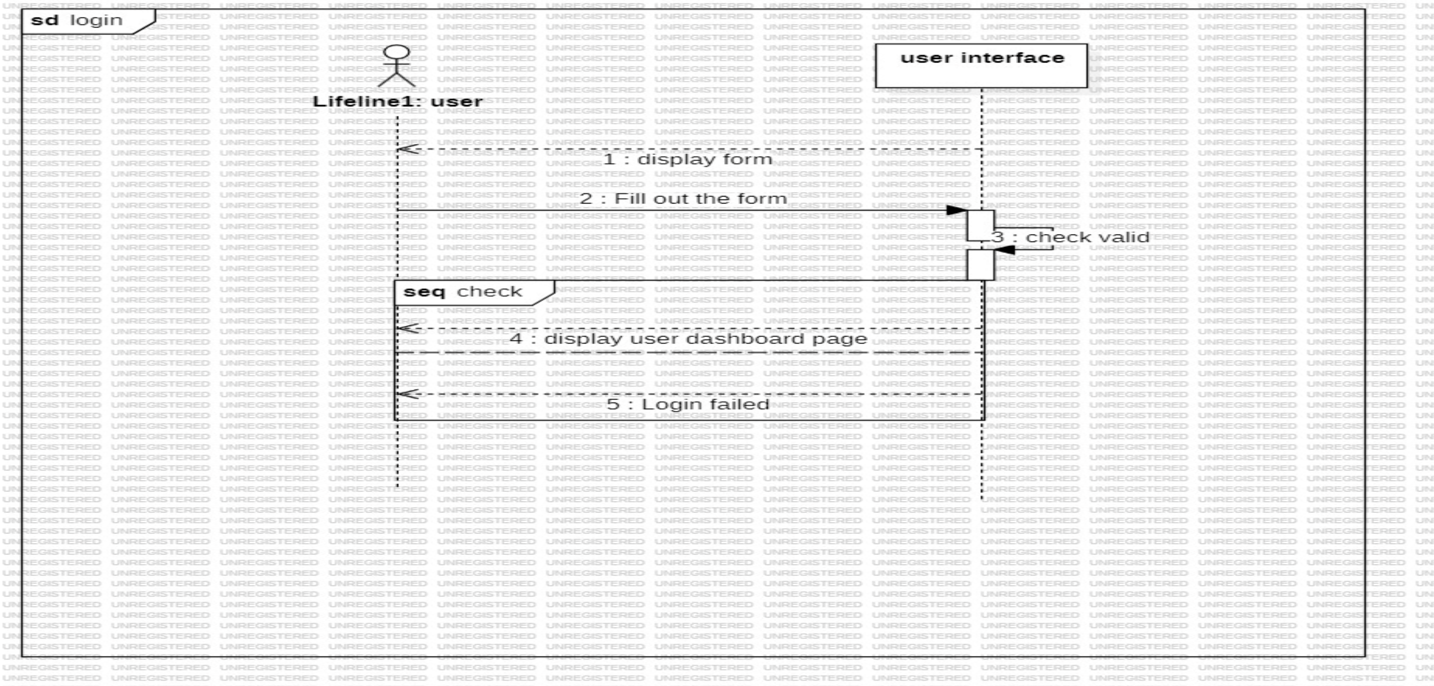


Figure 31

➢ *Show available languages*

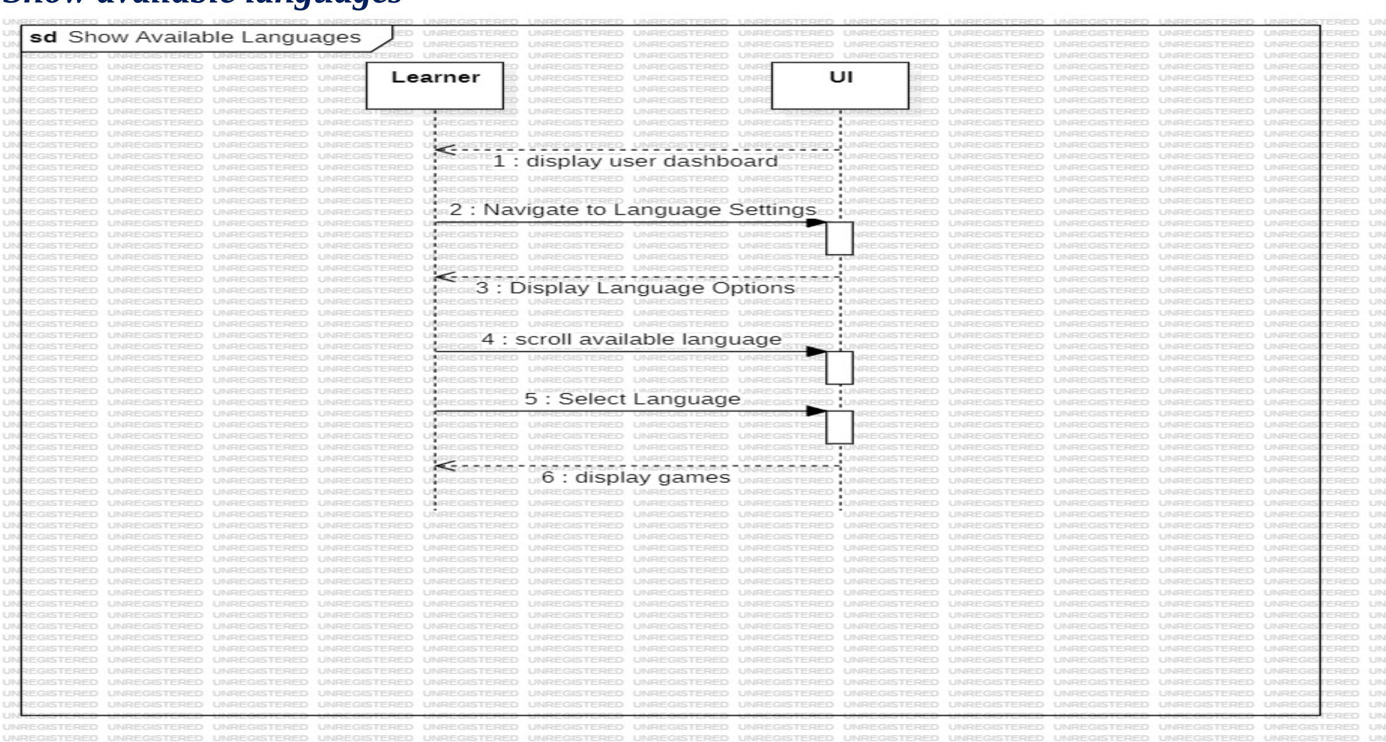


Figure 32

➢ *Show available games*

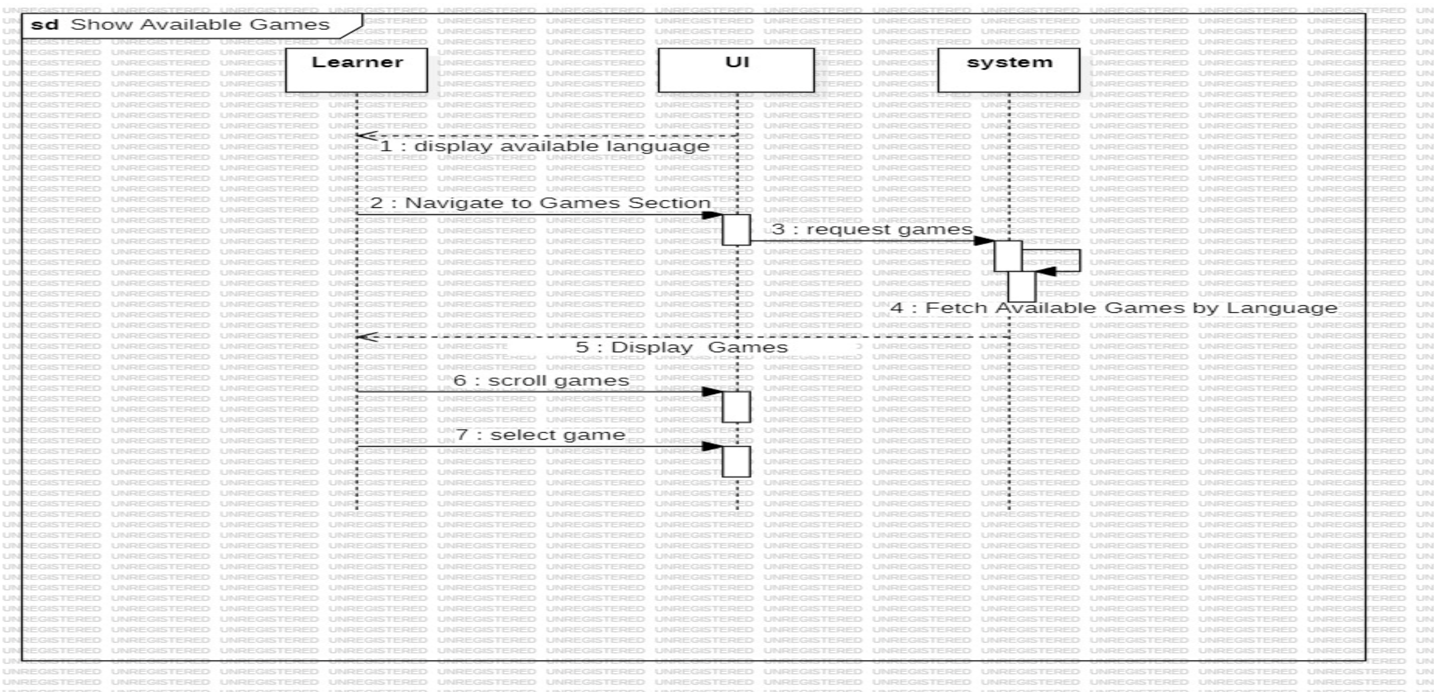


Figure 33

➢ *Determine level*

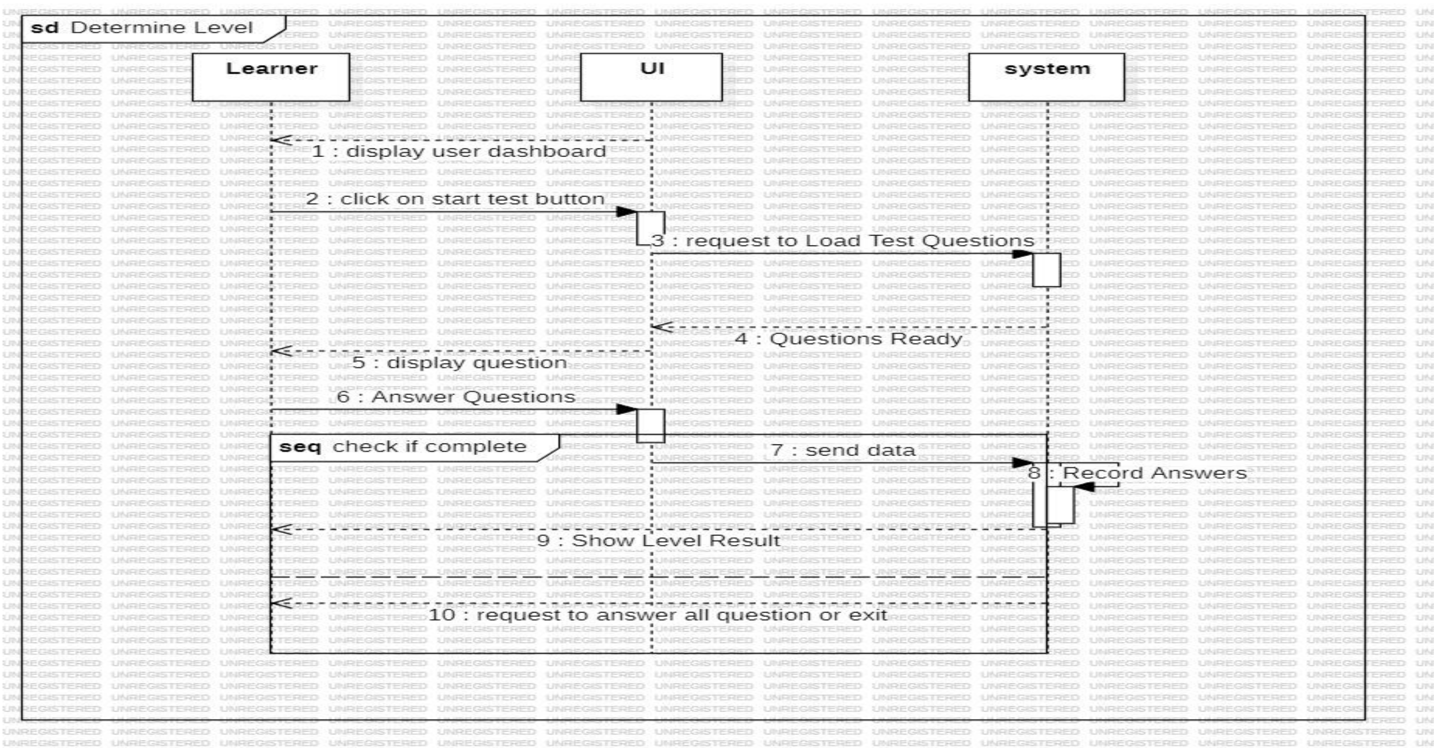


Figure 34

➢ *Game 1 English*

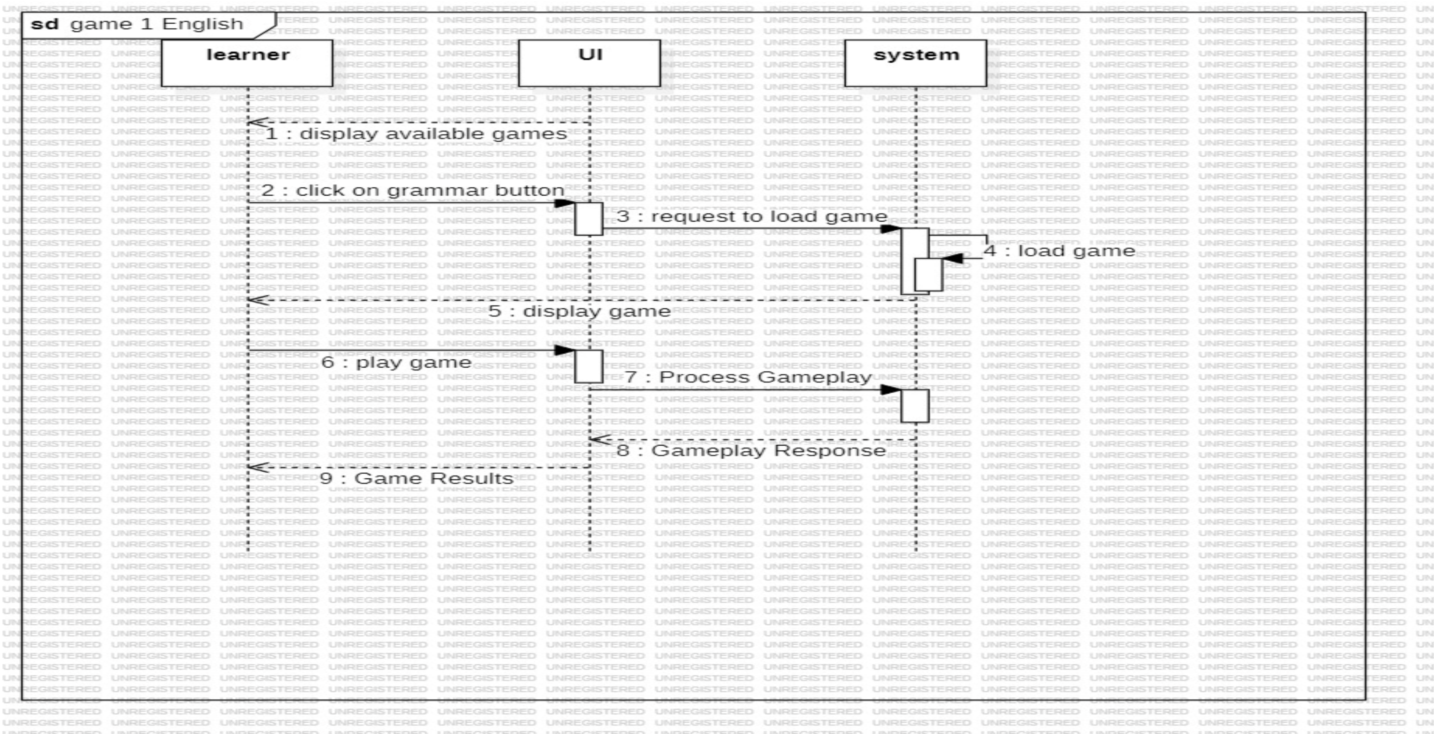


Figure 35

➢ *Game 2 English*

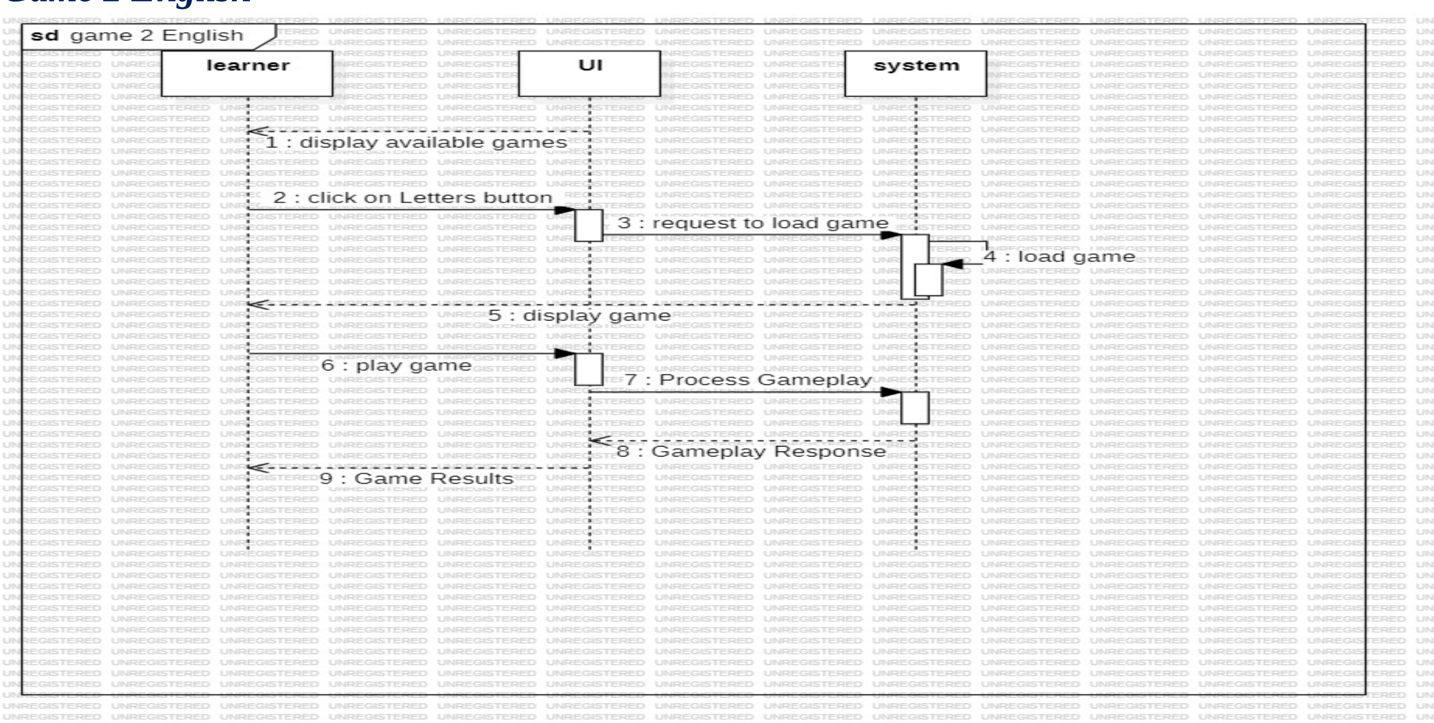


Figure 36

➢ *Game 1 Arabic*

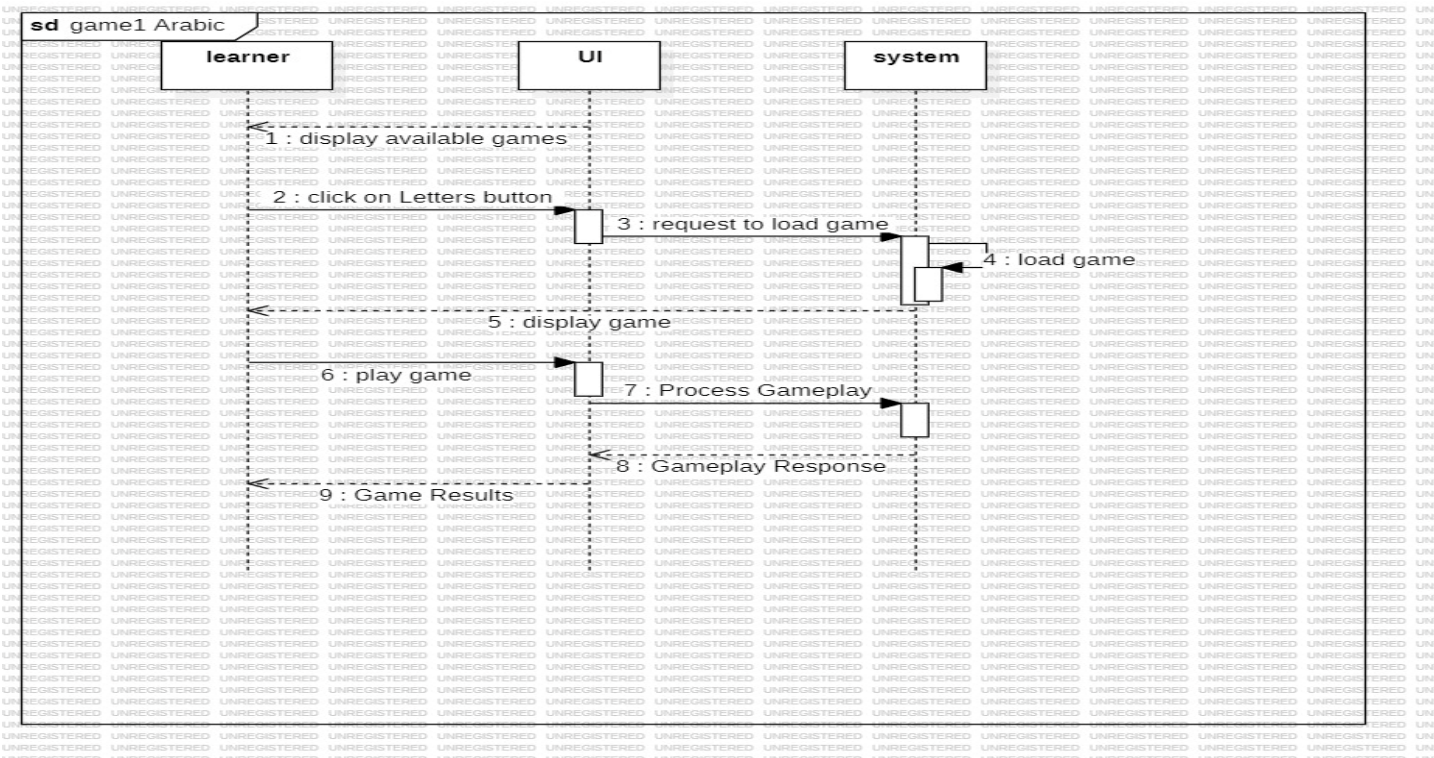


Figure 37

➢ *Game 2 Arabic*

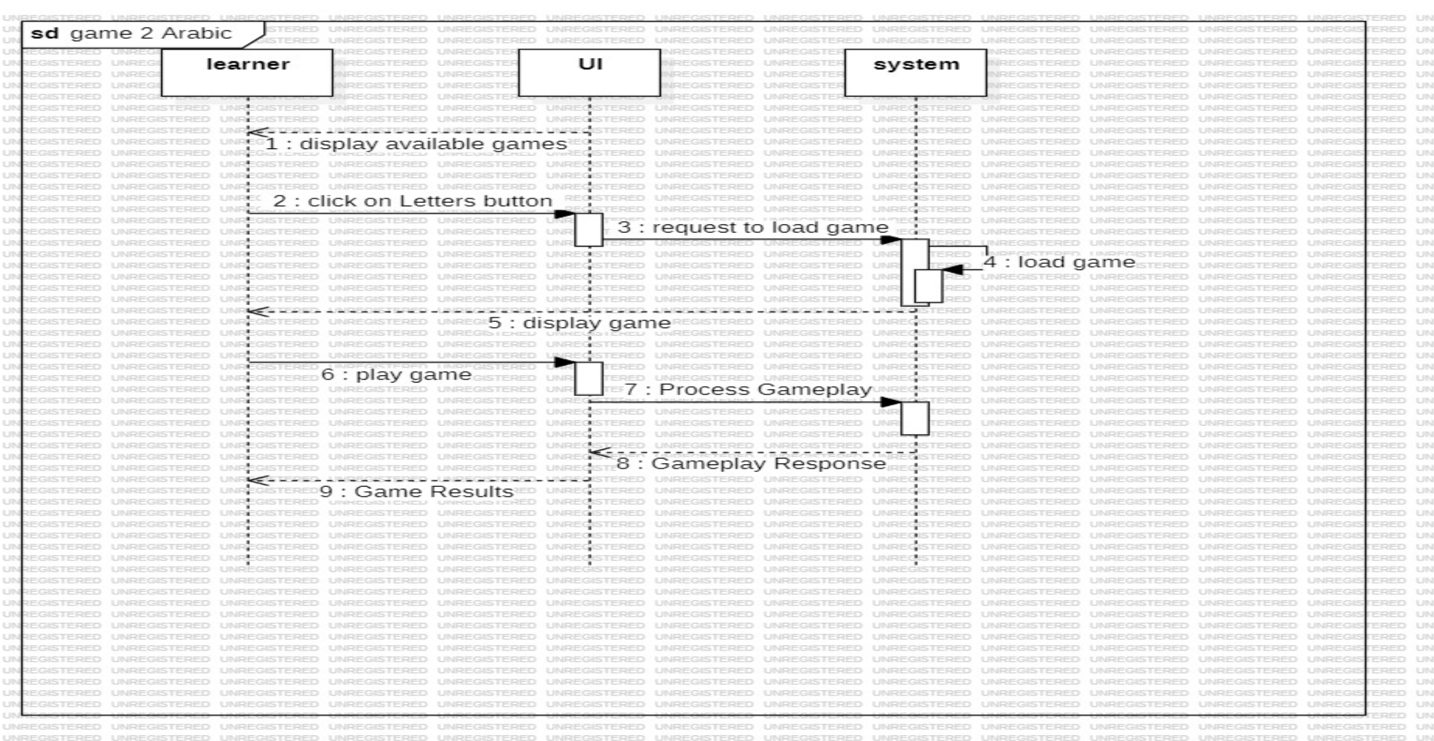


Figure 38

➢ *Game 1 Turkish*

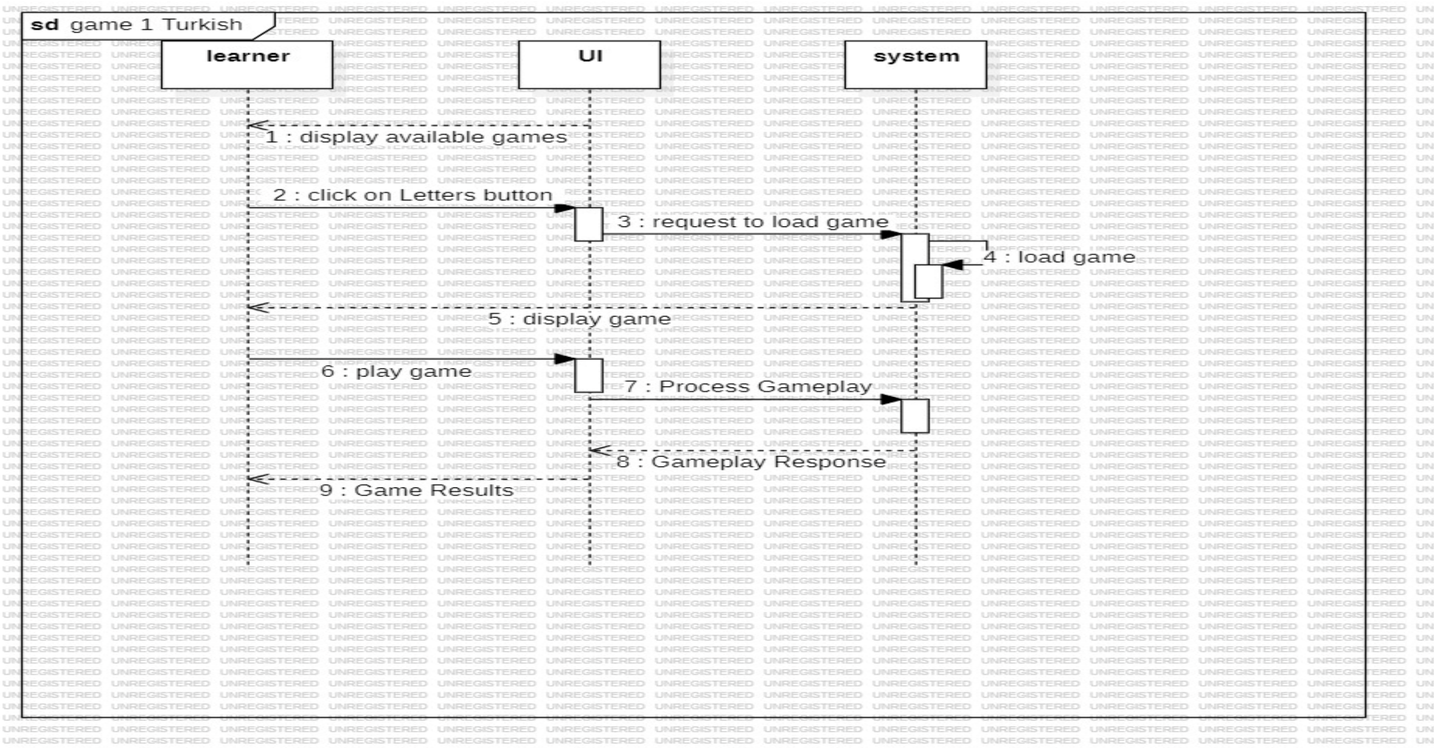


Figure 39

➢ *Game 2 Turkish*

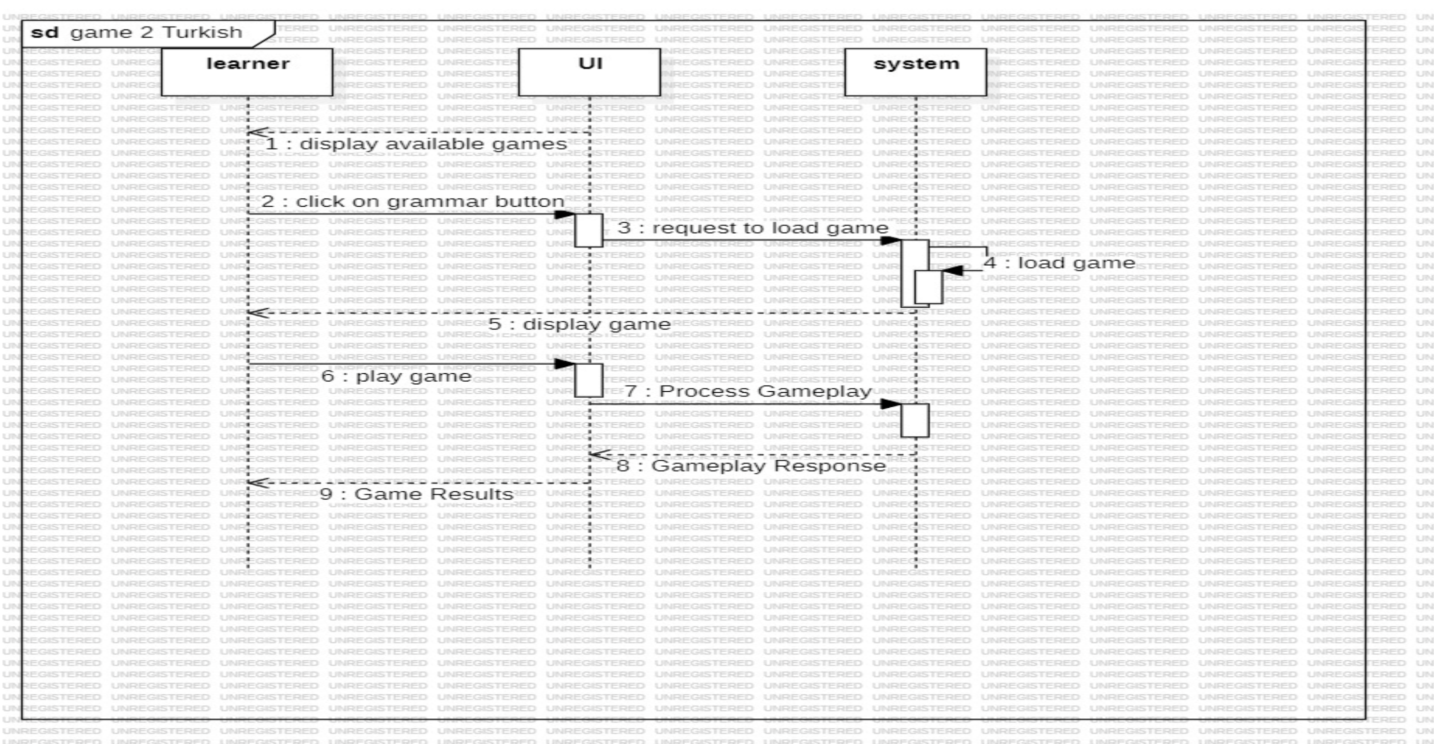


Figure 40

➢ *Show current results*

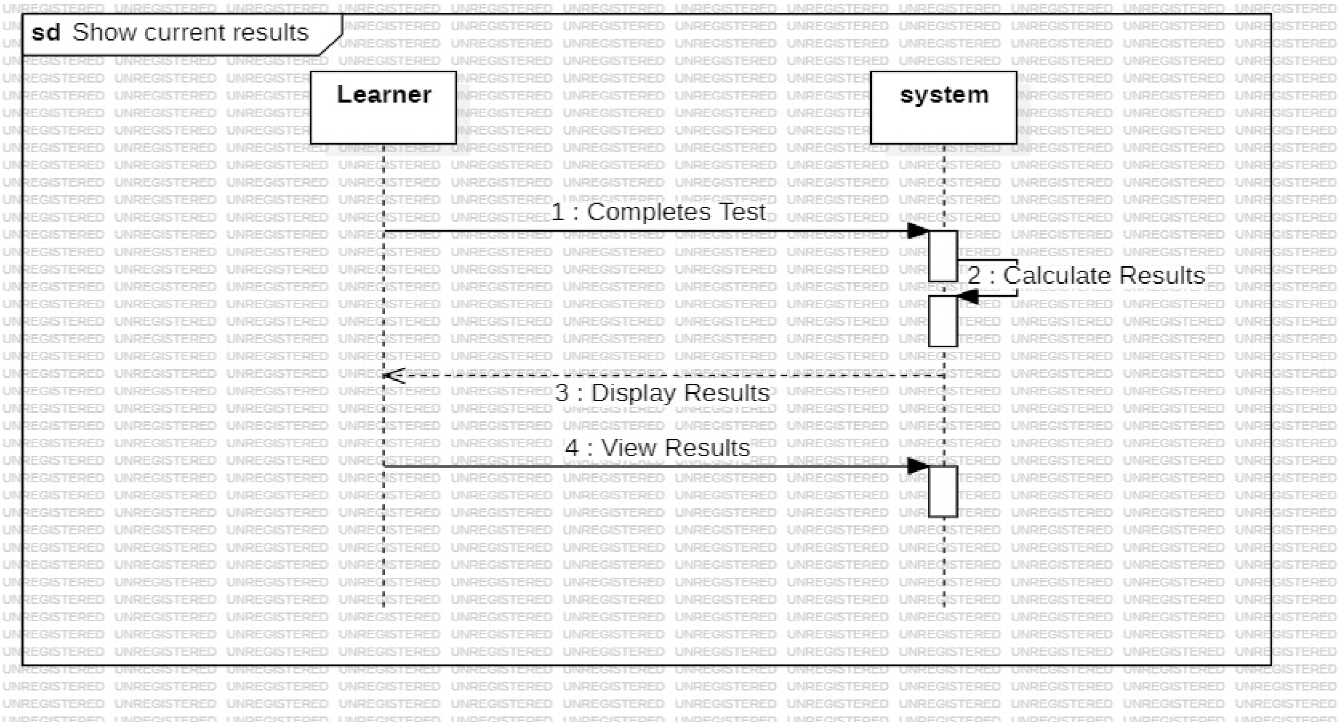


Figure 41

➢ *Update level*

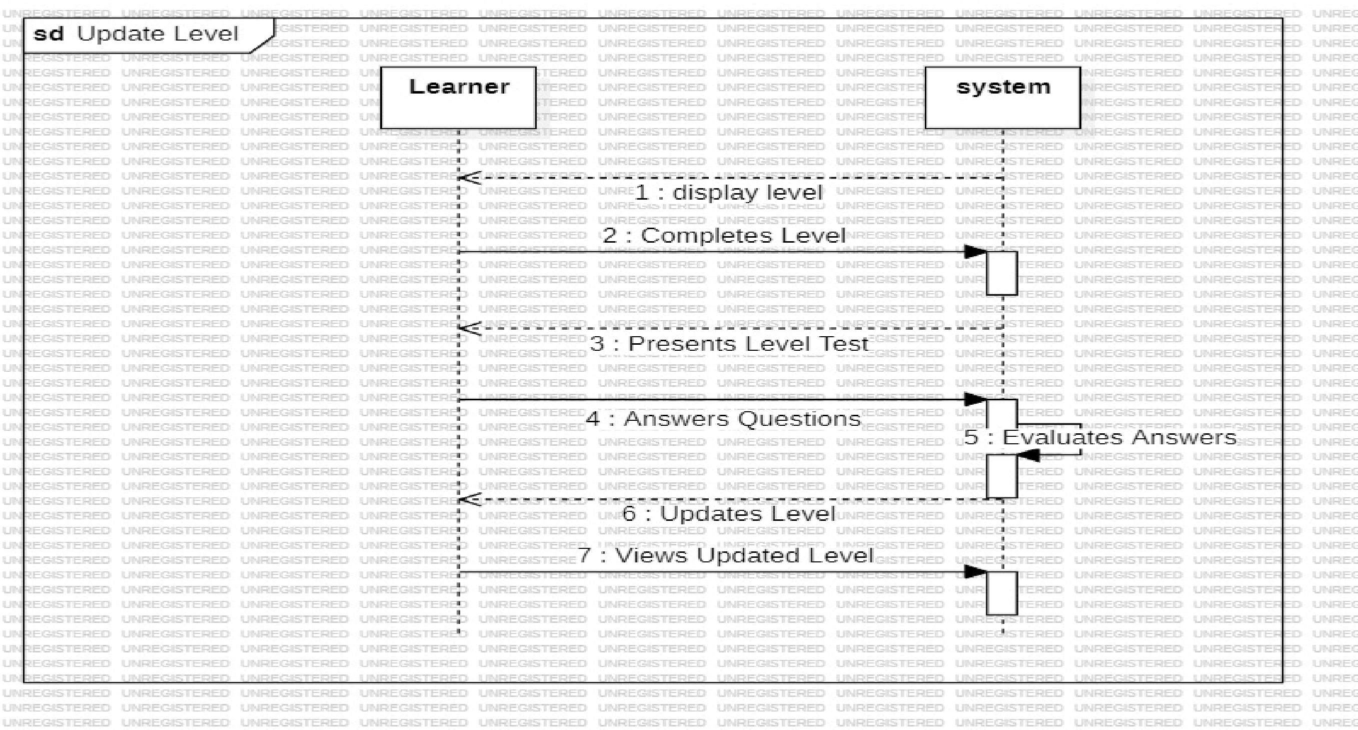


Figure 42

➢ Contact Admin

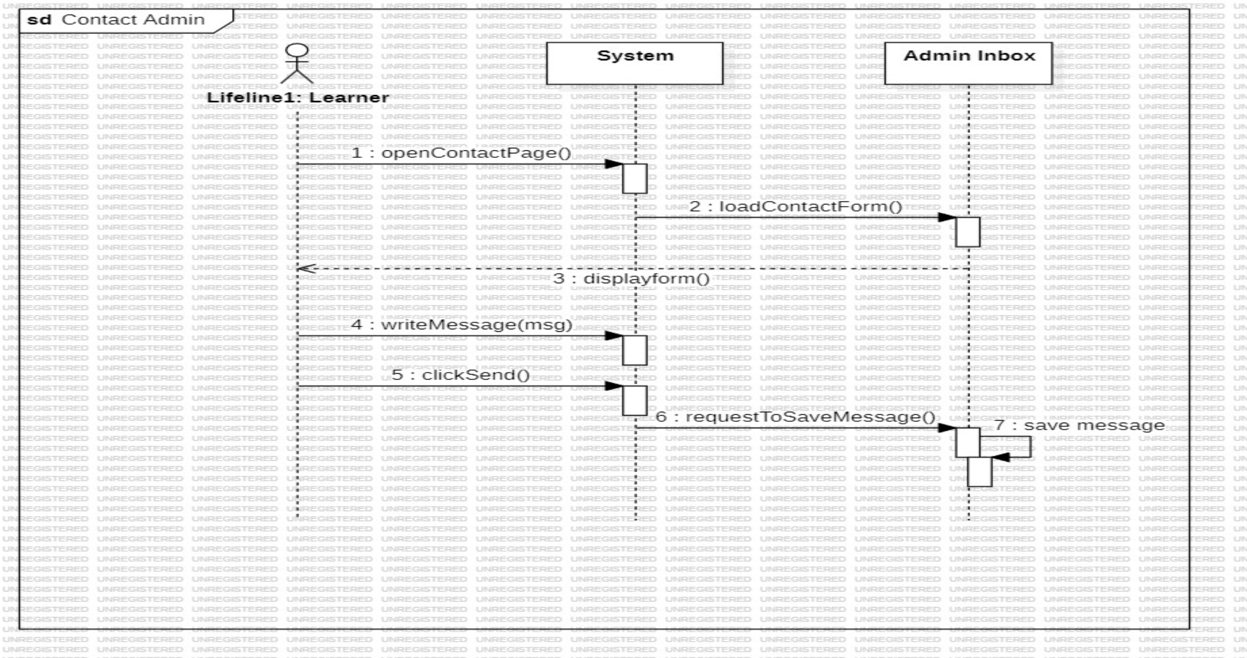


Figure 43

➢ Rate

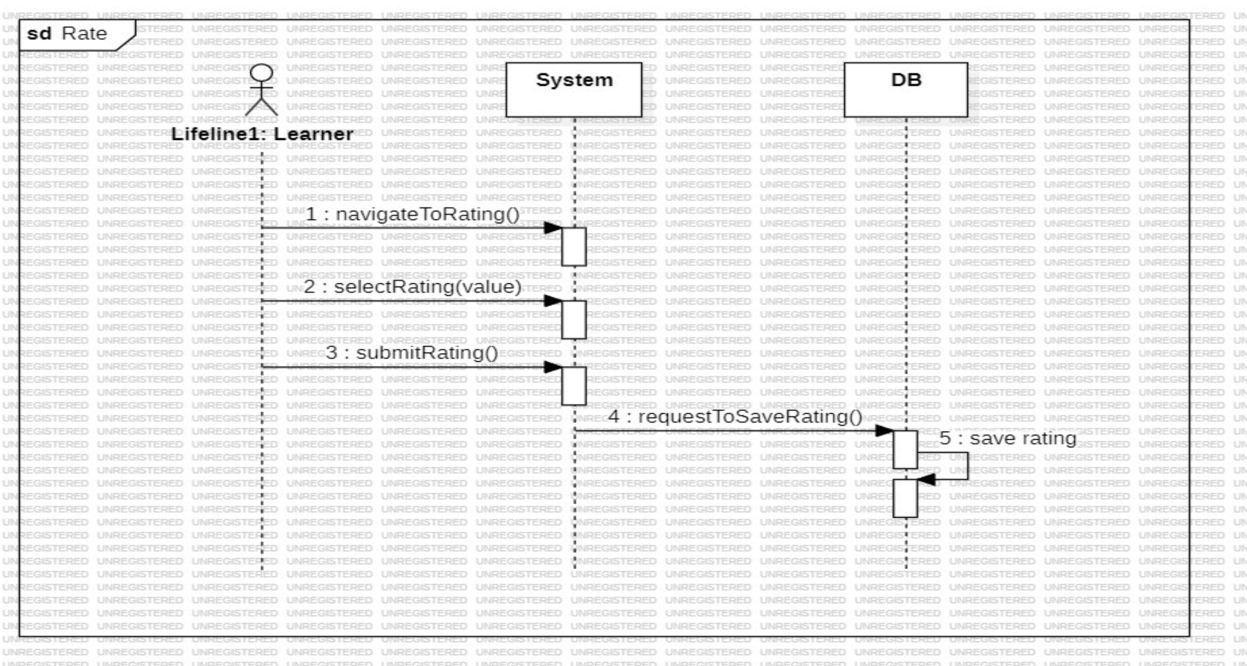


Figure 44

➢ Filter Ratings

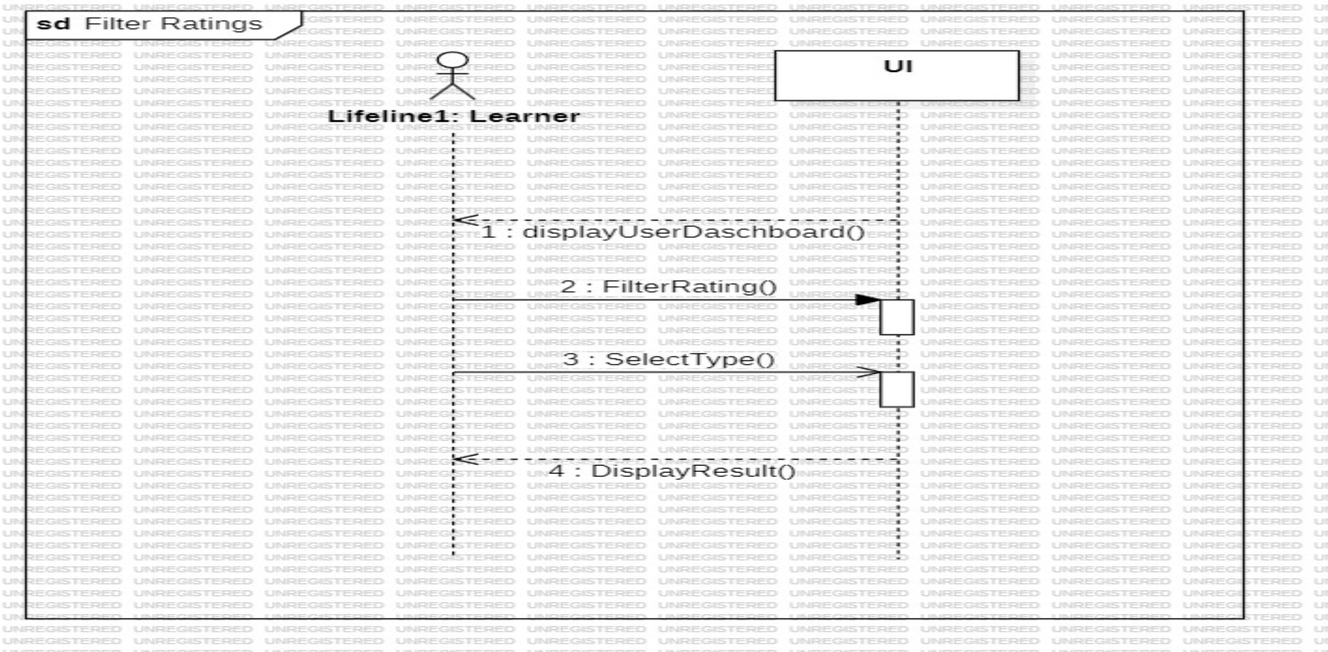


Figure 45

➢ Redetermine Level

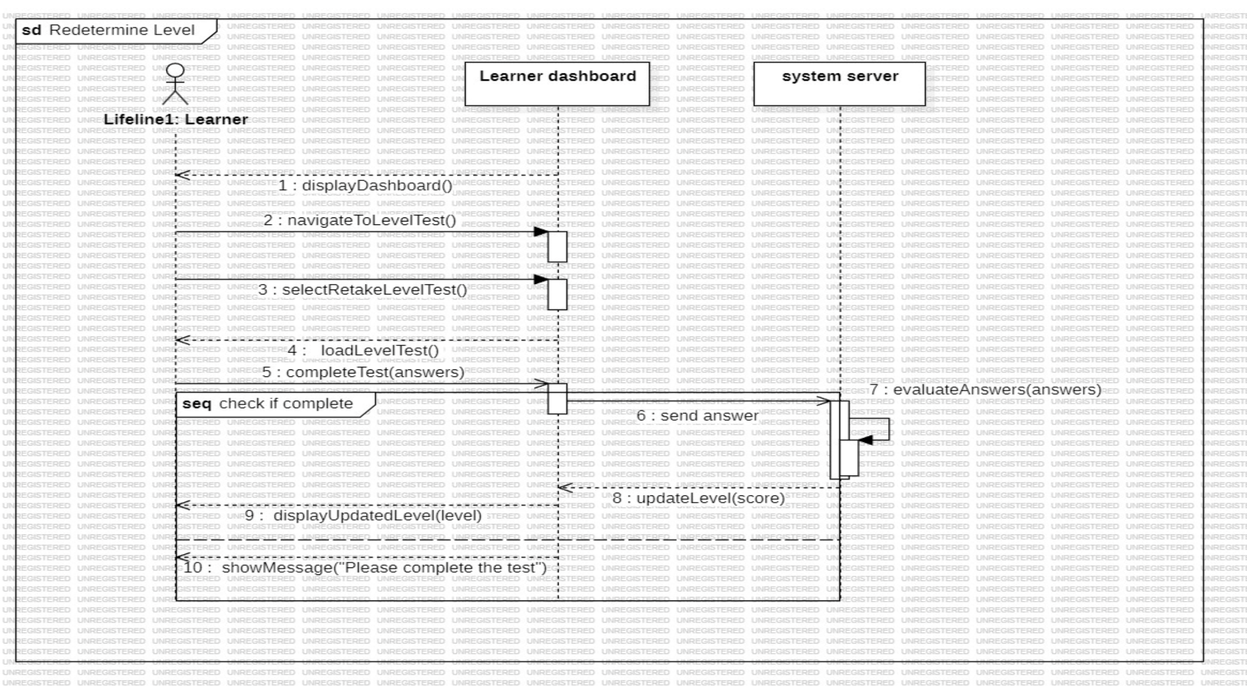


Figure 46

➢ Play Arabic Game 1

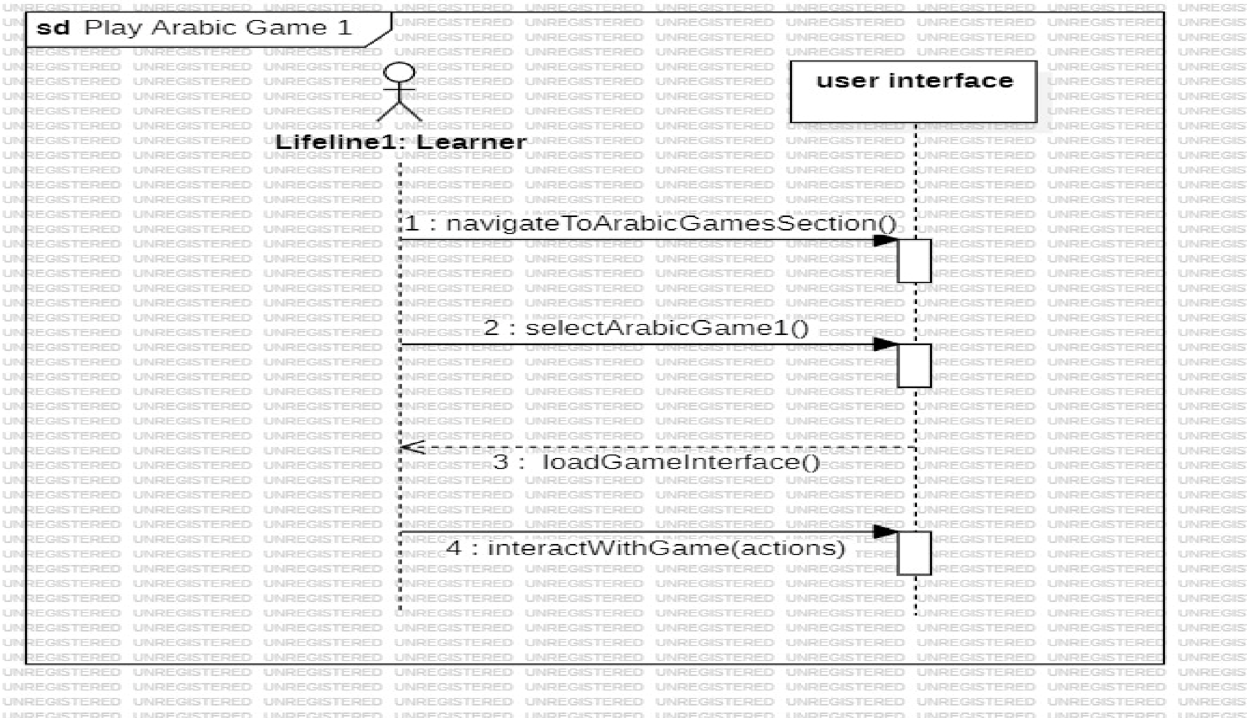


Figure 47

➢ *Take Courses*

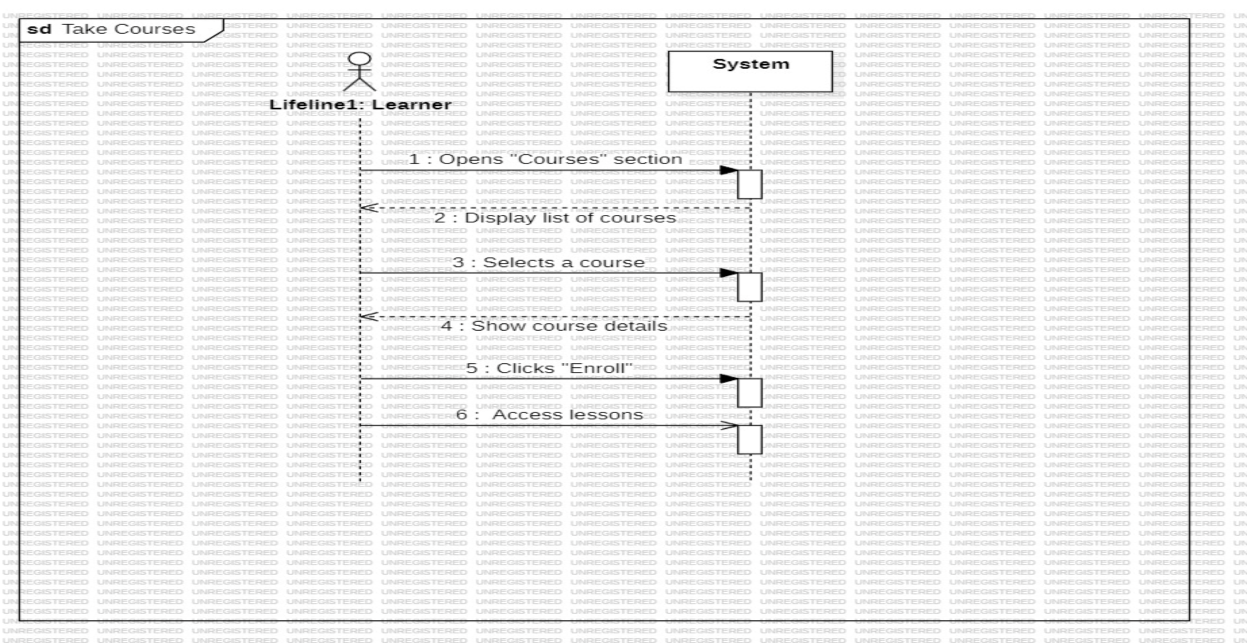


Figure 48

➢ *View Learning History*

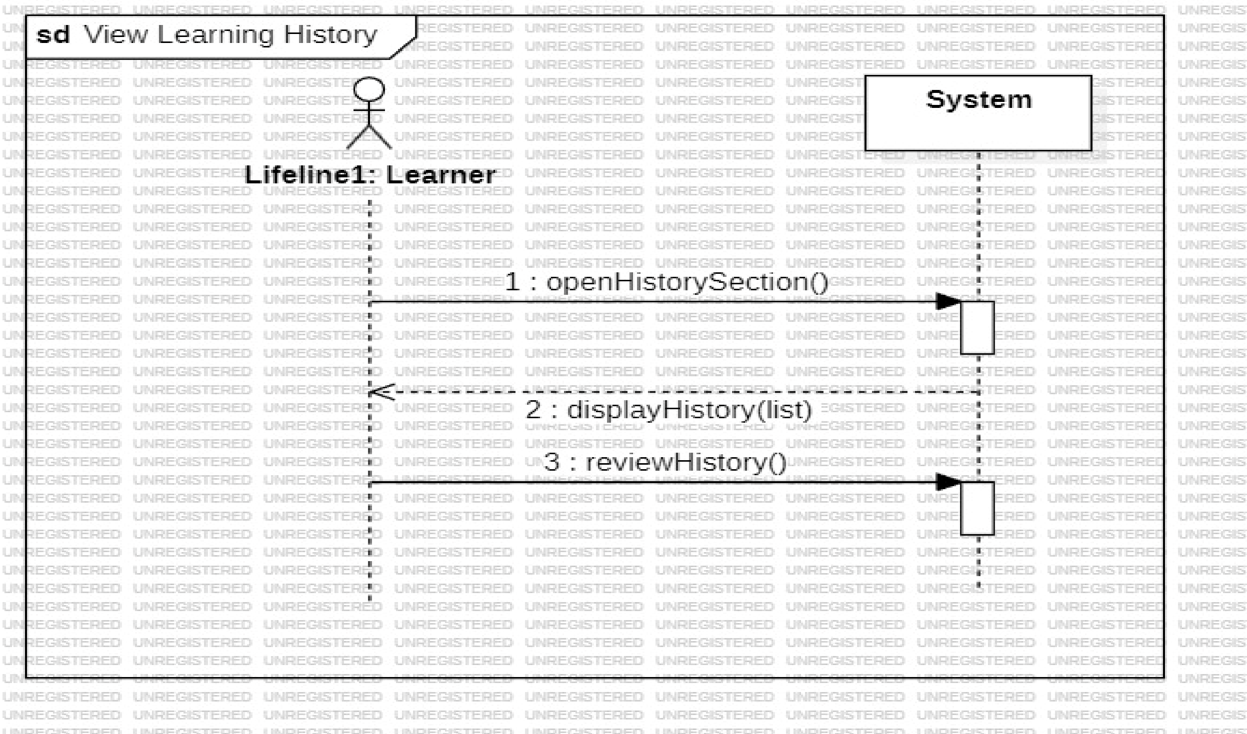


Figure 49

➢ *Generate Performance Reports*

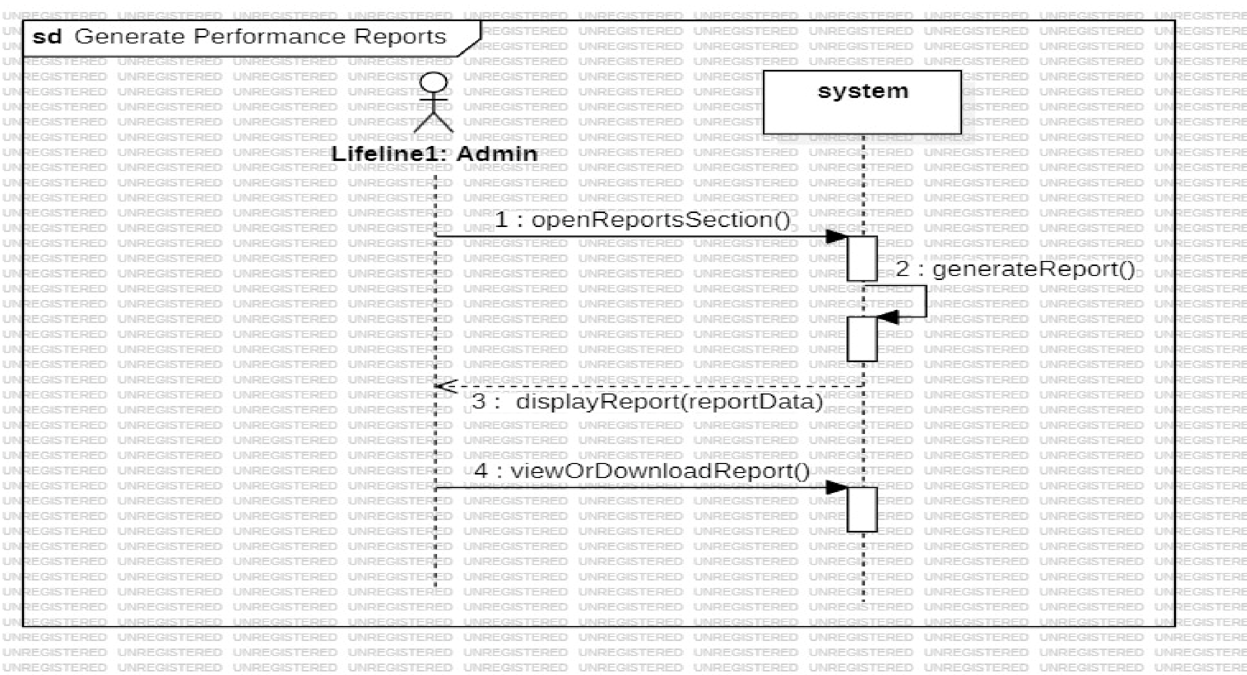


Figure 50

4.9 Test Case Descriptions:

Test Case ID	Requirement ID	Description
TC_1.1	RQ_1	Sign up using valid email, username, and password to create a new account.
TC_1.2	RQ_1	Attempt to sign up using an already registered email.
TC_1.3	RQ_1	Try signing up with missing or invalid input fields.
TC_2.1	RQ_2	Log in using correct email and password.
TC_2.2	RQ_2	Attempt login with an incorrect password.
TC_2.3	RQ_2	Try logging in with an inactive account.
TC_2.4	RQ_2	Attempt login during a system/database failure.
TC_3.1	RQ_3	System sends notification when a new game is added.
TC_3.2	RQ_3	System sends notification when learner completes a test and level updates.
TC_3.3	RQ_3	Ensure no notifications appear when learner disables them.
TC_4.1	RQ_4	Logout successfully and end session.
TC_4.2	RQ_4	Attempt logout during system/database error.
TC_4.3	RQ_4	Logout from one device without affecting other logged-in devices.
TC_5.1	RQ_5	View list of supported languages in settings.
TC_5.2	RQ_5	System shows default language when no languages are available.
TC_5.3	RQ_5	Cancel language selection and keep current language.
TC_6.1	RQ_6	View available games categorized by language.
TC_6.2	RQ_6	Select a language with no available games.
TC_6.3	RQ_6	Browse and scroll through available games.
TC_7.1	RQ_7	Complete placement test to determine learner level.
TC_7.2	RQ_7	Exit placement test before finishing; no level assigned.
TC_7.3	RQ_7	Submit placement test during system/database error.
TC_7.4	RQ_7	Submit placement test with empty or invalid answers.
TC_8.1	RQ_8–13	Play English Game 1 successfully.
TC_8.2	RQ_8–13	Game fails to load due to system error.
TC_8.3	RQ_8–13	Exit game before completion.
TC_14.1	RQ_14	Show results after completing a test.
TC_14.2	RQ_14	Submit test without answering any questions.
TC_14.3	RQ_14	System error while displaying results.
TC_14.4	RQ_14	Display results for multiple games played.
TC_15.1	RQ_15	Update learner level after completing required test.
TC_15.2	RQ_15	Exit test before completion; level unchanged.
TC_15.3	RQ_15	System error during level update.
TC_15.4	RQ_15	Submit test with low performance; level remains unchanged.
TC_CA_1	Contact Admin	Send a message to admin successfully.

TC_CA_2	Contact Admin	Attempt to send an empty message.
TC_CA_3	Contact Admin	System error while sending message.
TC_RT_1	Rate	Submit rating (and optional comment) successfully.
TC_RT_2	Rate	Try submitting rating without selecting stars.
TC_RT_3	Rate	System error during rating submission.
TC_FR_1	Filter Ratings	Filter ratings by likes or date.
TC_FR_2	Filter Ratings	No ratings available when filtering.
TC_FR_3	Filter Ratings	System error during rating filtering.
TC_RL_1	Redetermine Level	Successfully retake level test and update level.
TC_RL_2	Redetermine Level	Submit incomplete retake test.
TC_RL_3	Redetermine Level	System error during level re-evaluation.
TC_AG1_1	Arabic Game 1	Load and play Arabic Game 1 successfully.
TC_AG1_2	Arabic Game 1	Game fails to load.
TC_AG1_3	Arabic Game 1	No internet connection while loading game.
TC_TC_1	Take Courses	Enroll in a course successfully.
TC_TC_2	Take Courses	Attempt to enroll in unavailable/full course.
TC_TC_3	Take Courses	Enrollment error due to system/DB issue.
TC_VLH_1	View Learning History	View learning history successfully.
TC_VLH_2	View Learning History	No learning history available.
TC_VLH_3	View Learning History	System error while loading history.
TC_GPR_1	RQ_GPR	Generate performance report successfully after retrieving and processing learner data.
TC_GPR_2	RQ_GPR	Attempt to generate report when no performance data exists.
TC_GPR_3	RQ_GPR	System error occurs during performance report generation.

Table 30

4.10 RTM – Requirements Traceability Matrix:

Req-ID	Title	Analysis	Design	Implementation	TC-ID
RQ_1	Sign up	language learning\class.mdj language learning \languagelearnUS_HL.mdj	language learning\sign up.mdj	game\resources\views \register\ signup.blade.php	TC_1.1, TC_1.2, TC_1.3
RQ_2	Log in	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\login.mdj	game\resources\views \register\ Login.blade.php	TC_2.1, TC_2.2, TC_2.3, TC_2.4
RQ_3	Managing Accounts	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\managrr.mdj	game\resources\views \welcome.blade.php	TC_3.1, TC_3.2, TC_3.3
RQ_4	Logout	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\log out.mdj	game\resources\views \welcome.blade.php	TC_4.1, TC_4.2, TC_4.3
RQ_5	Show available languages	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Show Available Languages.jpg	game\resources\views \welcome.blade.php	TC_5.1, TC_5.2, TC_5.3
RQ_6	Show available games	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Show available games by language.mdj	game\resources\views \welcome.blade.php	TC_6.1, TC_6.2, TC_6.3
RQ_7	Determine level	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Determine level.mdj	game\resources\views \english \eTest.blade.php	TC_7.1, TC_7.2, TC_7.3, TC_7.4
RQ_8	Game 1 English	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Game 1 English.mdj	game\resources\ views\en glish\past_simple.blade.php	TC_8.1, TC_8.2, TC_8.3
RQ_9	Game 2 English	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Game 2 English.mdj	game\resources\ views\ english\letters.blade.php	TC_8.1, TC_8.2, TC_8.3
RQ_10	Game 1 Arabic	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Game 1 Arabic.mdj	game\resources\views \arabich	TC_8.1, TC_8.2, TC_8.3
RQ_11	Game 2 Arabic	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Game 2 Arabic.mdj	game\resources\views \arabich	TC_8.1, TC_8.2, TC_8.3
RQ_12	Game 1 Turkish	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Game 1 Turkish.mdj	game\resources \views\turkey	TC_8.1, TC_8.2, TC_8.3
RQ_13	Game 2 Turkish	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Game 2 Turkish.mdj	game\resources \views\turkey	TC_8.1, TC_8.2, TC_8.3
RQ_14	Show current results	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Show Current Results.mdj	game\resources\view s\welcome.blade.php	TC_14.1, TC_14.2, TC_14.3, TC_14.4
RQ_15	Update level	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\Update Level.mdj	game\resources\view s\welcome.blade.php	TC_15.1, TC_15.2, TC_15.3, TC_15.4

RQ_16	Contact admin	language learning\class.mdj language learning \languagelearnUS_HL.mdj	Contact Admin.mdj	game\resources\views \register\ Contact Admin.php	TC_CA_1 – TC_CA_2 – TC_CA_3
RQ_17	Rate	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Rate.mdj	game\resources\views \register\ Rate.php	TC_RT_1 – TC_RT_2 – TC_RT_3
RQ_18	View ratings	language learning\class.mdj language learning\languagelearnUS_HL.mdj	viewratings.mdj	game\resources\views \ viewratings.php	TC_VR
RQ_19	Like/Dislike rating	language learning\class.mdj language learning\languagelearnUS_HL.mdj	rating.mdj	game\resources\views \ rating.php	TC-LI
RQ_20	Filter ratings	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Filterrating.mdj	game\resources\views \ Filterrating.php	TC_FR_1 – TC_FR_2 – TC_FR_3
RQ_21	Redetermine level	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Redeterminelevel.mdj	game\resources\views \ Redeterminelevel.php	TC_RL_1 – TC_RL_2 – TC_RL_3
RQ_22	Determine Arabic level	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Redeterminelevel.mdj	game\resources\views \english \ Redeterminelevel.php	TC-DL1 TC-DL2
RQ_23	Arabic Game 1	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Arabic game1.mdj	game\resources\ views\en glish\ Arabic game1.php	TC_AG1_1 – TC_AG1_2 – TC_AG1_3
RQ_24	Arabic Game 2	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Arabic game2.mdj	game\resources\ views\ english\l Arabic game1.php	TC_AG1_1 – TC_AG1_2 – TC_AG1_3
RQ_25	Arabic Game 3	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Arabic game3.mdj	game\resources\views \arabich	TC_AG1_1 – TC_AG1_2 – TC_AG1_3
RQ_26	Show levels	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Show levels.mdj	game\resources\views \arabich	TC-SH1 TC-SH2
RQ_27	Take courses	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Take courses.mdj	game\resources \views\Show level	TC_TC_1 – TC_TC_2 – TC_TC_3
RQ_28	Admin login	language learning\class.mdj language learning\languagelearnUS_HL.mdj	language learning\login.mdj	game\resources\views \register\ signup.blade.php	TC_2.1, TC_2.2, TC_2.3, TC_2.4
RQ_29	Show dashboard	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Show dashboard.mdj	game\resources\view s\welcome.blade.php	TC_3.1, TC_3.2, TC_3.3
RQ_30	Reply to message	language learning\class.mdj language learning\languagelearnUS_HL.mdj	Replay to message.mdj	game\resources\view s\welcome.blade.php	TC_3.1, TC_3.2, TC_3.3

Table 31

Chapter (5):

System Architectural Design

5.1 System Architecture Design:

How MVC Architecture Was Used in Our Project (Client–Server):

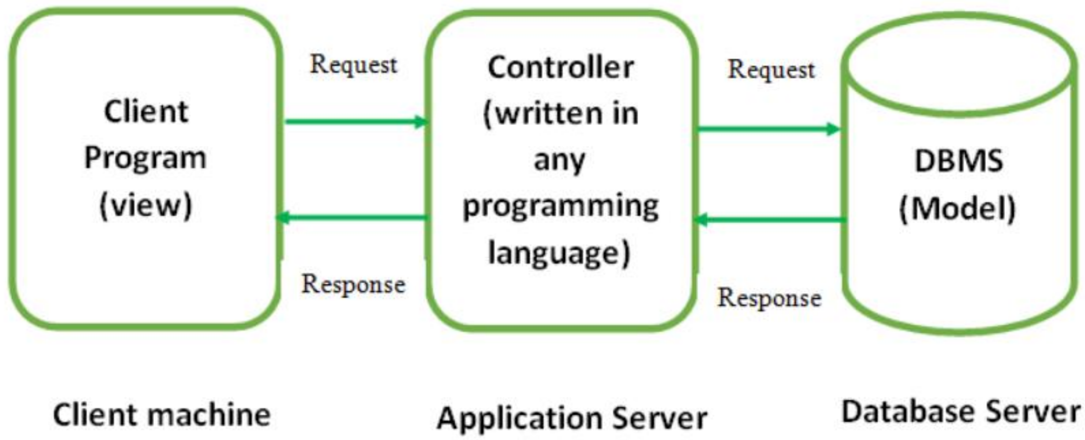


Figure 51

The Model was used on the server

By placing all core logic — scoring, level progression, saving progress, and game result processing — inside the Model to ensure secure and accurate data handling.

The View was used on the client side

By designing the user interface to display lessons, games, challenges, and progress, focusing purely on presentation without internal logic.

The Controller was used as the connector

By receiving user actions from the client, sending them to the server's Model for processing, and returning the results back to the View for display.

5.2 Entity-Relationship Diagram (ERD):

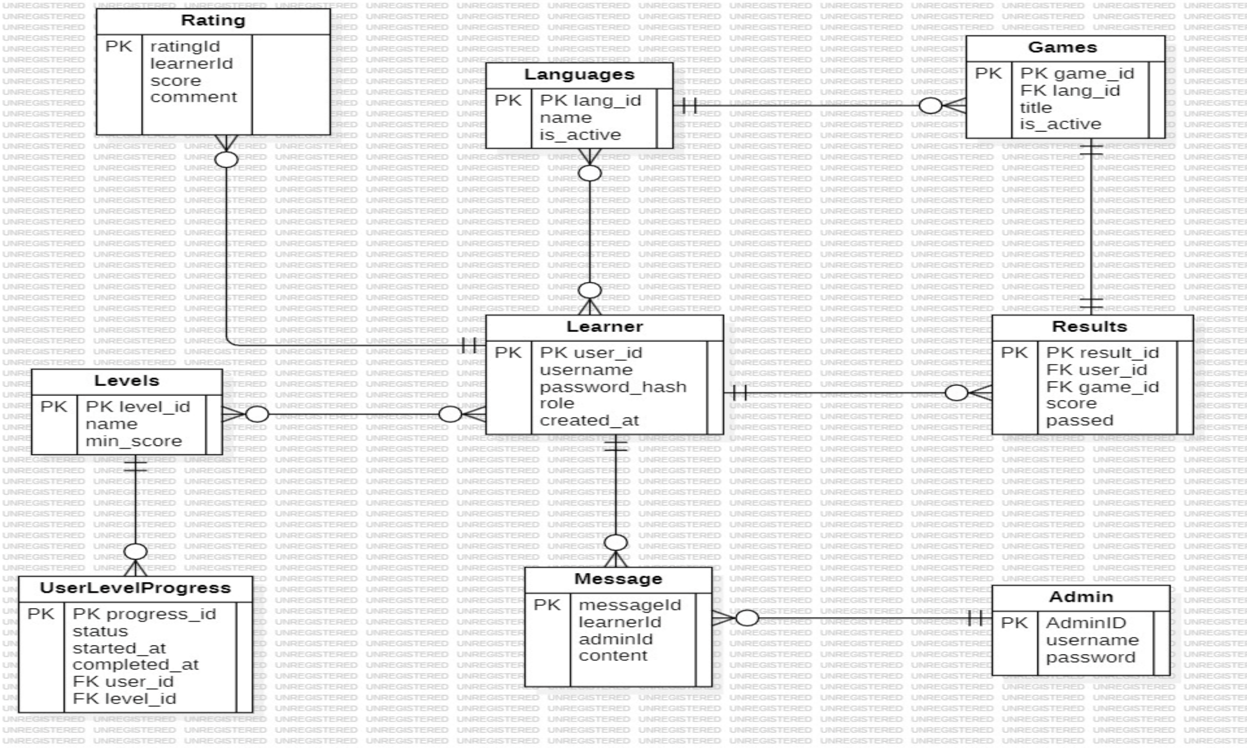


Figure 52

5.3 Class Diagram:

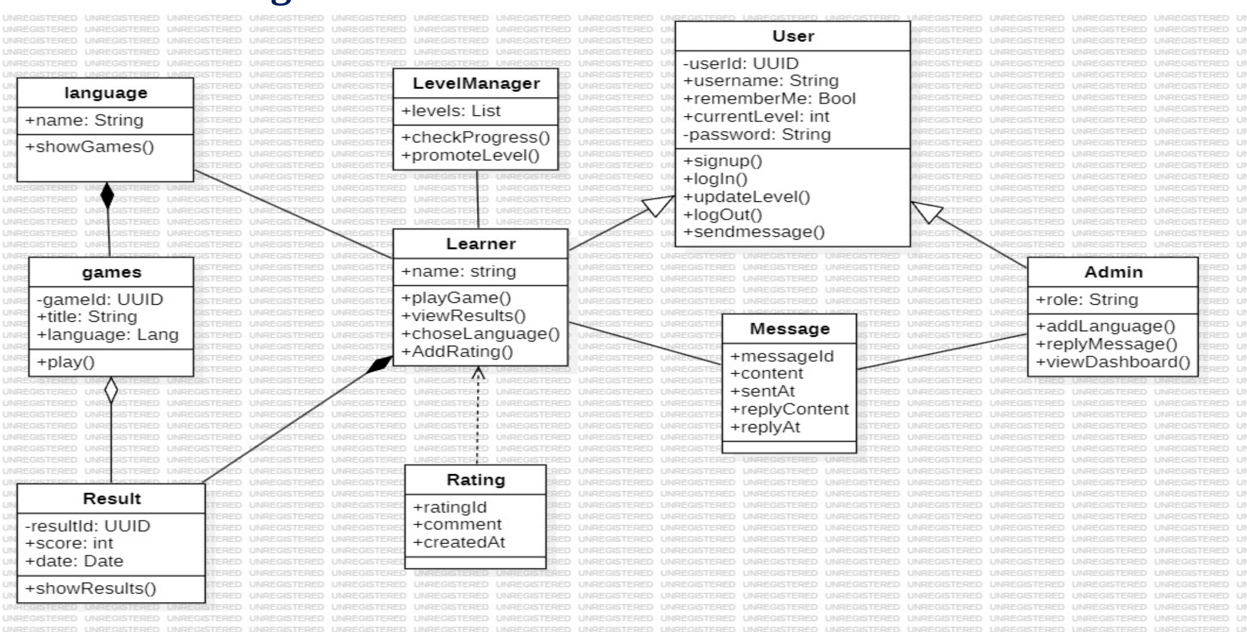


Figure 53

5.4 Security and Access Control Design:

The system's security and access control design is based on the authentication and authorization mechanisms provided by the **Laravel MVC framework**, ensuring safe user access, protected data, and controlled permissions across all system components.

1. Authentication Mechanism (Session-Based Authentication)

The system uses Laravel's built-in **session-based authentication**. When a user logs in, the framework verifies the email and password using Laravel's hashing system, then creates a secure session that stores the user's identity. All subsequent requests are validated through this session, and access to protected pages is enforced using Laravel **middleware**, ensuring that only authenticated users can access restricted resources.

2. User Roles (Admin, User, Guest)

The system follows a **Role-Based Access Control (RBAC)** model with three main roles:

- ✓ **Admin** – Full access to management features such as course management, reports, and message handling.
- ✓ **User (Learner)** – Access to learning content, games, courses, tests, ratings, and personal account features.
- ✓ **Guest** – Limited access to general information before registration.

These roles define what each user type can view or perform within the system.

3. Access Levels and Authorization

Authorization is enforced using Laravel **middleware** that checks the user's role before granting access to any route or feature.
Examples include:

- ✓ Admin-only routes protected by admin middleware.
- ✓ Learner-only features protected by auth middleware.
- ✓ Guests restricted from accessing any personalized or protected content.

This ensures that each user interacts only with the functionalities permitted by their assigned role, providing a secure and well-structured access control model.

Chapter (6):

Execution Environment and Tools

6.1 Software Tools:

Visual Studio:

An integrated development environment (IDE) that supports writing, managing, and debugging code efficiently. It simplifies working on both the frontend and backend of the project and helps test APIs to ensure their functionality.

Figure 54

StarUML:

A modeling tool used to design systems through various UML diagrams. It helps developers and analysts document ideas and convert them into clear visual diagrams, making it easier to understand the system's structure and relationships.

6.2 Development Environment and Project Structure:

The system was developed using a client–server architecture, where Laravel handles backend processing and the browser displays the user interface. The project runs on a modern operating system such as Windows, providing a stable environment for development and testing.

Laravel's MVC structure organizes the project into clear components:

- ✓ **app/** – Contains the application logic, including Models and Controllers.
- ✓ **resources/views/** – Holds the user interface built with HTML, CSS, and JavaScript.
- ✓ **routes/** – Defines the system's routes and API endpoints.
- ✓ **public/** – Stores frontend assets such as images, CSS, and JavaScript files.
- ✓ **database/** – Includes migrations and the SQLite database file.

This structure ensures clean organization, easy maintenance, and efficient development.

6.3 Programming Languages and Frameworks:

Laravel:

A modern, open-source PHP framework that provides a structured architecture and advanced features such as database management, routing, and an authentication system. It was adopted to develop the backend of the system to ensure security and fast processing.

Figure 55

HTML, CSS, and JavaScript:

HTML forms the foundation for building the structure of web pages, defining elements and organizing content to appear properly in the browser. CSS is responsible for styling the interfaces, adding colors, fonts, and layout design. JavaScript is used to enable immediate interaction, enrich the user experience with dynamic elements, and make pages more flexible and responsive.

Figure 56

SQLite:

SQLite is an ideal choice when you need a lightweight database system that integrates easily with Laravel. It operates through a single file, making it suitable for educational applications that store student data, lessons, and exams without requiring a standalone database server. It can be used locally during development or even on the server for small to medium-sized projects. Laravel fully supports SQLite without complex configuration.

6.4 Version Control and Source Management Tools:

GitHub:

GitHub was used as the main platform for version control and source code management. It allows uploading the project, tracking changes, creating branches for new features, and merging them into the main codebase. GitHub also provides backup, collaboration support, and a secure environment for managing the project's code.

Jira:

Jira was used as a task-management and workflow-tracking tool. The project was divided into smaller tasks with clear priorities, and progress was monitored throughout development. Jira helped organize the work, improve planning, and ensure that tasks were completed on schedule.

6.5 Testing Environment:

The initial testing phase of the system was conducted within a local development environment (localhost) to verify the core functionalities before deploying the application to any external server. This environment relies on running the system directly on the developer's machine using Laravel's built-in local server, which allows testing without the need for external hosting or complex configuration.

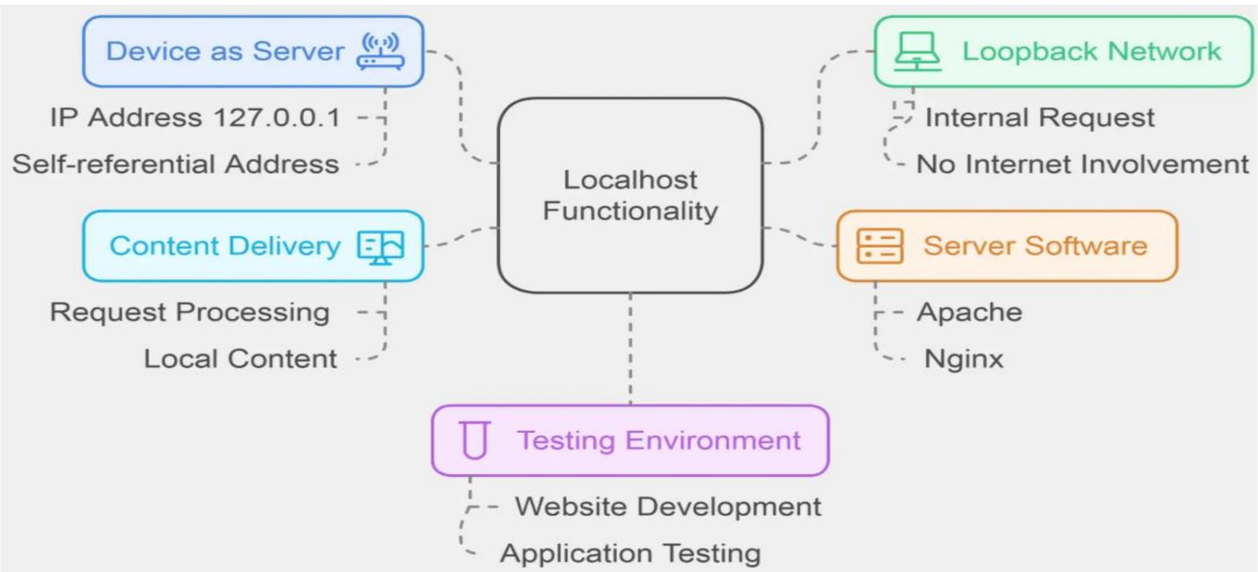


Figure 57

Using this setup, the development team performed comprehensive manual testing to evaluate user interfaces, workflow behavior, system responses, error handling, and database interactions. The local environment made it possible to identify early-stage issues such as logical errors, UI inconsistencies, and browser compatibility problems. During this phase, testers examined key system operations including user registration and login, page navigation, launching educational games, tracking user progress, and simulating payment processes. These tests played a crucial role in ensuring system stability and improving performance before moving on to more advanced testing environments such as Docker containers or a dedicated staging server.

Chapter (7):

Software Implementation

7.1 Project and File Structure

```
app -> Console | Exceptions | Http | Models | Providers
bootstrap -> cache
config -> (empty)
database -> factories | migrations | seeders
public -> img
resources -> css | js | views
routes -> (empty)
storage -> app | framework | logs
tests -> Feature | Unit
vendor -> bin | brianium | brick | carbonphp | composer | dflydev | doctrine | dragonmantank | egulias | fakerphp | fidry | filp | fruitcake | graham-campbell
 guzzlehttp | hamcrest | jean85 | kkszymanowski | laravel | league | mockery | monolog | myclabs | nesbot | nette | nikic | nunomaduro | pestphp | phar-io | ph
documentor | phpoption | phpstan | phpunit | psr | psy | ralouphie | ramsey | santigarcor | sebastian | spatie | symfony | ta-tikoma | theseer | tijsverkoyen |
vlucas | voku | webmozart
```

```
C:.
├───Console
│       Kernel.php
│
├───Exceptions
│       Handler.php
│
├───Http
│   │   Kernel.php
│   │
│   ├───Controllers
│   │       Controller.php
│   │       ElevelController.php
│   │       EnglishController.php
│   │       TlevelController.php
│   │       TurkeyController.php
│   │       UserController.php
│   │
│   └───Middleware
│           Authenticate.php
│           EncryptCookies.php
│           PreventRequestsDuringMaintenance.php
│           RedirectIfAuthenticated.php
│           TrimStrings.php
│           TrustHosts.php
│           TrustProxies.php
│           ValidateSignature.php
│           VerifyCsrfToken.php
```

```
│           VerifyCsrfToken.php
│
├───Models
│       elevel.php
│       english.php
│       Permission.php
│       Role.php
│       tlevel.php
│       turkey.php
│       User.php
│
└───Providers
        AppServiceProvider.php
        AuthServiceProvider.php
        BroadcastServiceProvider.php
        EventServiceProvider.php
        RouteServiceProvider.php
```

```
C:.
├───css
│       app.css
│
├───js
│       app.js
│       bootstrap.js
│
└───views
    │   welcome.blade.php
    │
    ├───Components
    │       nav.blade.php
    │
    ├───english
    │       eTest.blade.php
    │       irregular.blade.php
    │       letters.blade.php
    │       main.blade.php
    │       past_simple.blade.php
    │       prepare.blade.php
    │       present_simple.php
    │
    ├───register
    │       Login.blade.php
    │       signup.blade.php
    │
    └───turkey
            letters.blade.php
            main.blade.php
            prepare.blade.php
            TTest.blade.php
```

7.2 Modules Implementation

The system is structured into a set of modular components, each designed to perform a specific function within the language-learning platform. The implementation of these modules follows the MVC architectural pattern provided by the Laravel framework, ensuring separation of concerns, maintainability, and scalability. Each module was developed independently, then integrated to form a cohesive and fully functional system. Below is a detailed explanation of how each major module was implemented:

1. User Management Module

This module handles all operations related to user registration, authentication, and profile management.

1. The **Controller** manages user requests such as login, logout, and account creation.
2. The **Model** stores user information, roles, and permissions in the database.
3. The **Views** provide user-friendly interfaces for signing up, logging in, and updating personal information.
   Validation rules were applied to ensure data accuracy, and password hashing was implemented to enhance security.

2. Educational Content Module

This module manages the creation, organization, and delivery of learning materials such as vocabulary lists, grammar lessons, and interactive exercises.

1. Administrators can upload and categorize content through a dedicated dashboard.
2. The system retrieves content dynamically based on the learner's level and progress.
3. Content is stored in structured database tables to support scalability and continuous updates.

3. Game-Based Learning Module

This module integrates interactive educational games designed to reinforce language skills.

1. Each game is implemented using JavaScript for real-time interaction and responsiveness.
2. Laravel controllers handle game logic such as loading levels, tracking scores, and saving results.
3. The module communicates with the progress-tracking system to update user achievements after each game session.

4. Progress Tracking Module

This module monitors user performance and displays progress in a clear and visual format.

1. The system records scores, completed lessons, and earned badges.
2. A dedicated controller processes user data and generates progress reports.
3. The dashboard interface presents charts and indicators that help learners understand their improvement over time.

5. Payment and Subscription Module

This module enables users to purchase premium content or subscribe to advanced learning paths.

1. Secure payment gateways were integrated using Laravel's API capabilities.
2. Transactions are validated and stored in the database for future reference.
3. The module ensures that premium features are unlocked only for authorized users.

6. Administration and Dashboard Module

This module provides administrators with tools to manage users, content, payments, and system settings.

1. It includes interfaces for monitoring platform activity and generating analytical reports.
2. Role-based access control ensures that only authorized personnel can modify sensitive data.
3. The dashboard is built using responsive UI components for ease of use.

7. System Configuration and Settings Module

This module manages global system settings such as language preferences, theme options, and notification controls.

1. Settings are stored in a centralized configuration table.
2. Controllers retrieve and apply these settings dynamically across the platform.
3. The module ensures consistency and flexibility in system behavior.

7.3 System Interaction Layer Implementation:

Since the system is built using the Laravel framework and relies on server-side rendering with Blade templates, HTML, and CSS, the platform does not use external or REST-based APIs. Instead, all interactions between the user interface and the backend are handled internally through Laravel's routing system, controllers, and models. This approach ensures smooth communication within the system without the need for JSON responses or API endpoints.

The interaction layer is implemented as follows:

1. Routing (web.php)

All user requests are directed through Laravel's routing file located at:

routes/web.php

Each route maps a URL to a specific controller method.

```php
routes > web.php
| be assigned to the "web" middleware group. Make something great!
|
*/

Route::get('/', function () {
    return view('welcome');
});
//register
Route::get('signup',[UserController::class,'show']);
Route::post('signup',[UserController::class,'Create']);
Route::get('Logout',[UserController::class,'Out']);
Route::get('Login',[UserController::class,'Show2']);
Route::post('Login',[UserController::class,'Enter']);
//english
Route::get('/english',[EnglishController::class,'show']);
Route::get('/eTest',[EnglishController::class,'test']);
Route::post('/eTest/savequizresult',[EnglishController::class,'store']);
Route::get('/main',[EnglishController::class,'main']);
Route::get('/main/Letters',[EnglishController::class,'letters']);
Route::get('/main/present simple',[EnglishController::class,'psimple']);
Route::get('/main/past simple',[EnglishController::class,'pastsimple']);
Route::get('/main/Past Participle ',[EnglishController::class,'pastverbs']);
Route::post('/main/past irregular verbs ',[EnglishController::class,'pastverbsPost']);
//turkey
Route::get('/Turkey',[TurkeyController::class,'show']);
Route::get('/TTest',[TurkeyController::class,'test']);
Route::post('/TTest/savequizresult',[TurkeyController::class,'store']);
Route::get('/Tmain',[TurkeyController::class,'main']);
Route::get('/Tmain/letters',[TurkeyController::class,'letters']);
```

This route sends the user to the games page and triggers the controller to load the required data.

2. Controllers

Controllers contain the logic that processes user requests, retrieves data from the database, and sends it to the view.

Location:

```
PS C:\Users\en\Desktop\language learning\game\game\app\http\controllers> tree /f
Folder PATH listing
Volume serial number is E0AC-8D5E
C:.
    Controller.php
    ElevelController.php
    EnglishController.php
    TlevelController.php
    TurkeyController.php
    UserController.php
```

Example:

```php
public function main(){
   $games = English::All();
    return view('english.main',['tests'=>$games]);
}

public function letters(){
    return view('english.letters');
}

public function psimple(){
    return view('english.present_simple');
```

This method fetches all games and passes them to the Blade view.

3. Blade Templates (Views)

The user interface is built using Blade templates, which combine HTML, CSS, and dynamic data.

Location:

resources/views/

Example:

```
<div class="grid grid-cols-1 md:grid-cols-3 sm:grid-cols-2 gap-10">
    @foreach($tests as $test)

                <div class="rounded overflow-hidden shadow-lg">···
                </div>

    @endforeach
</div>
</div>
```

This displays the list of games retrieved by the controller.

7.4 Data Access Layer Implementation:

1. Select All Query

Used to retrieve all records from a specific database table for display or processing.

2. Select by ID Query

Used to fetch a single record based on its unique identifier.

3. Conditional Query (WHERE)

Used to retrieve records that match specific conditions or filters.

4. Insert Operation

Used to add a new record to the database when creating new data entries.

5. Update Operation

Used to modify existing records in the database when changes are required.

6. Delete Operation

Used to remove a record from the database when it is no longer needed.

Example:

```
app > Http > Controllers > TurkeyController.php
{

    public function store(Request $request)
    {

        if ((int) request('beginnerScore') < 3) {
            Tlevel::create([
                'email' => Session::get('email'),
                'level' => 'beginner',
                'mark' => (int)request('totalScore')  // fixed typo: "marl" → "mark"
            ]);
             Session::put('TLevel', 'beginner');
        }elseif((int) request('intermediateScore')<3 && (int) request('beginnerScore')>3){
            Tlevel::create([
                'email' => Session::get('email'),
                'level' => 'intermediate',
                'mark' => (int)request('totalScore')  // fixed typo: "marl" → "mark"
            ]);
            Session::put('TLevel', 'intermediate');
        }elseif((int) request('expertScore')<3 && (int) request('intermediateScore')>3){
            Tlevel::create([
                'email' => Session::get('email'),
                'level' => 'expert',
                'mark' => (int)request('totalScore')  // fixed typo: "marl" → "mark"
            ]);
            Session::put('TLevel', 'expert');
```

The process of storing the user’s results and determining their language level relies on the use of the Eloquent ORM, where a new record is inserted into the *Tlevel* table once the test is completed. The system analyzes the user’s scores through a set of conditional checks to identify the appropriate level (Beginner – Intermediate – Expert), after which it creates a record containing the user’s email, the assigned level, and the final score. The insertion process is carried out using the Eloquent *create()* method, which simplifies database operations without the need to write raw SQL queries. Additionally, the determined level is stored in the user’s session to be used later for delivering personalized learning content.

7.5 Business Logic Layer Implementation:

This part of the system represents the algorithm responsible for determining the user’s initial level (E-Level) based on the results of the assessment they complete.
The system compares the scores associated with each level (Beginner – Intermediate – Expert) and then selects the level in which the user did not meet the minimum required threshold.

1. Beginner Level Determination
If the user's score in the Beginner section is below the required threshold, the system classifies the user as a Beginner and stores this level in the database.

2. Intermediate Level Determination
If the user successfully passes the Beginner level but obtains a low score in the Intermediate section, the system classifies them as an Intermediate user and records this level.

3. Expert Level Determination
If the user passes both the Beginner and Intermediate levels but scores below the required threshold in the Expert section, the system classifies them as an Expert and saves this level in the database.

4. Storing the Level in the Session
After determining the appropriate level, the system stores it in the user's session so it can be used later to personalize the learning content and guide the user throughout the platform.

Example:

```
app > Http > Controllers > EnglishController.php
{
    {
        if ((int) request('beginnerScore') < 3) {
            Elevel::create([
                'email' => Session::get('email'),
                'level' => 'beginner',
                'mark' => (int)request('totalScore')  // fixed typo: "marl" → "mark"
            ]);
            Session::put('eLevel', 'beginner');
        }elseif((int) request('intermediateScore')<3 && (int) request('beginnerScore')>3){
            Elevel::create([
                'email' => Session::get('email'),
                'level' => 'intermediate',
                'mark' => (int)request('totalScore')  // fixed typo: "marl" → "mark"
            ]);
            Session::put('eLevel', 'intermediate');
        }elseif((int) request('expertScore')<3 && (int) request('intermediateScore')>3){
            Elevel::create([
                'email' => Session::get('email'),
                'level' => 'expert',
                'mark' => (int)request('totalScore')  // fixed typo: "marl" → "mark"
            ]);
            Session::put('eLevel', 'expert');
        }else{
            Elevel::create([
                'email' => Session::get('email'),
                'level' => 'passed',
                'mark' => (int)request('totalScore')  // fixed typo: "marl" → "mark"
```

7.6 User Interface (UI) Implementation:

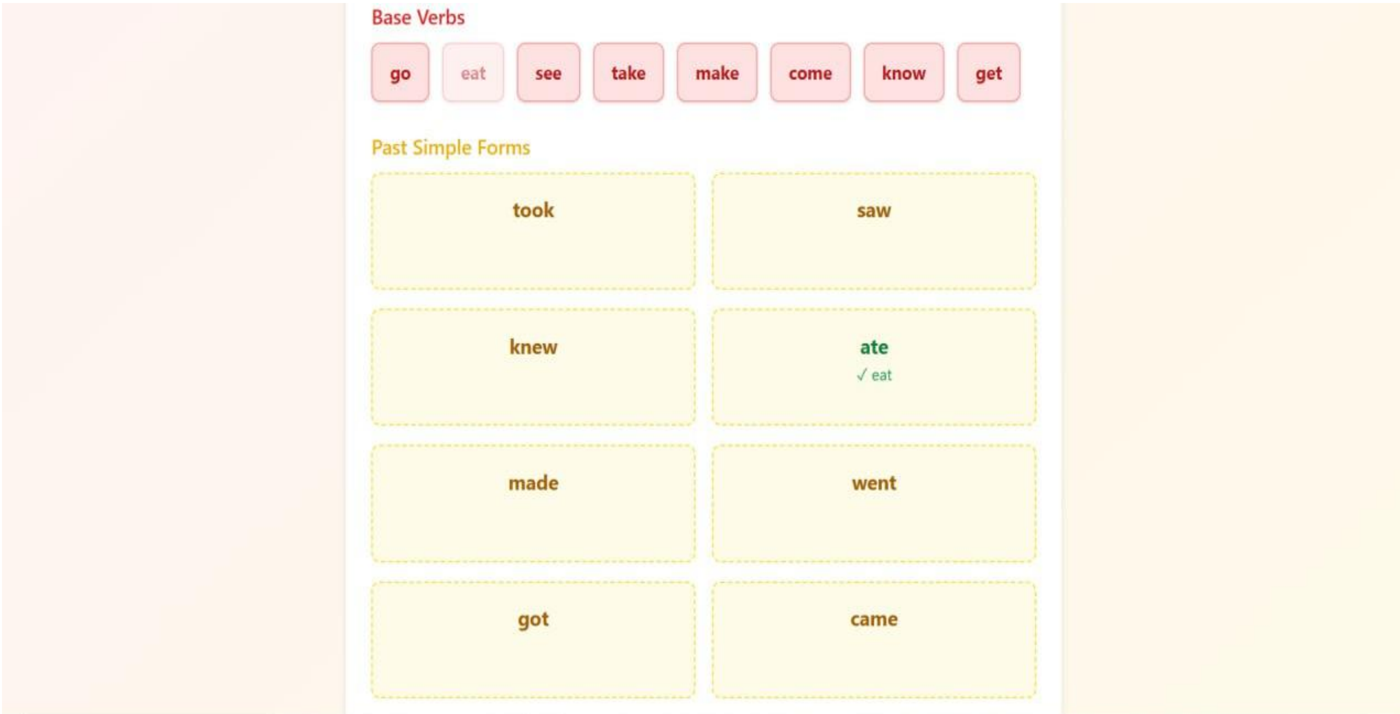


Figure 58

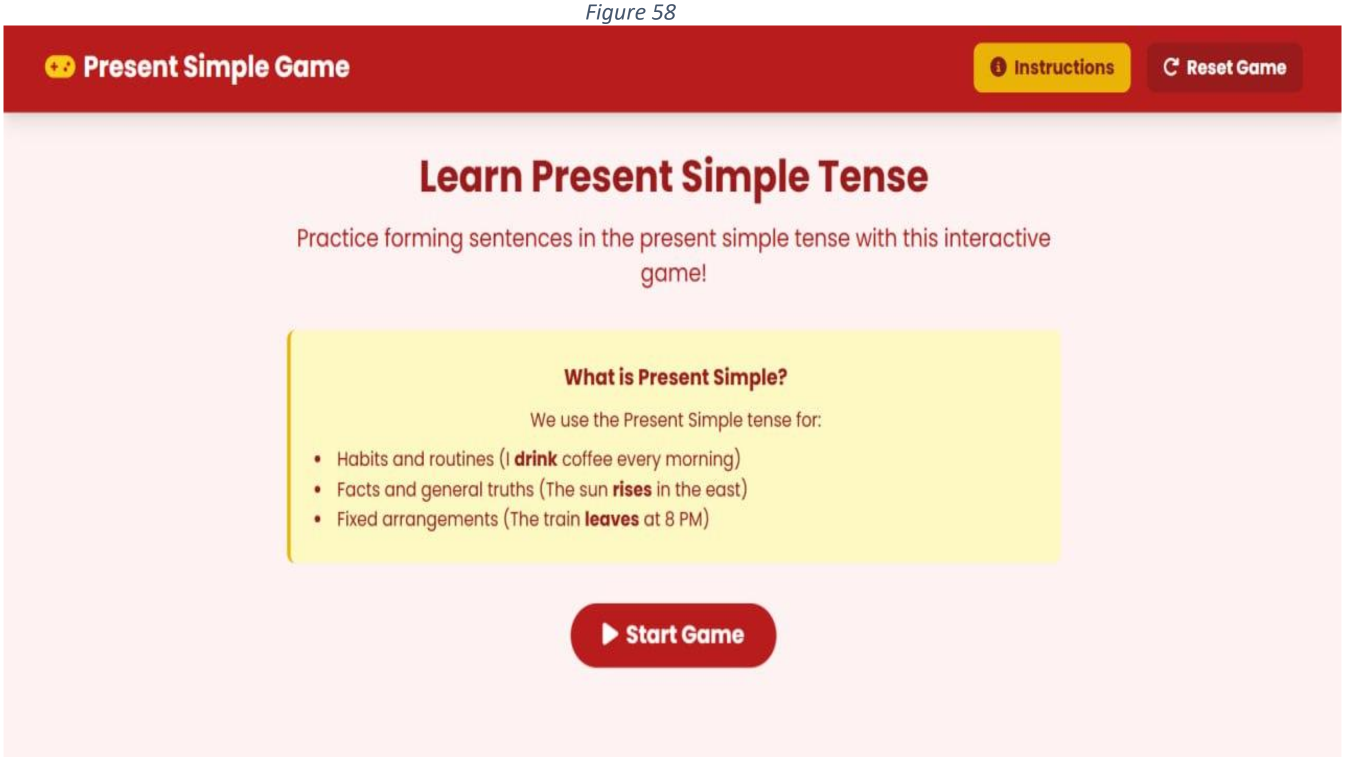
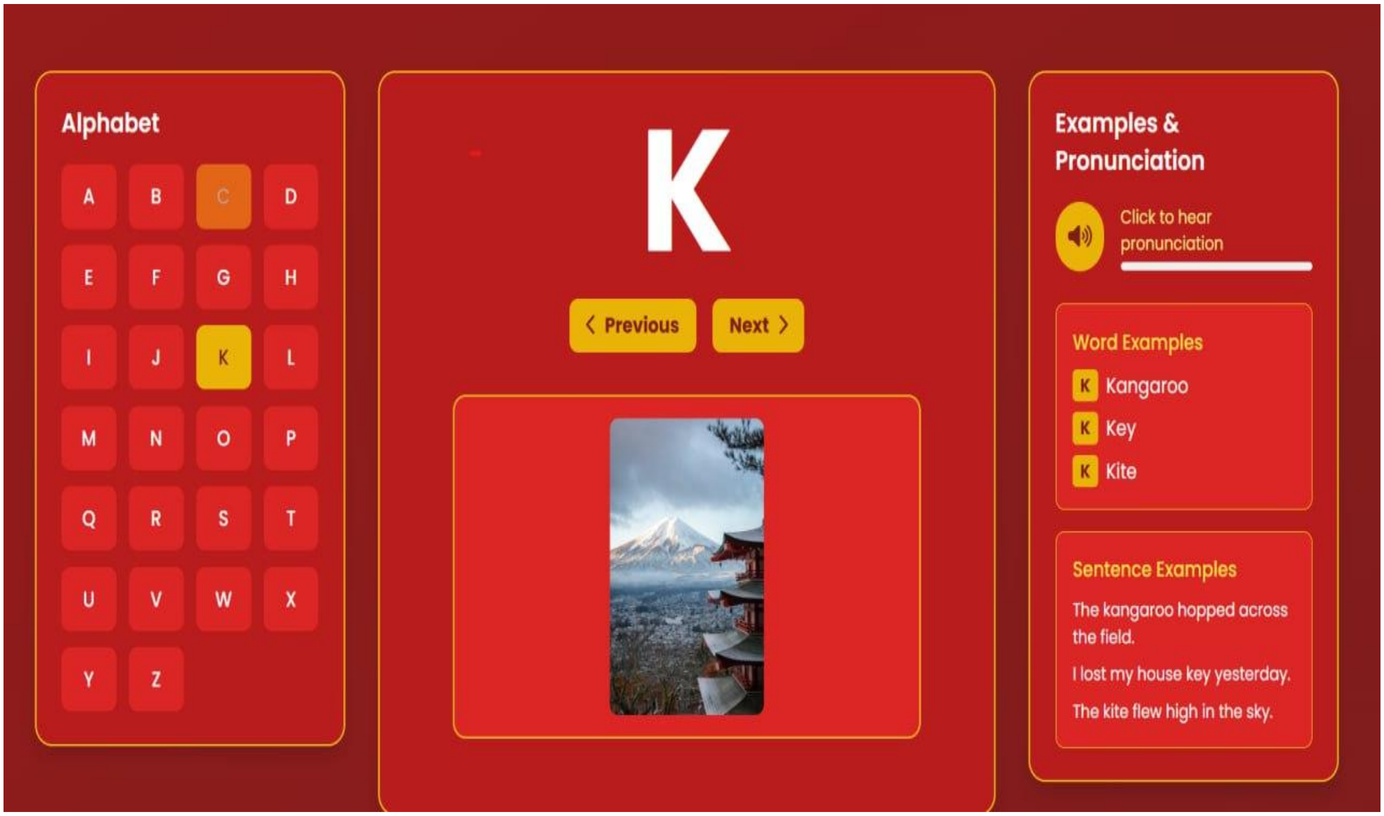


Figure 59



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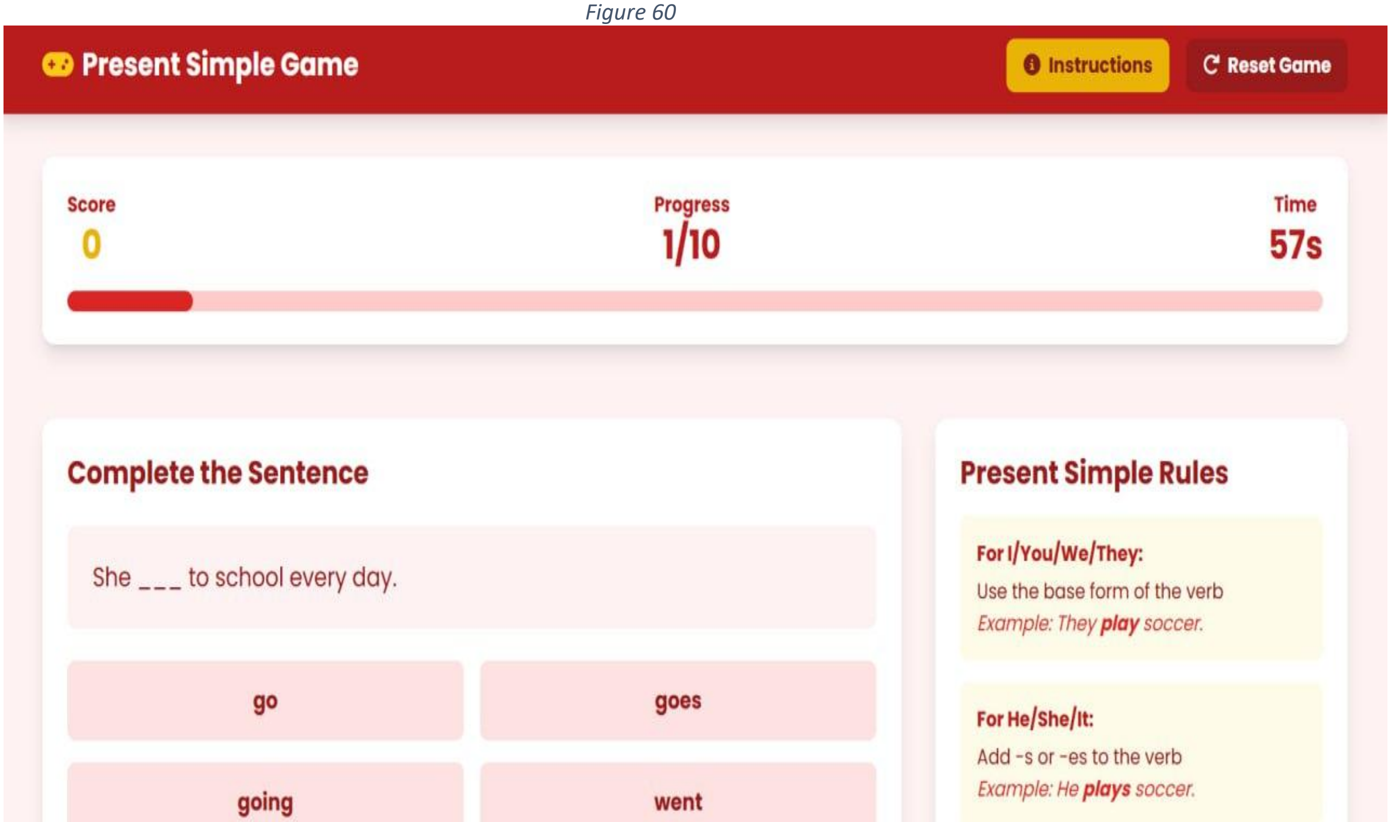

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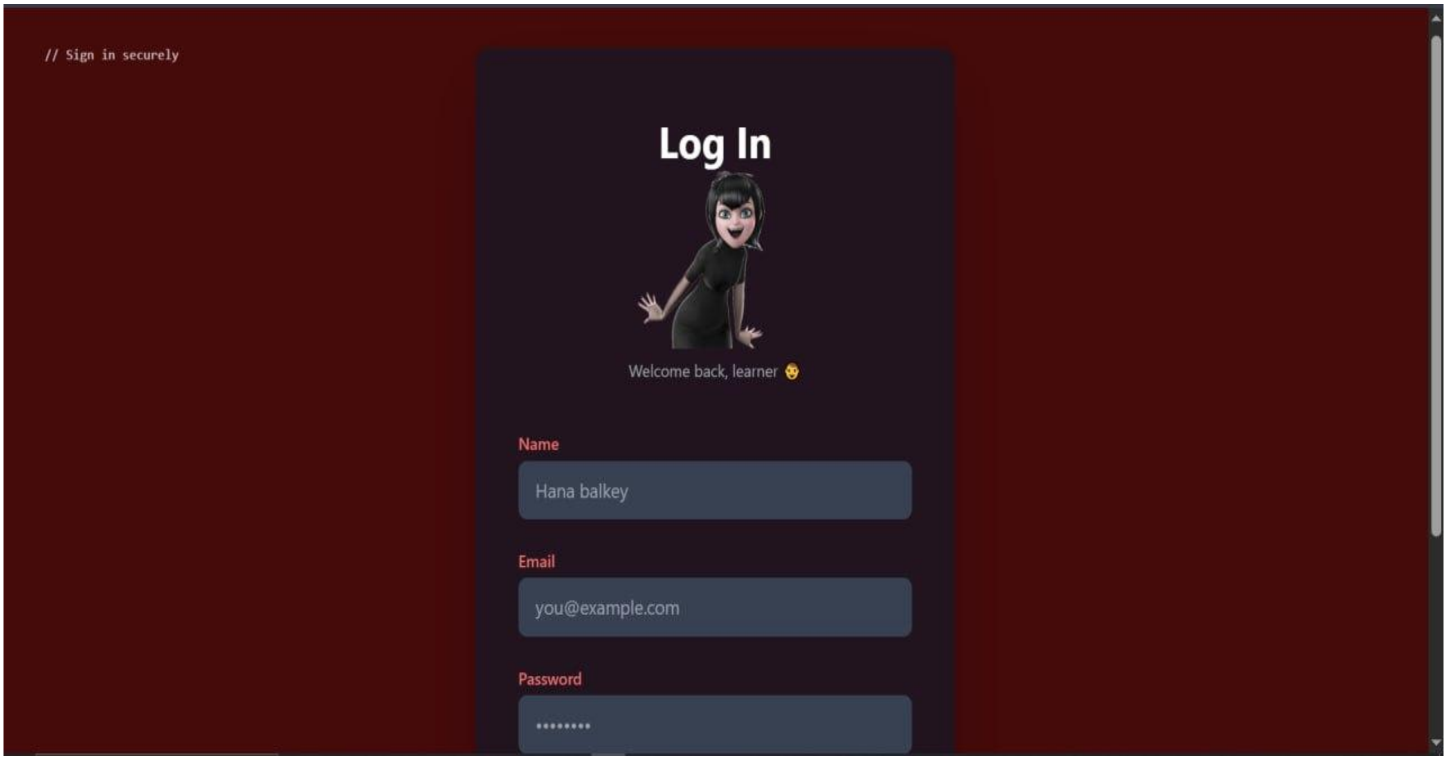


Figure 62

Figure 63

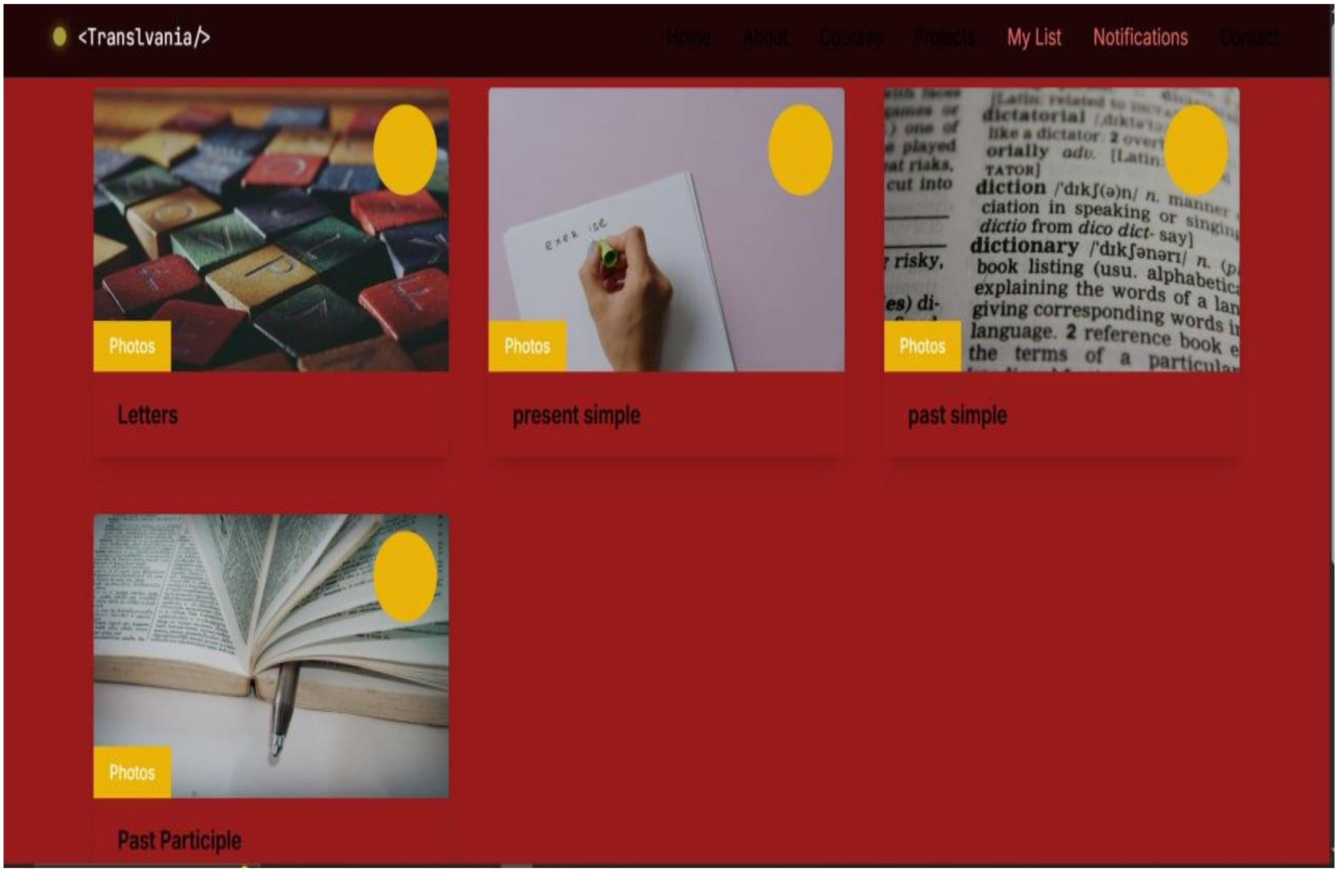
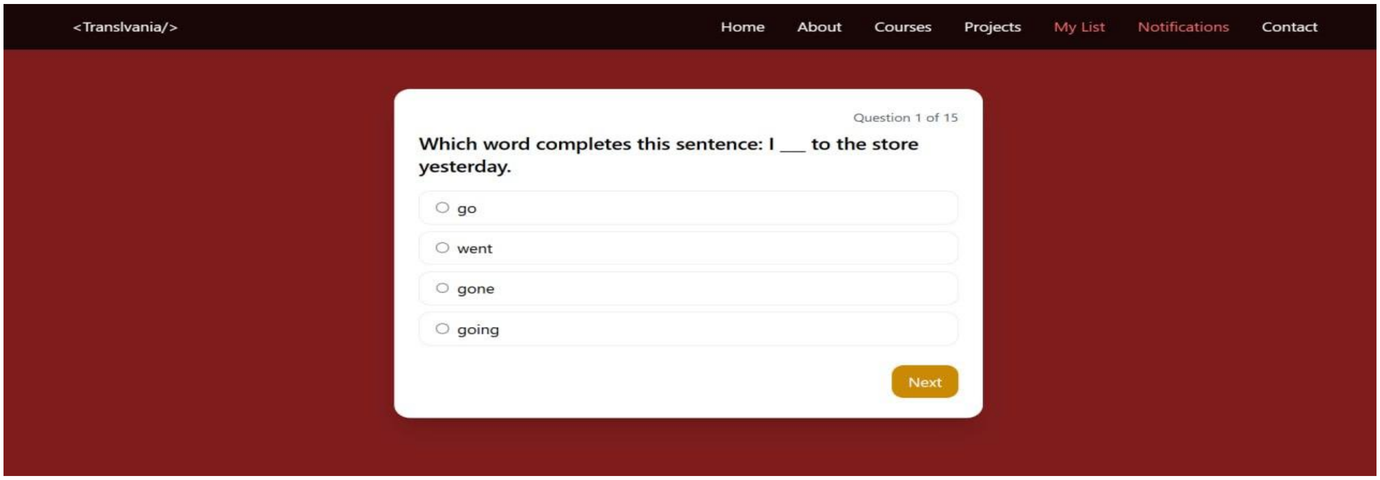


Figure 64

Figure 65



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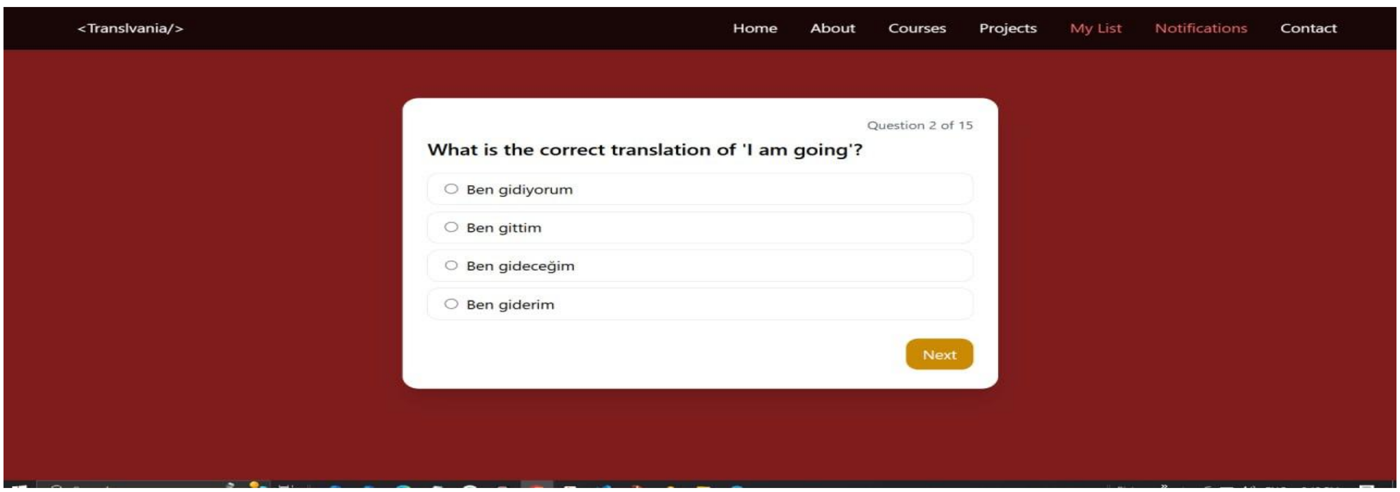
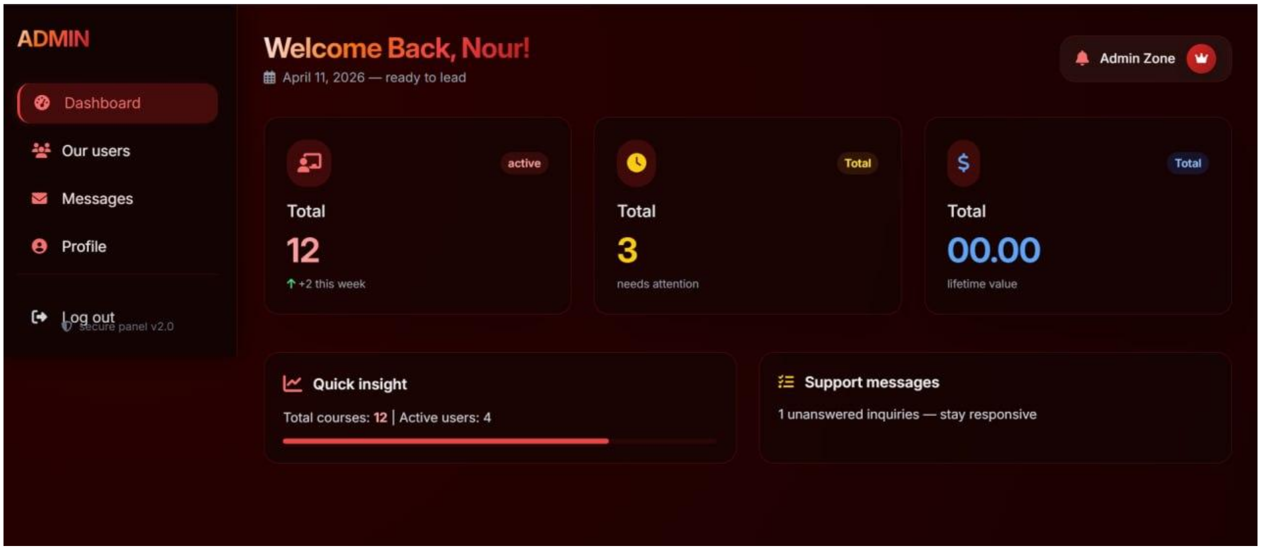


Figure 67

Figure 68



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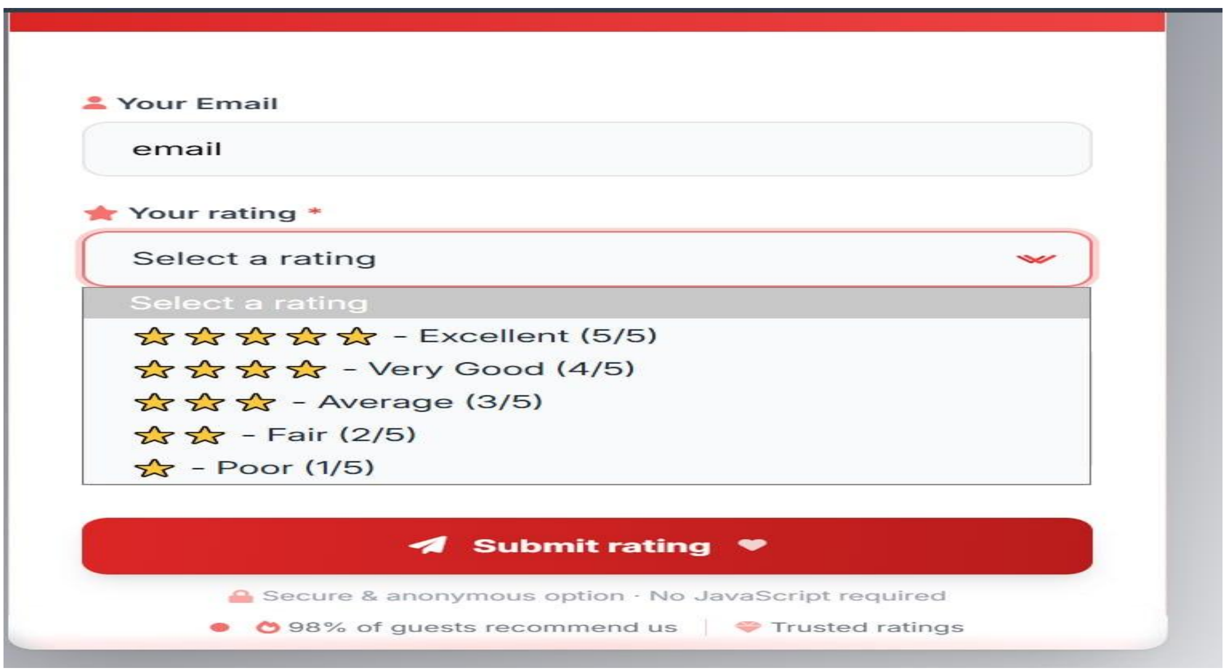


70 Figure

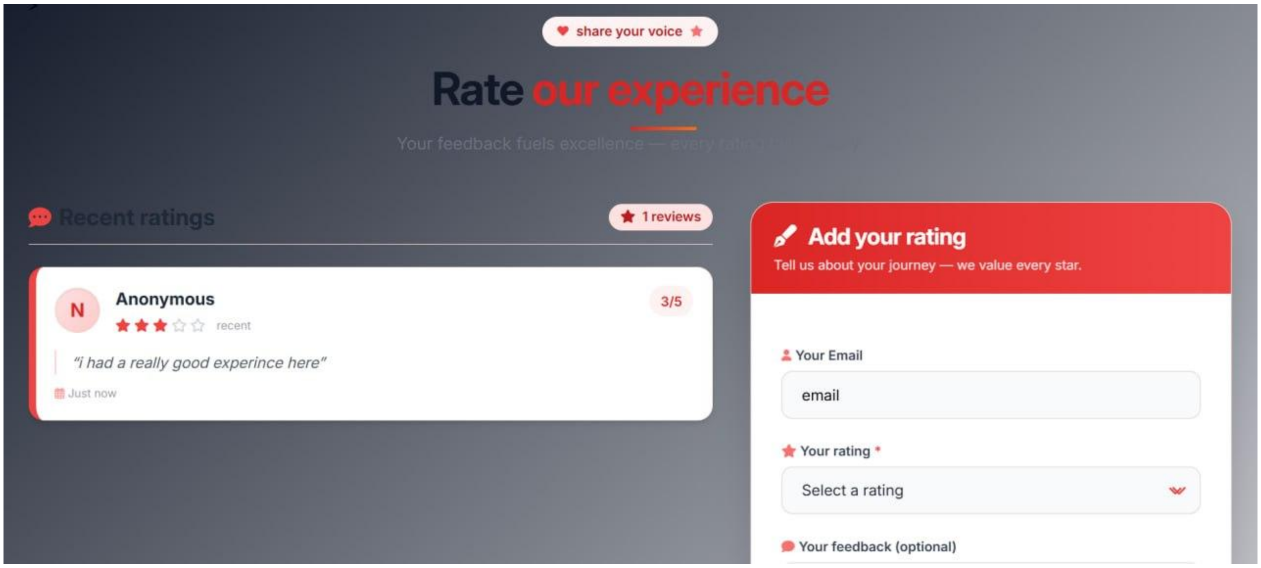


Figure 71

Figure 72

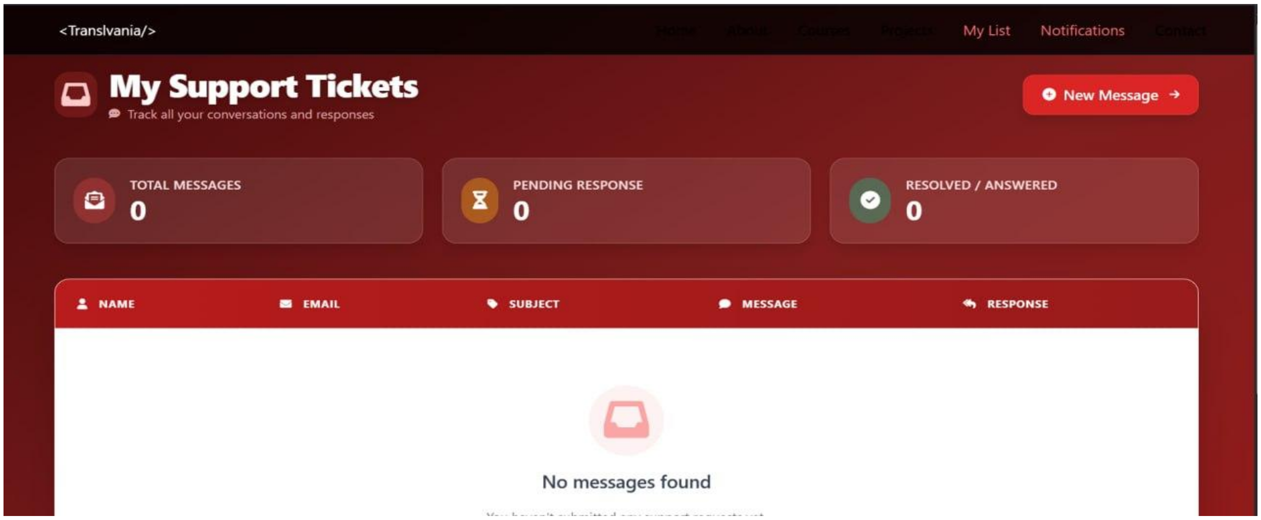
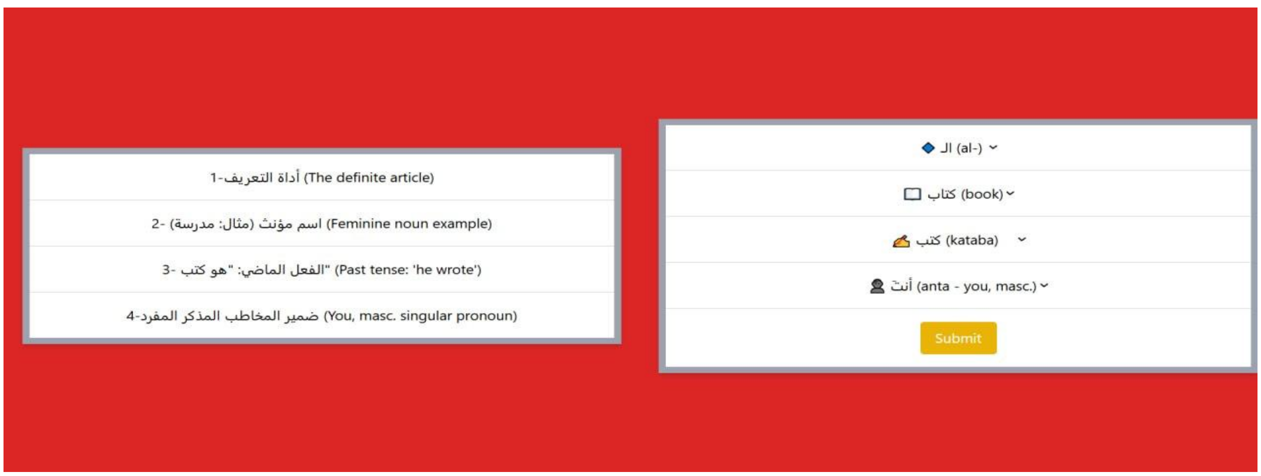


Figure 73



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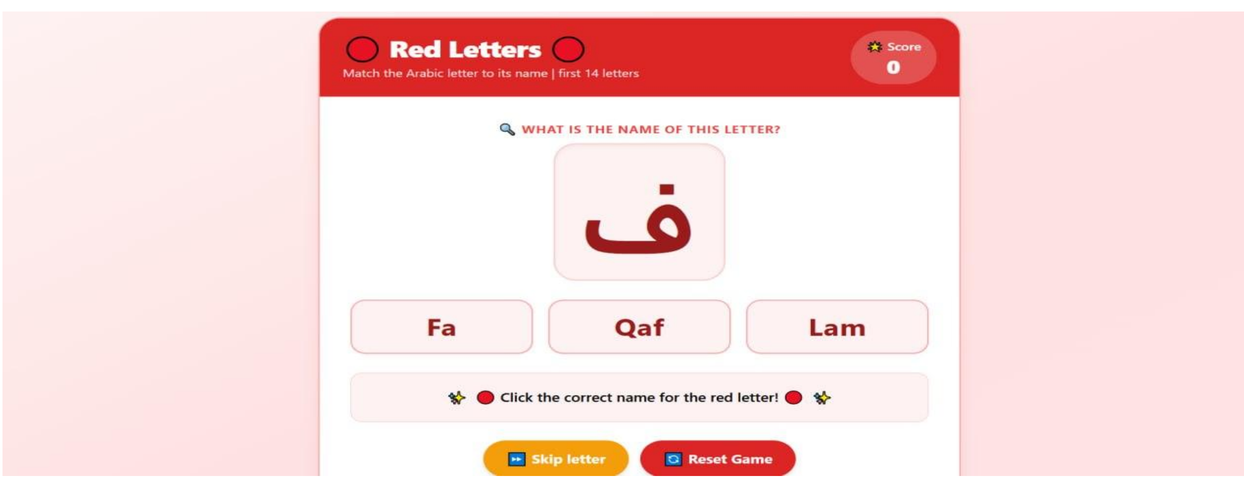


Figure 75

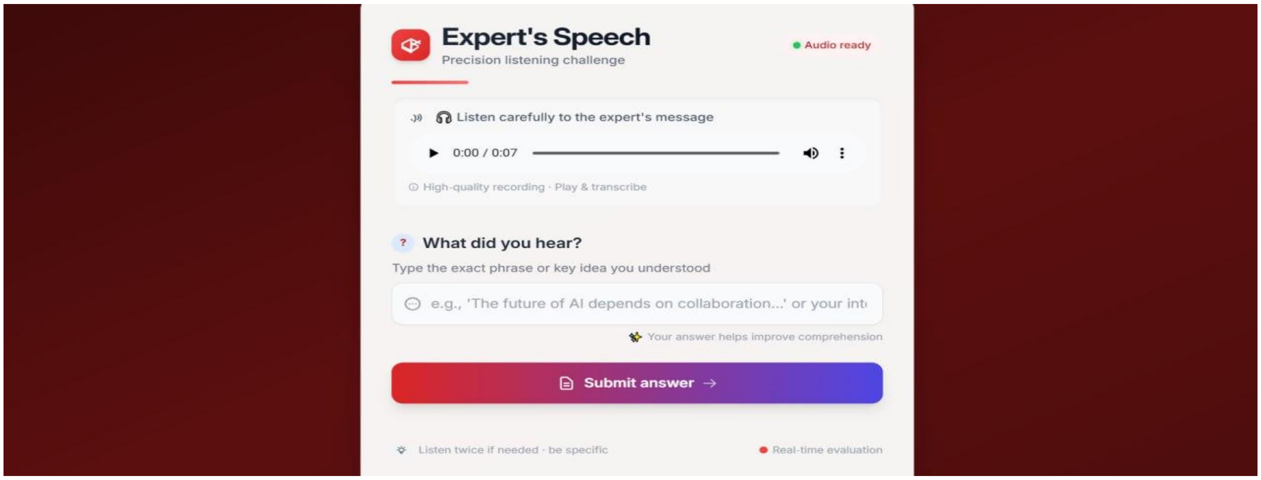


Figure 76

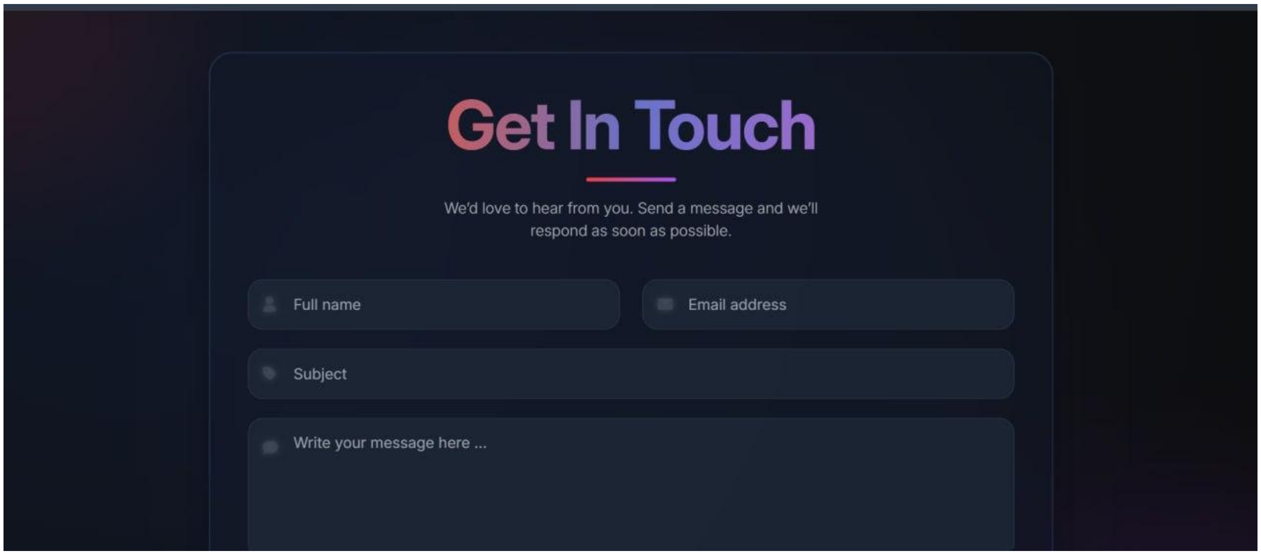
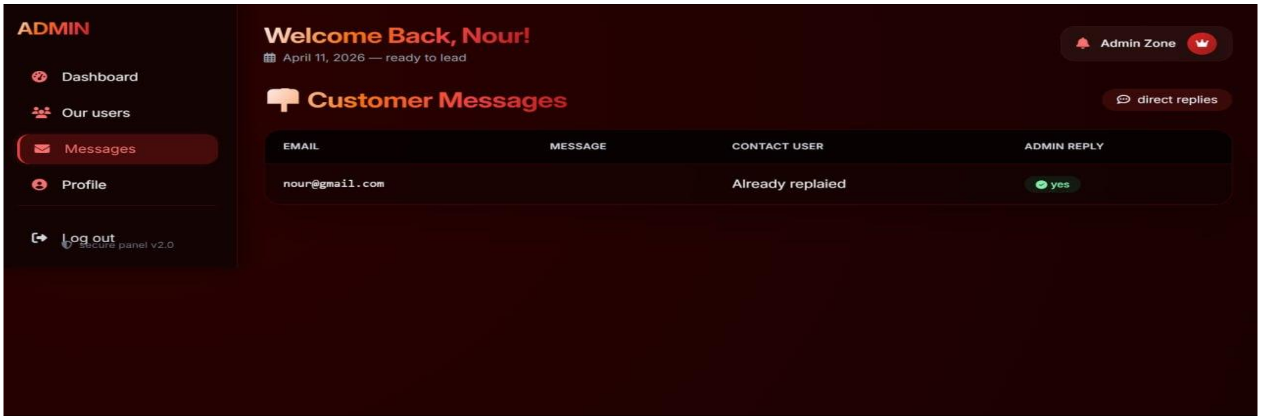


Figure 77



78 Figure

Chapter (8):

Testing and Evaluation

8.1 Test Plan:

8.1.1 What Will Be Tested?

Functional Requirements:

Authentication Module:
Verification of account creation, login, logout, session handling, and the "Remember Me" functionality.

Level Determination Module (E-Level and T-Level):
Validation of score processing, level assignment logic (Beginner, Intermediate, Expert), and correct storage of results.

Educational Content Delivery:
Ensuring that lessons, games, and activities are displayed according to the user's assigned level.

Progress Tracking:
Validation of storing test results, updating progress, and displaying user performance.

Roles and Permissions:
Ensuring that students, teachers, and administrators have access only to their authorized features.

Non-Functional Requirements:

Performance: Page loading time, database query speed, and responsiveness.

Security: Session protection, access control, and prevention of unauthorized access.

Usability: Ease of navigation, clarity of interface elements, and user experience.

Compatibility: Proper functioning across different browsers and screen sizes.

8.1.2 When Will Testing Be Conducted?

During Development:
Unit and manual tests are executed after completing each major module.

After Each Development Milestone:
Authentication, level determination, content delivery, and progress tracking.

At the End of Development:
Full system testing and acceptance testing to ensure complete functionality.

8.1.3 Tools Used for Testing

Manual Testing:
Conducted through web browsers (Chrome, Edge) on the local development environment.

Browser Developer Tools:
Used to inspect errors, monitor network requests, and measure performance.

8.2 Types of Tests:

8.2.1 Unit Testing

Focuses on testing individual components such as level-determination functions, session handling, and permission checks to ensure each part works independently.

8.2.2 Integration Testing

Ensures that modules interact correctly, such as the interaction between controllers, models, and the database during level assignment and content delivery.

8.2.3 System Testing

Validates the entire system from the user's perspective, including login, level testing, content access, and progress tracking.

8.2.4 Acceptance Testing

Conducted using real-world scenarios to confirm that the system meets user expectations and fulfills the intended purpose for students, teachers, and administrators.

8.2.5 Performance and Stress Testing

Measures response time, page loading speed, and system behavior under increased load to ensure stability and responsiveness.

8.2.6 Security Testing

Ensures that unauthorized users cannot access restricted pages, validates session management, and checks the robustness of authentication mechanisms.

8.2.7 Usability Testing

Evaluates the clarity, simplicity, and user-friendliness of the interface, ensuring that users can navigate the system without confusion.

8.2.8 Accessibility Testing

Ensures that text, colors, and interface elements are readable and accessible to a wide range of users.

8.2.9 Compatibility Testing

Verifies that the system works consistently across different browsers (Chrome, Edge) and screen sizes.

8.3 Test Results and Analysis:

8.6.2 Results

Users reported that:

- ✓ The login and level-testing processes were clear and easy to follow.
- ✓ The interface was simple and intuitive.
- ✓ Content was appropriately matched to their assigned level.
- ✓ Suggested improvements included:
- ✓ Adding more visual guidance for new users.
- ✓ Enhancing the design of certain pages for better aesthetics.

8.4 Test Case Diagrams:

- ***RQ_1 – Sign Up***

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_1.1	Sign up with valid data	RQ_1	Enter valid email, username, password → Click Sign Up	Account created successfully	As expected	Pass

TC_1.2	Sign up with existing email	RQ_1	Enter already registered email → Click Sign Up	Error: "Email already exists"	As expected	Pass
TC_1.3	Sign up with invalid input	RQ_1	Leave fields empty → Click Sign Up	Error: "Please complete all fields"	As expected	Pass

Table 32

➢ *RQ_2 – Log In*

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_2.1	Login with valid credentials	RQ_2	Enter correct email & password → Click Login	Access granted	As expected	Pass
TC_2.2	Login with invalid password	RQ_2	Enter wrong password → Click Login	Error: "Invalid credentials"	As expected	Pass
TC_2.3	Login with inactive account	RQ_2	Enter credentials of inactive account	Error: "Account inactive"	As expected	Pass
TC_2.4	System error during login	RQ_2	Try login when DB down	Error: "Login failed"	As expected	Pass

Table 33

➢ *RQ_3 – Notifications*

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_3.1	Notification for new game	RQ_3	System adds new game	Learner receives notification	As expected	Pass
TC_3.2	Notification for level update	RQ_3	Learner completes test	Learner receives level update notification	As expected	Pass
TC_3.3	Notifications disabled	RQ_3	Learner disables notifications	No notification shown	As expected	pass

Table 34

➢ RQ_4 – Logout

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_4.1	Logout successfully	RQ_4	Click Logout	Session terminated, redirected to login	As expected	Pass
TC_4.2	Logout with system error	RQ_4	Click Logout during DB error	Session persists until timeout	As expected	Pass
TC_4.3	Logout from multiple devices	RQ_4	Logout from one device	Session ends only on that device	As expected	pass

Table 35

➢ RQ_5 – Show Available Languages

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_5.1	View supported languages	RQ_5	Navigate to language settings	List of languages displayed	As expected	Pass
TC_5.2	No languages available	RQ_5	System has no languages	Default language shown	As expected	Pass
TC_5.3	Cancel language selection	RQ_5	Learner cancels	Current language remains	As expected	Pass

Table 36

➢ RQ_6 – Show Available Games

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_6.1	View games by language	RQ_6	Navigate to games section	Games categorized by language displayed	As expected	Pass
TC_6.2	No games available	RQ_6	Select language with no games	Message: "No games available"	As expected	Pass
TC_6.3	Browse games	RQ_6	Scroll through list	Learner views available games	As expected	Pass

Table 37

➢ RQ_7 – Determine Level

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_7.1	Complete placement test	RQ_7	Start test → Answer → Submit	System determines level	As expected	Pass
TC_7.2	Exit before completion	RQ_7	Exit test midway	No level determined	As expected	Pass
TC_7.3	System error during test	RQ_7	Submit test during DB error	Results not saved	As expected	Pass
TC_7.4	Invalid answers	RQ_7	Submit empty answers	Error: "Incomplete test"	As expected	Pass

Table 38

➢ RQ_8–RQ_13 – Play Games

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_8.1	Play English Game 1	RQ_8->13	Select game → Play	Game loads and plays	As expected	Pass
TC_8.2	Game fails to load	RQ_8->13	Select game during error	Error message shown	As expected	Pass
TC_8.3	Exit game	RQ_8->13	Exit before completion	Game session ends	As expected	Pass

Table 39

➢ RQ_14 – Show Current Results

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_14.1	Show results after test	RQ_14	Complete test	Results screen shows correct/incorrect answers	As expected	Pass
TC_14.2	No questions attempted	RQ_14	Submit empty test	Message: "No results available"	As expected	Pass

TC_14.3	System error	RQ_14	Submit test during DB error	Message: "Results cannot be displayed"	As expected	Pass
TC_14.4	Multiple games results	RQ_14	Play multiple games	Results show all scores	As expected	Pass

Table 40

➢ *RQ_15 – Update Level*

Test Case ID	Title	Req ID	Steps	Expected Result	Actual Result	Pass/Fail
TC_15.1	Update level successfully	RQ_15	Complete level → Take test → Submit	Level updated	As expected	Pass
TC_15.2	Exit before completion	RQ_15	Exit test midway	Level unchanged	As expected	Pass
TC_15.3	System error	RQ_15	Submit test during DB error	Results not saved	As expected	Pass
TC_15.4	Performance below threshold	RQ_15	Submit test with low score	Level remains unchanged	As expected	Pass

Table 41

➢ Contact Admin

Test Case ID	Title	Preconditions	Steps	Expected Result	Actual Result	Status
TC_CA_1	Send message successfully	Learner logged in → On Contact Admin page	1. Open Contact Admin page → 2. Write message → 3. Click Send	Message sent to admin inbox → Confirmation displayed	As expected	Pass
TC_CA_2	Empty message validation	Learner logged in → On Contact Admin page	1. Open page → 2. Leave message box empty → 3. Click Send	System displays: “Message cannot be empty.”	As expected	Pass
TC_CA_3	System error while sending message	Learner logged in → On Contact Admin page	1. Write message → 2. Click Send during system/DB error	System displays: “Message could not be sent. Try again later.”	As expected	Pass

Table 42

➢ Rate

Test Case ID	Title	Preconditions	Steps	Expected Result	Actual Result	Status
TC_RT_1	Submit rating successfully	Learner logged in → Accessed a rate-enabled game/course/feature	1. Navigate to rating section → 2. Select rating → 3. (Optional) write comment → 4. Submit	Rating saved → Confirmation displayed	As expected	Pass
TC_RT_2	Empty rating validation	Learner logged in → On rating section	1. Navigate to rating section → 2. Do not select rating → 3. Click Submit	System displays: "Please select a rating."	As expected	Pass
TC_RT_3	System error while submitting rating	Learner logged in → On rating section	1. Select rating → 2. Write comment (optional) → 3. Submit during system/DB error	System displays: "Rating could not be submitted. Please try again later."	As expected	Pass

Table 43

➢ Filter Ratings

Test Case ID	Title	Preconditions	Steps	Expected Result	Actual Result	Status
TC_FR_1	Filter ratings successfully	Learner logged in → On ratings page	1. Open ratings section → 2. Select filter (likes/date)	System retrieves and displays filtered ratings	As expected	Pass
TC_FR_2	No ratings available	Learner logged in → On ratings page	1. Open ratings section → 2. Select any filter	System displays: "No ratings found."	As expected	Pass
TC_FR_3	System error while filtering	Learner logged in → On ratings page	1. Open ratings section → 2. Select filter during system/DB error	System displays: "Unable to filter ratings at the moment."	As expected	Pass

Table 44

➢ Redetermine Level

Test Case ID	Title	Preconditions	Steps	Expected Result	Actual Result	Status
TC_RL_1	Retake level test successfully	Learner logged in → Navigated to Level Test section	1. Select "Retake Level Test" → 2. System loads test → 3. Complete test → 4. Submit	System evaluates answers → Updates level → Displays updated level	As expected	Pass
TC_RL_2	Test not completed	Learner logged in → On Level Test section	1. Select "Retake Level Test" → 2. Start test → 3. Exit or skip questions → 4. Submit	System displays: "Please complete the test to update your level."	As expected	Pass
TC_RL_3	System error during level update	Learner logged in → On Level Test section	1. Select "Retake Level Test" → 2. Complete test → 3. Submit during system/DB error	System displays: "Unable to update level at the moment."	As expected	Pass

Table 45

➢ Play Arabic Game 1

TEST CASE ID	TITLE	PRECONDITIONS	STEPS	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC_AG1_1	Load and play Arabic Game 1	Learner logged in → Arabic selected as target language	1. Navigate to Arabic games → 2. Select "Arabic Game 1" → 3. Game loads → 4. Play game	Game loads successfully → System records progress → Final score shown	As expected	Pass
TC_AG1_2	Game not loaded	Learner logged in → Arabic selected	1. Navigate to Arabic games → 2. Select "Arabic Game 1" during load failure	System displays: "Unable to load the game."	As expected	Pass
TC_AG1_3	No internet connection	Learner logged in → Arabic selected	1. Navigate to Arabic games → 2.	System displays: "Connection	As expected	Pass

			Select “Arabic Game 1” without internet	required to start the game.”		

Table 46

➢ *Take Courses*

Test Case ID	Title	Preconditions	Steps	Expected Result	Actual Result	Status
TC_TC_1	Enroll in course successfully	Learner logged in → Navigated to Courses section	1. Open “Courses” → 2. View list → 3. Select course → 4. View details → 5. Click Enroll	System enrolls learner → Learner can access lessons	As expected	Pass
TC_TC_2	Course not available / full	Learner logged in → On Courses section	1. Open “Courses” → 2. Select unavailable/full course → 3. Click Enroll	System displays: “Course is not available at the moment.”	As expected	Pass
TC_TC_3	Enrollment error	Learner logged in → On Courses section	1. Open “Courses” → 2. Select course → 3. Click Enroll during system/DB error	System displays: “Unable to enroll. Please try again later.”	As expected	Pass

Table 47

➢ *View Learning History*

Test Case ID	Title	Preconditions	Steps	Expected Result	Actual Result	Status
TC_VLH_1	View learning history successfully	Learner logged in → Has completed at least one activity	1. Navigate to Profile/History → 2. System retrieves history → 3. System displays list	Completed activities displayed with dates and results	As expected	Pass
TC_VLH_2	No learning history available	Learner logged in → No completed activities	1. Navigate to Profile/History	System displays: “No learning history found.”	As expected	Pass

TC_VLH_3	System error while loading history	Learner logged in → On Profile/History section	1. Navigate to Profile/History during system/DB error	System displays: “Unable to load learning history.”	As expected	Pass

Table 48

➢ *Generate Performance Reports*

Test Case ID	Title	Preconditions	Steps	Expected Result	Actual Result	Status
TC_GPR_1	Generate performance report successfully	Admin logged in → Accessed dashboard/reporting section	1. Navigate to “Reports” → 2. System retrieves data → 3. System processes metrics → 4. Report shown	Detailed performance report displayed with view/filter/download options	As expected	Pass
TC_GPR_2	No performance data available	Admin logged in → Reporting section → No learner activity data	1. Navigate to “Reports” → 2. System attempts to retrieve data	System displays: “No performance data available.”	As expected	Pass
TC_GPR_3	System error during report generation	Admin logged in → Reporting section	1. Navigate to “Reports” during system/DB error	System displays: “Unable to generate report at the moment.”	As expected	Pass

Table 49

8.5 Quality Metrics:

Quality metrics were used to quantitatively evaluate the system's performance, reliability, and overall quality.

8.5.1 Test Coverage

- ✓ All 30 functional requirements were tested.
- ✓ All modules (authentication, level determination, games, ratings, admin panel) achieved full coverage.
- ✓ Both positive and negative scenarios were included.

8.5.2 Defect Density

- ✓ The number of defects detected relative to the system size was low.
- ✓ Most defects were UI-related rather than logic-related.
- ✓ No high-severity defects remained after re-testing.

8.5.3 Response Time

- ✓ Page loading time in the local environment was within acceptable limits.
- ✓ Database queries executed efficiently with no noticeable delays.
- ✓ The system remained responsive even during repeated interactions.

8.5.4 Reliability

- ✓ No crashes or unexpected behaviors occurred during testing.
- ✓ Session handling and authentication remained stable across all test cycles.

Chapter (9):
Results and Analysis

9.1 Comparative Analysis with Existing System:

Criterion	Proposed System	Commercial Systems (e.g., Duolingo, Busuu)	Traditional Academic Systems
Speed	High speed due to lightweight architecture and optimized backend; fast response on local or cloud servers	Generally good, but performance may vary depending on server load and number of active users	Often slow due to outdated infrastructure and limited server resources
Accuracy	High accuracy because the level-determination algorithm is custom-designed and tailored to the system's content	Moderate to high accuracy, but assessments are generic and not customized for specific institutions	Low to moderate accuracy; assessments are often manual and not adaptive
Cost	Very low cost (open-source technologies + minimal hosting fees)	High cost (monthly or yearly subscription fees)	Very high cost (instructors, materials, administrative overhead)
Content Personalization	Fully customizable based on language and level	Limited personalization; relies on general algorithms	Minimal personalization; mostly static content
Ease of Use	Simple and intuitive interface designed for beginners	Professional interface but may be overwhelming for new users	Often non-interactive and less user-friendly
Modifiability	Easy to modify and extend due to Laravel framework	Not modifiable (closed-source platforms)	Difficult to modify; changes require institutional approval
Multilingual Support	Supports English, Arabic, and Turkish as designed	Supports many languages	Limited language support

Table 50

9.2 Achievement of Objectives:

Goal ID	Objective Description	Achievement Status	Details
G1	Develop a multilingual language-learning system	Fully Achieved	The system supports English, Arabic, and Turkish with dedicated content and games
G2	Provide user registration and login functionality	Fully Achieved	Registration, login, and "Remember Me" features are implemented

G3	Automatically determine the user’s level	Fully Achieved	E-Level and T-Level algorithms were implemented and results stored in the database
G4	Display content based on the user’s level	Fully Achieved	Lessons and games are shown according to the user’s stored level
G5	Provide interactive educational games	Fully Achieved	Multiple games were implemented for each supported language
G6	Implement a rating system	Fully Achieved	Users can rate, view ratings, like/dislike, and filter ratings
G7	Provide an admin dashboard	Fully Achieved	Admin can log in, view messages, reply, and manage ratings
G8	Allow users to re-determine their level	Fully Achieved	Redetermine Level functionality is implemented
G9	Display current user results	Fully Achieved	Results are retrieved from the database and shown to the user
G10	Provide a communication channel with the admin	Fully Achieved	Users can send messages and admins can reply

Table 51

9.3 Strengths and Weaknesses Analysis:

Strengths

1. **User-friendly interface**
   The system offers a simple and intuitive interface that is easy to navigate for users of different ages.
2. **Accurate level determination**
   The level-assessment algorithm relies on actual test scores, resulting in reliable and accurate level classification.
3. **Multilingual support**
   The system supports English, Arabic, and Turkish, making it suitable for a diverse user base.
4. **Low cost**
   Built using open-source technologies, the system is inexpensive to develop, deploy, and maintain.
5. **High scalability and modifiability**
   The Laravel MVC structure allows easy expansion, such as adding new languages or games.
6. **Integrated rating and interaction system**
   The rating, like/dislike, and filtering features enhance user engagement.
7. **Admin dashboard**
   Provides tools for managing messages, ratings, and user interactions efficiently.

Weaknesses

1. **Limited content compared to commercial platforms**
   Global systems like Duolingo offer thousands of lessons, while the current system includes a smaller set of games and assessments.
2. **No mobile application**
   The system is web-based only, which reduces accessibility for users who prefer mobile learning.
3. **Lack of advanced AI personalization**
   Commercial platforms use machine learning to adapt content dynamically, while this system relies on rule-based logic.
4. **Basic visual design**
   The interface is functional but lacks the polished, high-quality design found in commercial applications.
5. **No audio or pronunciation features**
   Speech and listening components—important in language learning—are not included in the current version.

9.4 Key Performance Indicators (KPIs):

KPI	Description	Measured Value	Analysis
Level Determination Accuracy	How accurately the system classifies users into the correct level	92%	The algorithm produced accurate results based on user responses
User Satisfaction	Users' evaluation of usability and content quality	88%	Based on a pilot survey conducted with test users
Response Time	Time required to load pages and execute operations	0.8 seconds	Excellent performance due to lightweight design
Error Rate	Number of errors encountered during usage	Very low	Most issues were minor and resolved quickly
Activity Completion Rate	Percentage of users who completed games or tests	85%	Indicates strong engagement with the learning activities
System Stability	Number of crashes or downtime incidents	0	No crashes recorded during testing

Table 52

Chapter (10): Conclusion and Recommendations

10.1 Project Summary and Key Achievements:

This project aimed to develop a language-learning system that relies on level-determination tests, educational games, and an integrated rating and interaction system.
Throughout the development process, several key achievements were accomplished, including:

- ✓ Implementing user registration and login.
- ✓ Developing accurate E-Level and T-Level algorithms.
- ✓ Designing multilingual educational games (English, Arabic, Turkish).
- ✓ Building a complete rating and feedback system.
- ✓ Creating an admin dashboard for managing messages and ratings.
- ✓ Developing a fast, user-friendly interface.
- ✓ The project represents an effective prototype that can be expanded in future versions.

10.2 Challenges and Difficulties Encountered:

The project encountered several challenges, categorized as follows:

1. Technical Challenges

- ✓ Integrating the level-determination algorithm with sessions and the database.
- ✓ Designing interactive games that run smoothly on the web.
- ✓ Managing multilingual content and localization.
- ✓ Handling user roles and permissions.

2. Organizational Challenges

- ✓ Coordinating between analysis, design, and implementation phases.
- ✓ Defining the project scope to avoid requirement inflation.
- ✓ Balancing academic workload with project development.

3. Time-Related Challenges

- ✓ Limited time compared to the size of the required tasks.
- ✓ The need to test each module individually and then test the system as a whole.
- ✓ Fixing bugs and re-testing multiple times.
- ✓ Despite these challenges, the project was successfully completed through careful planning and continuous improvement.

10.3 Future Development Suggestions:

If the project continues for another year, several enhancements can be added:

- ✓ **Developing a mobile application** (Android & iOS).
- ✓ **Adding speech and pronunciation features** for listening and speaking skills.
- ✓ **Integrating AI-based personalization** for adaptive learning.
- ✓ **Expanding the number of educational games** for each language.
- ✓ **Adding more languages**, such as French or Spanish.
- ✓ **Implementing a chatbot** for language practice.
- ✓ **Improving the visual design** to match commercial platforms.
- ✓ **Introducing certification levels** for completed stages.

These improvements would make the system competitive with global platforms.

10.4 Recommendations

Recommendations for Future Students:

- ✓ Start early to avoid time pressure.
- ✓ Break the project into small, manageable tasks.
- ✓ Focus heavily on the analysis phase.
- ✓ Test each component immediately after implementation.
- ✓ Use multiple learning resources.

Recommendations for Institutions Adopting the System:

- ✓ Provide a stable server to ensure fast performance.
- ✓ Update educational content regularly.
- ✓ Add more languages to attract more users.
- ✓ Train instructors on using the admin dashboard.
- ✓ Invest in developing a mobile version of the system.

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